

Spectrum Power 5
Communication Front-End
(CFE)

5.30

Technical Specification

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NOTE

For your own safety, observe the warnings and safety instructions contained in this document, if available.

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Revision Record

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Preface

Notes on Safety

This manual is not a complete index of all safety measures required for operation of the equipment (module or device). However, it comprises important information that must be followed for personal safety, as well as to avoid material damage. Information is highlighted and illustrated as follows according to the degree of danger:



WARNING

WARNING means that death or severe injury **may** result if the measures specified are not taken.

- ✧ Comply with all instructions, in order to avoid death or severe injuries.



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NOTE

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Scope

This document provides specific and detailed information on how to use a particular product or product component.

**NOTE**

Note that the screenshots used in this document contain sample data which may not be available in some systems.

Documentation Conventions

Example	Meaning
File > Open	Selecting the menu item Open on the File menu
F1	Pressing the F1 function key on the keyboard
Control+o	Holding down the Ctrl key and pressing the letter o on the keyboard
Control+Shift+o	Holding down the Ctrl key and the Shift key and pressing the letter o on the keyboard
Left-click	Clicking the left mouse button
Right-click	Clicking the right mouse button
Shift-click	Holding down the Shift key on the keyboard and clicking the left mouse button

Questions?

This document is part of a set of manuals that describes the complete product.

Various other product modules may be mentioned or discussed in this document. For more detailed information or if you have any questions about these products, please contact your Siemens representative.

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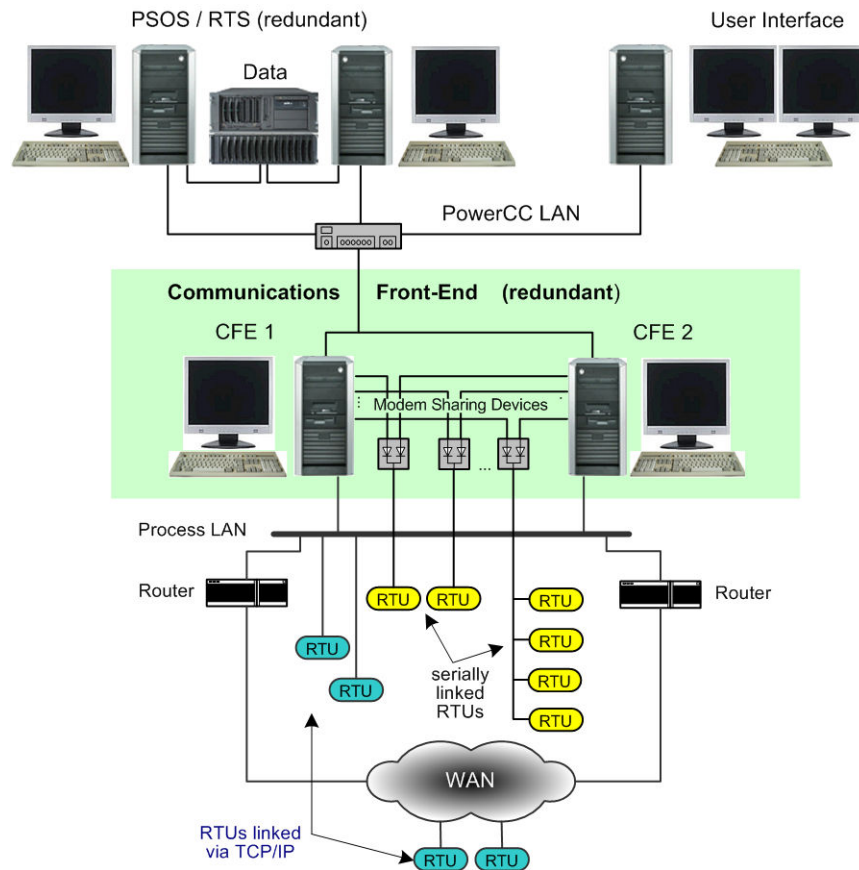
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1 Introduction

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1.1 Basics

The Communication Front End (CFE) is a PC-based process interface for coupling RTU of different manufacturers to the Spectrum Power 5 distributed control center system. It operates as a server of the control center system and communicates with the remaining Spectrum Power 5 servers via the Spectrum Power 5 LAN. For additional information, refer to Spectrum Power 5 Configuration with Redundant CFE Subsystems and Different RTU Connections



[cfets007-040509-01.tif, 1, en_US]

Figure 1-1 Spectrum Power 5 Configuration with Redundant CFE Subsystems and Different RTU Connections

The CFE consists of a standard PC. Communication with the RTU is carried out via standard COM ports or Ethernet. Depending on the number of RTU to be connected, boards with 8, 16 or 32 COM ports are used. CFE is a fully integrated Spectrum Power 5 subsystem. Data modeling and parameter entry is carried out in the Spectrum Power 5 environment. All data and settings are stored in the central Spectrum Power 5 database.



NOTE

CFE is the only front-end system used in Spectrum Power 5. If the number of COM ports required is more than what one server can accommodate, additional CFE servers are configured.

2 Functionality

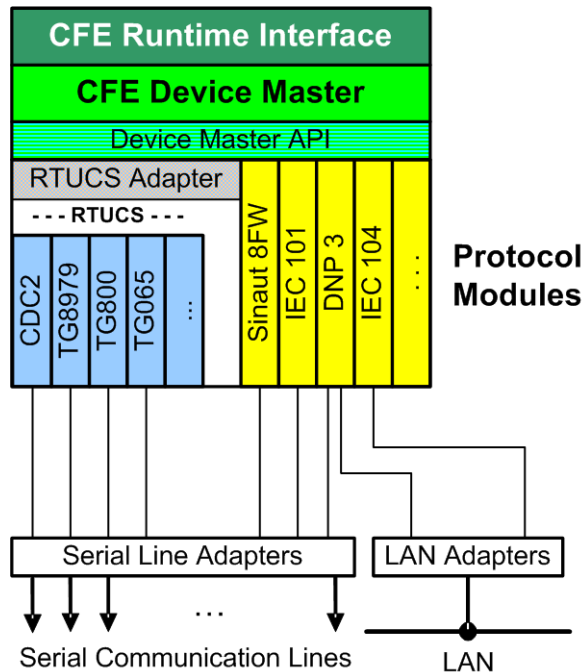
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2.1 Components for Communication and Processing

Major software components of CFE are the **Protocol Modules**, the **Device Master** and the **Runtime Interface**. The **Protocol Module** performs all protocol-related communication services like frame check, acknowledgments on link layer, supervision of communication links and similar. Some of the Protocol Modules are interfaced through a special adapter type **RTUCS**.

The **CFE Device Master** carries out data preprocessing routines. For example General Interrogation, Completeness Check, Old/New Check, Intermediate State Suppression, Scaling of analogs etc.

The CFE Runtime Interface (CfeRti) is responsible for the communication (IPC) between CFE and higher ranking system components.



[cfets026-140509-01.tif, 1, en_US]

Figure 2-1 Software Components for RTU Communication: Device Master and Protocol Modules

Sinaut 8FW and other bit-oriented protocols like CDC II are currently not supported by Power 5 because no communication hardware exists of the shelf which is compatible with the delivered Windows 2008 Server hardware and software. Nevertheless, the protocols remain in this documentation for reference and future use.

2.2 Communication Services

Reception and transmission of character-serial messages (RS232) are carried out on the COM board and processed by the Protocol Module. This software package performs all protocol-specific communication procedures.

In case of protocol IEC 60870-5-101 the main tasks comprise:

- Checking the incoming message frames (parity, block check)
- Forming the check code for outgoing message frames.
- Acknowledging received message frames in balanced mode (spontaneous data transmission)
- Repetition of message frames that are not acknowledged by the RTU (in time)
- Requesting data classes in unbalanced mode (polling mode)
- Management of Frame Counts and Data Flow Control flag.
- Periodic Status of Link Request (in case of broken links)
- Resetting Status of Link (when the link comes up)
- Scanning of RTUs in unbalanced mode.
- Time triggered transmission of synchronization messages.

For the communication service, the following IEC 60870-5-101 protocol attributes are set or modified by the user for each communication channel individually.

Attribute	Range
Acknowledge time-out	from 1 ms to n [ms]
Acknowledge type	Single character / short telegram
Baud rate	from 300 to 38400
Data length	Max. 255
Serial port	Depends on the hardware of the CFE server.
Transmission mode	Balanced / unbalanced

For the communication link:

Attribute	Range
Link address	from 1 to 65535
Priority	from 1 to 256
Successive errors for RTU fail	from 1 to 10
Time synchronization interval	from 0 (disabled) to n ¹⁾ [ms]
General Interrogation cycle time	from 0 (disabled) to n [ms]
Polling time	from 0 (disabled) to n [ms]
Polling time-out	from 0 (disabled) to n [ms]

¹⁾ specifies max. value for n is 2 147 483 647 (long integer)



NOTE

For definition of protocol-specific terms, like **Link Address**, **Polling Time** and others, refer to the dedicated protocol literature.

For other protocols than IEC 60870-5-101, suitable procedures and parameters are implemented to enable the interface to communicate with the connected RTUs.

2.3 Characteristic Curve

The characteristic curves serve to transform raw values, received from RTU, into physical values (engineering values) as required in graphical diagrams or in the archive. It depends on the protocol and message format how the analog is encoded and what the range of valid values is. Characteristic curves are used in the opposite direction for analog set points and for forwarded analogs. In this case, the conversion is from engineering value coming from any application server (for example RTS) into a protocol specific raw value.



NOTE

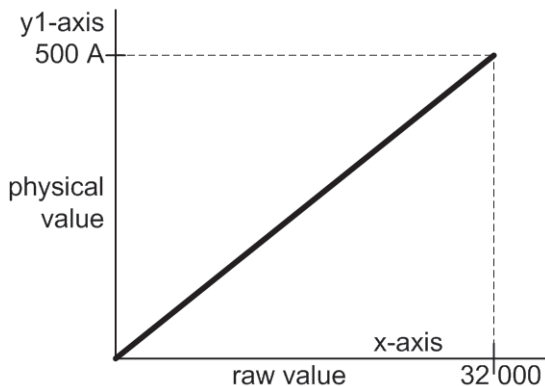
A characteristic curve is not a real curve: it is a straight line or consists of 2 or more straight segments as shown in Figure.

There is no need to define a characteristic curve for every single analog value in the system. One characteristic is allocated to different analog measurements having the same transducer in the substations bay and the same type of ADC (analog to digital converter) in the RTU.

The following 4 basic types of characteristic curves are configured in Spectrum Power 5:

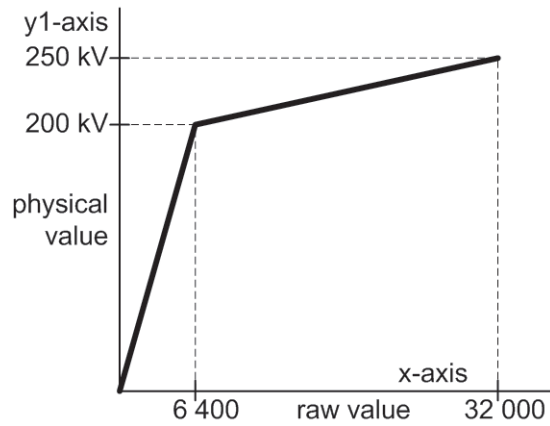
- Linear, unipolar
- Non-linear unipolar
- Linear, bipolar
- Non-linear, bipolar

The following 3 diagrams show typical characteristics to convert raw values of protocol IEC60870-5-101, type ID 9 (Normalized Value) into physical value. In case of IEC101 (or IEC104) protocol the raw value ranges between -215 up to 215 (-32 767 ... 32 768), the physical range and unit are any one.



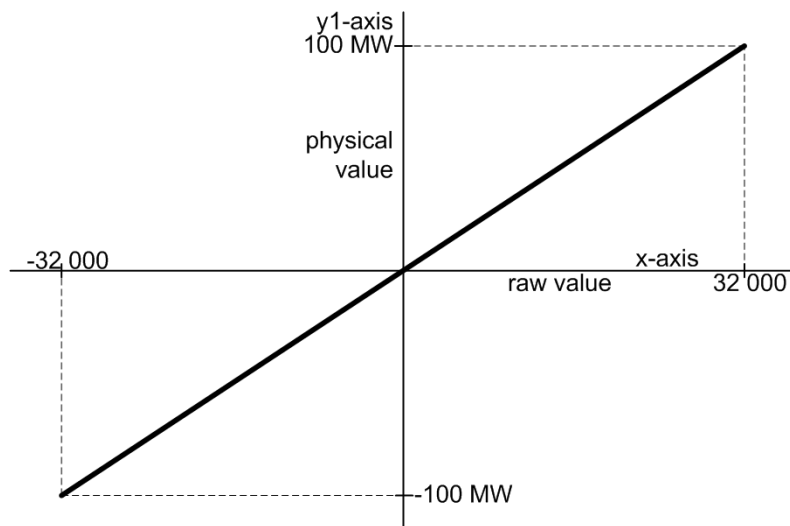
[cgcf126-230609-01.tif, 1, en_US]

Figure 2-2 Linear, Unipolar Characteristic



[cgcf127-230609-01.tif, 1, en_US]

Figure 2-3 Non-Linear, Unipolar Characteristic



[cgcf128-230609-01.tif, 1, en_US]

Figure 2-4 Linear, Bipolar Characteristic

2.4 CFE / RTU Status Information and Communication Statistics

Status information and communication statistics is system information. Unlike process information, that is gathered from switchgear by the RTU, this type of information is generated in CFE or in RTU by software routines. Status information is seen as optional information. There is no real need to take care on this type of data. Without any configuration work they are listed in the Runtime Explorer shows their current value. However, if the customer wants to display status information or statistic counters in customized diagrams or put them into a log or into the archive, some configuration work is done. This is described in the following sections.

Status Information

Status information is available for RTUs and CFE components, like CFE interfaces, links to RTUs. They provide information about the availability of every component.

Values for CFE Status information points are:

- FAILURE
- PREFERRED
- ACTIVE
- LOCKED

The following tables summarize the status information for CFE and RTU:

CFE Status	Information Interface (COM Port)	Link to RTU
FAILURE	x	x
ACTIVE (process controlling)	x	x
LOCKED	x	x
PREFERRED	Only for dependent communication channels	Only for independent communication channels
LISTEN MODE	x	-
CLOCKSYNC ENABLE	-	x

CFE RTU Status Information	Remark
RTU_FAILURE derived in CFE	derived in CFE
GENERAL INTERROGATION RUNNING	derived in CFE
GENERAL INTERROGATION INCOMPLETE	derived in CFE
GROUP INTERROGATION RUNNING	derived in CFE
GROUP INTERROGATION INCOMPLETE	derived in CFE
RENEWAL CHECK	derived in CFE



NOTE

System information that is generated and transmitted by the RTU is not considered as a type CFE RTU status information. It is treated like process information.

Typical RTU system information for example:

- RTU DISTURBED.
- RTU BUFFER OVERFLOW.
- RTU HARDWARE FAILURE.

Communication Statistics

Communication Statistics is obtained from system counters integrated in some of the protocol software packages

Typical communication statistics are:

- Number of sent messages
- Number of received messages
- Number of received corrupted messages
- Number of scan requests

Check the protocol of interest, whether communication statistics are implemented and if so, what types of counters are available.

2.5 Test and Diagnosis

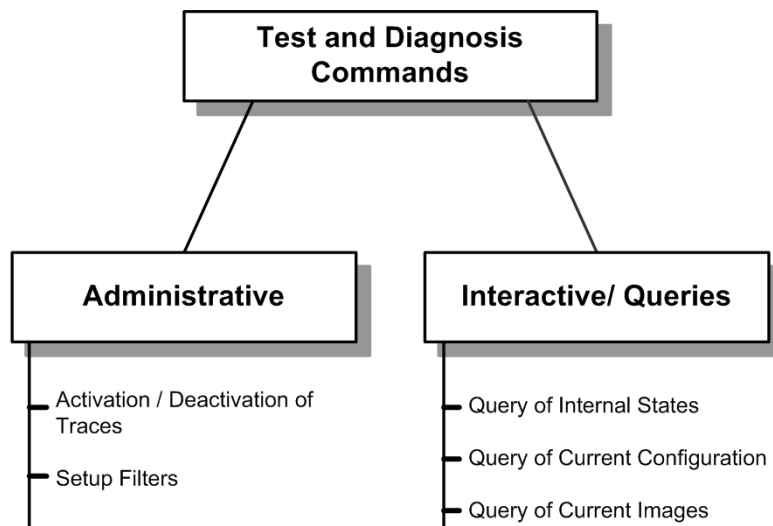
2.5.1 General

Purpose

Test and Diagnostic Tools are used to analyze and solve problems during configuration and operation of the Communications Front-End (especially during commissioning).

Test and Diagnosis Commands Provided

CFE provides a set of test and diagnostic utilities to get an overview of the current configuration of the devices and measurement relevant for the CFE. It is also possible to get information about communication statistics and to query internal status information. In order to achieve a high user acceptance the approach to Test and Diagnosis capability is completely done by means of a graphics user interface. The following figure depicts the different kinds of Test and Diagnosis commands.



[cgcfe152-230609-01.tif, 1, en_US]

Figure 2-5 Test and Diagnosis Commands - Overview

Trace Commands

Trace Commands are used to control the trace output for a selected CfeInterface respectively its corresponding CFE process. In that context it is possible to activate the trace and setup filters.

For additional information refer Using Test and Diagnosis Tools

Query Commands

Query Commands provide a direct interaction between the user and the CFE application so it is possible to query internal states of devices (interfaces and links). It is also possible to read out the current status of the redundancy management and (if available) to query internal process images of a whole remote terminal unit or selected values. Furthermore Query Commands offer the facility to get information about pre-processing. For example the result of the last general interrogation (in order to find out what GI-relevant information objects is not transmitted) or the status of the renewal control

Operational Commands

Operational commands used for instance to lock or unlock a CfeInterface or CfeInterfaceLink or setting a preference are not part of Test and Diagnosis User Interface. Commands of that kind are integrated in the Runtime Explorer and not subject of this description.

2.5.2 Facilities Overview

As mentioned in earlier section, there are 2 kinds of Test & Diagnostic Tools to help users to get knowledge about the internal status of devices and communication links and the data flow from the Remote Terminal Unit to the RTS processing. These are the trace facilities and the queries.

Trace Facilities

- Set individual trace level and filters adjustable per CfeInterface.
- Execute technological trace in order to track
 - A single information object (specified by the path name of a Measurement) or
 - The complete transferred information of a selected Remote Terminal Unit from the received protocol specific telegram up to the SCADA interface. The processing of that information through all layers of the CFE is visible.
- The amount of trace information is scalable due to adjustable debug levels.

Query of Internal States

- Getting redundancy information on:
 - The selected interface,
 - One or all links to the selected interface.
- Getting preprocessing state
 - Result of the last General Interrogation in order to identify missing information objects those raised a failed Completeness Check.
 - Detailed result of last Renewal Check.

Query of Communication Statistics

Getting data throughput and Communication errors

Query Process Image

Reading the image of an individual substation or of selected substations of the same party line.

2.6 Management of Redundant Links

2.6.1 Overview

If the Communication Front-End System is redundant, each RTU is connected: at CFE 1 and at CFE 2. The modem lines in control direction are de-coupled through active or passive modem sharing devices (modem splitters). Such a Y-formed link is described in the data model as 2 dependent links (link 1, link 2), which are combined in a Dependency Group.

If the RTUs have 2 independent communication interfaces, completely redundant and independent links are established to the RTUs. In this case, neither modems sharing devices nor the definition of Dependency Groups are required. In each case, the existence of more than one link to an RTU at the front-end system, a criterion is needed to decide whether a link becomes active or standby at system run-up and during operation.

The following settings serve as a criterion:

- In IMM: setting priority attribute to high or low.
- In operation: setting the status to **PREFERRED** or **LOCKED**.

Setting Priority

By setting a higher or lower priority it is defined, which CFE interface and, which link (to the RTU) becomes active at run-up. Interfaces and links with lower priority operate in standby mode; those with high priority become active.

Status **PREFERRED**

By the controllable status information **PREFERRED** the operator changes the preset configuration and thus invoke a change of roles between active and standby interfaces or links. The modified configuration is retained until it is made undone manually. Even a system shutdown and run-up will not reset the configuration.

The immediate switch-over only takes place if the preference is issued to an intact standby channel. If the standby channel is out of operation when setting the preference the switch-over takes place at a later time when it is available again.



NOTE

Status **PREFERRED** on an interface is only applicable for dependent interfaces (connected to single ported RTUs). Status **PREFERRED** on a link is only applicable for independent interfaces (connected to dual ported RTUs).

Status **LOCKED**

An interface or a link is completely deactivated by means of the status information **LOCKED**. If locking is set to an active interface or link a fail-over to the (intact) standby component takes place.



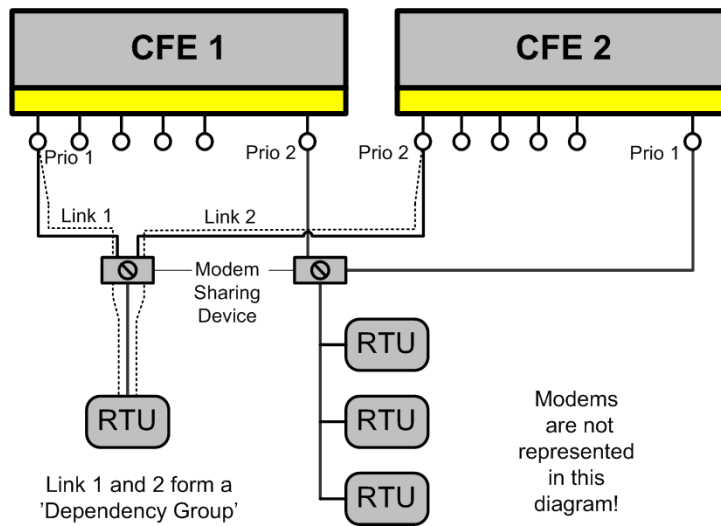
NOTE

Locking an active interface or link results in a complete breakdown of the RTU communication when the standby component fails or has already failed when setting the locking.

2.6.2 Dependent Links

The diagram below shows dependent links (with only one communication port per RTU). The links are dependent as they use the same physical way to the RTU and thus, they have an identical parameter setting, matching to the RTU interface. Logical links to the same RTU interface and thus having identical communication parameters are called Dependency Group.

This figure shows dependent links (point-to-point and party line)



[cgcfe133-230609-01.tif, 1, en_US]

Figure 2-6 Dependent Links

The following rules are observed for configuration:

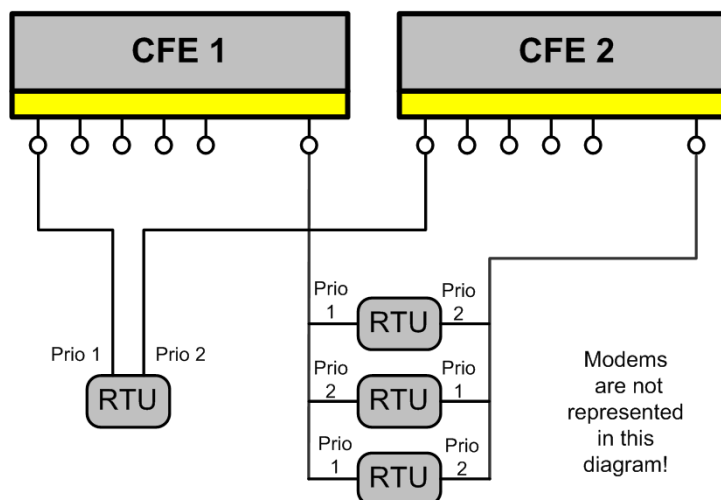
- The (2) links to an RTU have different priority, for example link1 (at CFE 1) => priority 1, link 2 (at CFE 2) => priority 2 (or either ways).
- For party lines, all links of a CFE interface have the same priority.

The status information **PREFERRED** is set to the CFE interface and not to the links.

2.6.3 Independent Links

The following diagram explains independent links that is with 2 independent communication interfaces per RTU. The links are independent as they are physically separated and thus they have different communication parameters. Depending on the type of RTU, also configurations with more than 2 links are feasible.

This figure shows independent links (point-to point and party line)



[cgcfe134-230609-01.tif, 1, en_US]

Figure 2-7 Independent Links

The following rules are observed for configuration:

- The (t2) links to an RTU have different priority, for example link1 (at CFE 1) => priority 1, link 2 (at CFE 2) => priority 2 (or vice versa).
- For party lines, the links of a CFE interface have different priorities.

However, whether the communication load is better shared between both CFE interfaces by this method depends on the features of the communication protocol and the ability of the RTU. The status information **PREFERRED** is set to the CFE links and not to the CFE interface.

2.7 Data Acquisition

2.7.1 Overview

There are 2 basic ways of controlling the data flow from RTU to the CFE in the Control Centre:

- Data transmission controlled by the RTU:
Data is sent by the RTU (substation) either spontaneously on change, or (in some cases) in periodical intervals depending on the configuration of the **Transmission Mode Control** of the RTU.
- Data transmission controlled by the Communication Front End:
Data is sent by the RTU only on request from Communication Front End. This covers:
 - Single data object requests.
 - Requests of groups of data objects (like General Interrogation)
 - Requests of events saved in RTU data buffer.

Requests are generated periodically or based upon events, including user intervention. These methods are known as Scan Control.

For additional information, refer to [2.7.6 Scan Control, Scan Groups, Scan Volumes](#).

The CFE allows both types of data acquisition to occur in parallel. However, some RTU protocols only permit one type.

2.7.2 General Interrogation

General Interrogation commands (GI) are used to update the system database or areas of the database. It is done after startup of a system component, timer-controlled if configured so, manually by operator intervention or triggered by any received message if configured accordingly and supported by the protocol implementation. For example on **RTU buffer overflow** message.

- Startup General Interrogation
The CFE automatically initiates and monitors for completeness when:
 - The CFE is started up
 - A RTU is started up
 - The GI is automatically released as soon as the link is re-established (disappearing link failure). In addition, an optional run-up message from RTU releases an automatic GI. It depends on protocol and settings. For example TID70 in protocol IEC 60870-5-101.
- General Interrogation after line switchover
After a failover or a manual switchover of communication links CFE issues automatically a GI over the new active link. This functionality is protocol specific implemented and available for the protocols IEC 101 and IEC 104.
- General Interrogation triggered by a message
An automatic GI command is issued on reception of a dedicated (service-) message from RTU. The GI command is sent either on appearing or on disappearing of the message information. Typical RTU service messages are:
 - Buffer Overflow
 - Information Loss
 - Hardware Failure
- Periodic General Interrogation
In addition to the previously mentioned GIs, the CFE offers the possibility of periodically repeated GI. The GI periodicity can be configured for each RTU individually.

Both, the criteria for a GI initiation as well as the returned information that is included in completeness check, are configured.

For additional information on completeness check, refer to [2.7.4 Completeness Check](#).

2.7.3 IEC 101 IEC104 Group Interrogation

General

In addition to the General Interrogation, some protocols allow to specify a number of smaller data subsets that are interrogated from RTU.

The implementation of the Group Interrogation for the protocols IEC 101 and IEC 104 features as following:

- Up to 16 Interrogation Groups is configured
- Each Interrogation Group is monitored for completeness
- For each Interrogation Group, individual Status Information **Group Interrogation Running** and **Group Interrogation Incomplete** is available
- Group Interrogation commands are configured to run periodically at predefined time ranges or on operator intervention. The configuration of an individual offset is also possible.
- Data points are allocated to Interrogation Groups in a flexible way:
They are allocated to one group, to several groups, to all groups or to none.

Protocol Details

The standard protocols IEC 101 and IEC 104 use a dedicated message type to organize the General Interrogation as well as the Group Interrogation:

Type ID: 100 Code:	C_IC_NA_1
Information Object address:	0
Cause of Transmission (CoT):	Refer the following table
Qualifier of Interrogation (QoI):	Refer the following table

Activation	CFE	6	GI: 20	-
Activation Confirmation	RTU	7	Group 1 16: 21 ...36	Activation Confirmation not evaluated
Data Transmission	RTU	Refer succeeding section		-
Activation Termination	RTU	10		Option Activation Termination is recommended

Preferred message types for interrogated data are ASDUs without time stamp. For example type 1 (single-point indications), type 3 (double-point indications), type 9 (Measured values) or similar. Interrogated messages have a CoT of 20 (GI) or 21 (Group 1) up to 36 (Group 16), respectively.

In order to increase the transmission effectivity, information objects of the same type are combined in one message block by the RTU.

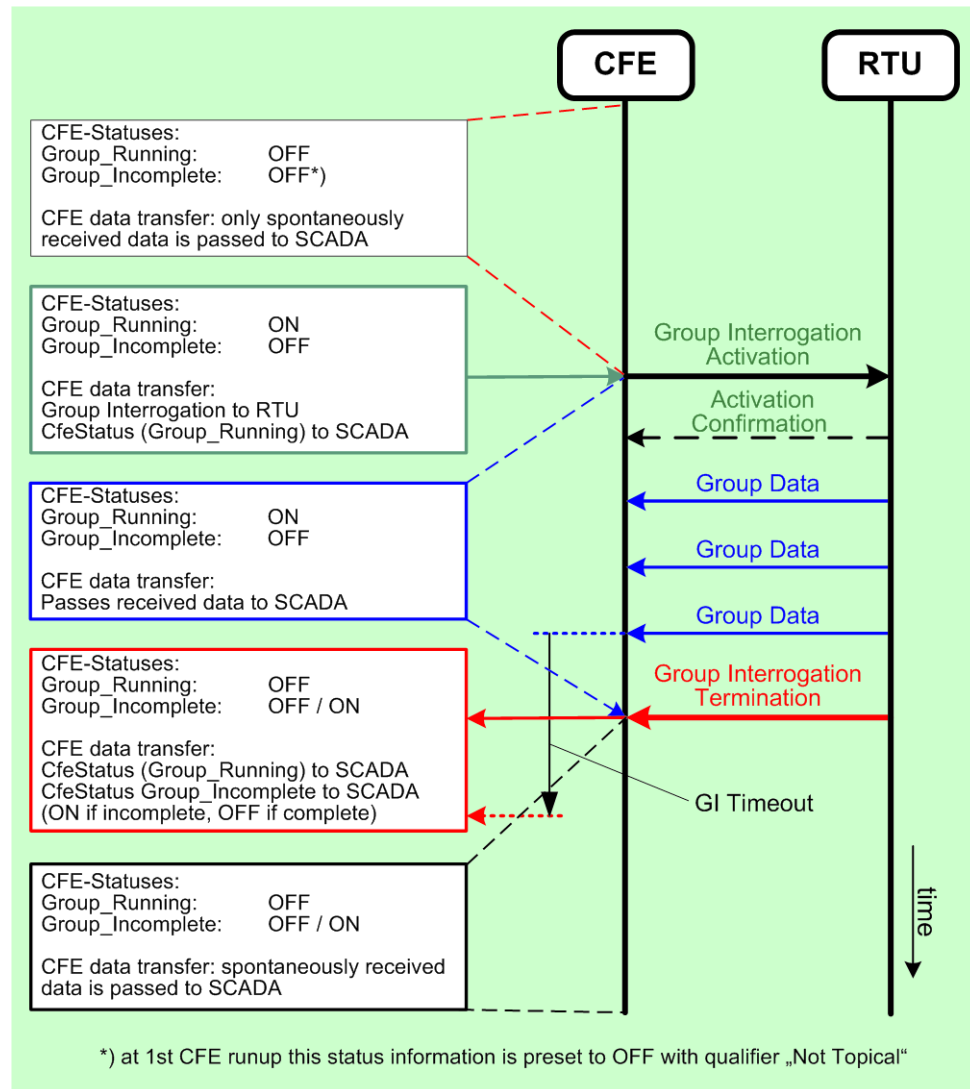
Activation Confirmation: The RTU sends an **Activation Confirmation** message after receiving an **Activation** command. If so, CFE acknowledges it by the link layer as appropriate. However, there is no further processing by the application layer. As such, this message is omitted by the RTU without adverse effect.

Cause of Transmission: The Cause of Transmission (CoT) for return messages as per IEC Standard is 20 for GI messages, and 21 to 36 for Group 1 to Group 16 messages. If the RTU sends data with an incorrect CoT, the data is passed on to SCADA, but not included in completeness check.

Activation Termination: After the requested Interrogation Group is transmitted, the RTU is to send an **Activation Termination** message. This message causes the CFE to start the completeness check without further delay and to terminate the interrogation process. If the termination message is missing, the GI time-out expires and triggers eventually the completeness check at a later time. The GI time-out parameter (**Application time-out**) is configured for each RTU individually.

CFE – RTU Dialog and Status Information

The following diagram shows the data flow between CFE and RTU as well as corresponding CFE Status information that is generated by CFE and transferred to SCADA.



[cfets027-140509-01.tif, 1, en_US]

Figure 2-8 Group Interrogation: CFE - RTU Dialog and Associated Status Information

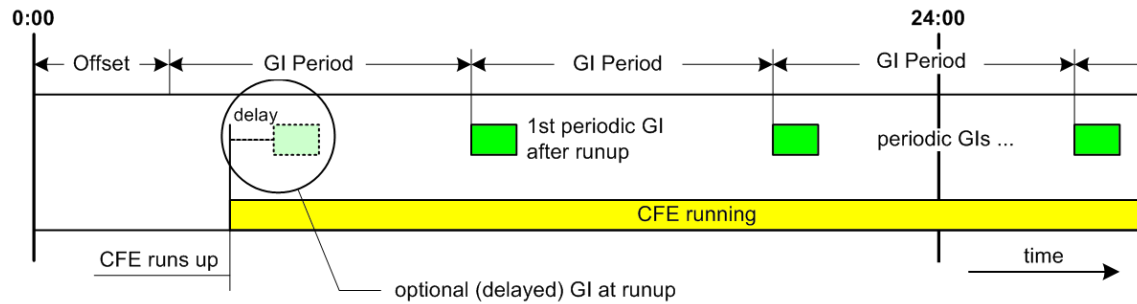
Triggering a Group Interrogation

Manual Trigger: If the Man-Machine Interface provides interactive **Group Interrogation** elements, the operator is in a position to trigger a Group Interrogation command at any time and for any given reason. As long as no interrogation process is just running, the manual interrogation is executed without any delay.

Periodical Trigger: It is possible to interrogate groups of data periodically by CFE in fixed intervals. For each data group the following parameters can individually be configured.

- Period (any reasonable value in ms)
- Offset (delay in ms related to 0:00 clock UTC)
- Delay Time (runup delay in ms after CFE runup)

The following diagram shows a periodical Interrogation Group with offsets and (optional) runup delay.



[cfets028-150509-01.tif, 1, en_US]

Figure 2-9 Periodical Group Interrogation with Offset and Delayed Runup-GI

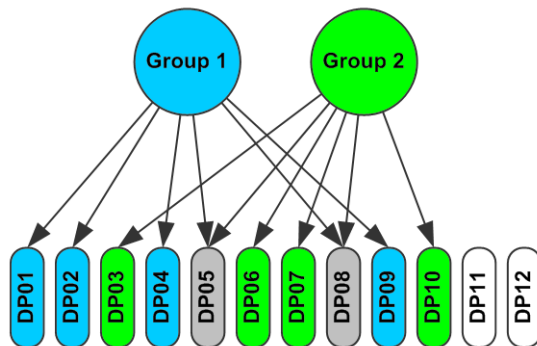


NOTE

Interrogations are executed one by one. As long as an interrogation is not terminated all further interrogation requests (periodical or manual) are queued.

Allocating Data Points to Interrogation Groups

Each data point can be allocated to interrogation groups in a flexible way, as shown by the following example. Data points DP01, DP02, DP04, DP09 are allocated to Group 1 (blue colored). Data points DP03, DP06, DP07, DP10 are allocated to Group 2 (green colored). Data points DP05, DP08 are allocated to both interrogation groups (gray colored). Data points DP11 and DP12 are not included in any interrogation group (white).



[cfets029-150509-01.tif, 1, en_US]

Figure 2-10 Example: Allocating Data Points to Different Interrogation Groups

2.7.4 Completeness Check

With a certain delay, **Completeness Check** follows automatically to the interrogation command. The check is initiated either on receiving a termination message (IEC 101 IEC 104) or on time-out. Time-out is always the criterion if the RTU or the protocol does not support any Interrogation Termination indication, or if data acquisition fails in RTU. The interrogation supervision time is retriggered every time an interrogation message is received.

The result of the completeness check is complete or incomplete

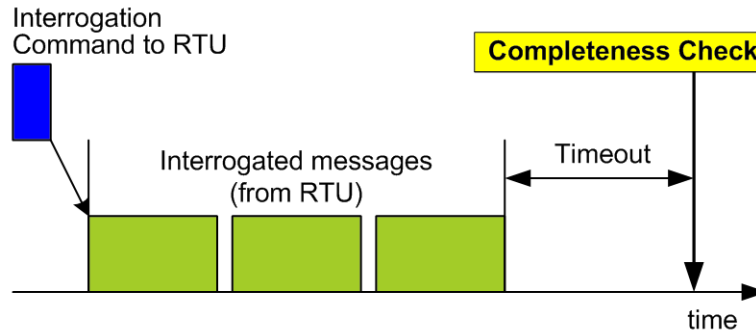
- **Complete**

The interrogation is considered complete, if all expected data points were received when the completeness check is carried out.

- **Incomplete**

The interrogation is considered incomplete, if any (one, several, all) of the expected data points was/ were not received when the completeness check is carried out. In case of General Interrogation (CoT=20), the interrogation is repeated for a pre-configured number of times. Appropriate status information is generated by CFE and sent to SCADA.

As soon as missing information is received from RTU within an interrogation period, or the missing data points are taken out from Completeness Check (via IMM), the status information **Interrogation Incomplete** is set to OFF at next Completeness Check.



[cfets030-150509-01.tif, 1, en_US]

Figure 2-11 Completeness Check

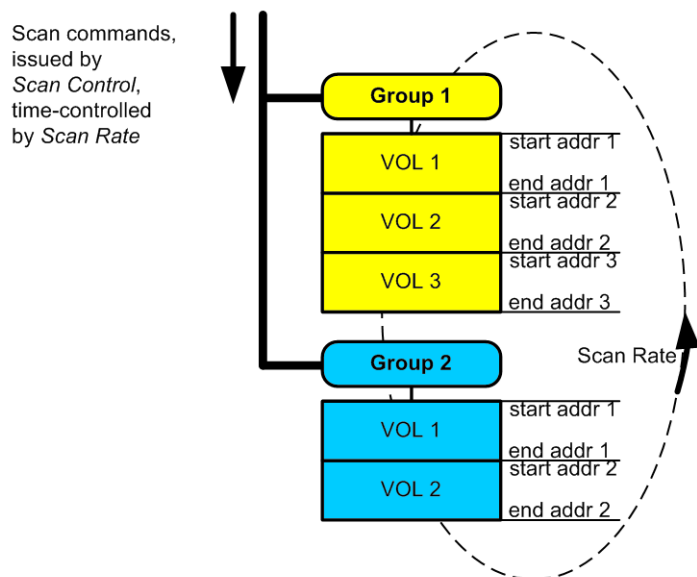
2.7.5 Integrity Scan

Some protocols such as DNP3, CDC2, provide so called **Integrity Scans** to update the database. Like GI, Integrity Scan consists of a command to the RTUs, response messages from RTU and a completeness check carried out through CFE. The difference between GI and integrity scan is the fact that GI is triggered automatically after specific events, like RTU runup, CFE runup. Unlike GI, start and response of Integrity Check depends on the configuration of the Scan Control (Scan Group, Scan Rate, Scan Offset).

2.7.6 Scan Control, Scan Groups, Scan Volumes

Scan Control is implemented for the protocols IEC 101, IEC 104, DNP3, CDC2, TG800, TG8979, TG065, Sinaut 8FW, and MODBUS.

Scan groups are defined within each RTU. Each scan group has characteristics including assigned scan rate and scan type within each protocol. Depending on the protocol, each scan group contains one or more scan volumes (IEC101/104: only one volume per group is possible). Each volume provides protocol-specific attributes. For example Data Class Type, Request Mode, Start, and End Address of data as described in the following figure.



[cfets031-150509-01.tif, 1, en_US]

Figure 2-12 Scan Groups, Scan Volumes, Scan Rates

2.8 Data Preprocessing - Incoming Data Processing

2.8.1 Overview

Data preprocessing is performed in the Device Master module. The goals of data preprocessing in the front-end system are:

- Transforming the process data into a unified protocol independent format.
This is a precondition for transporting the data to the Real Time Processing Domain.
- Reducing the processing load of the Real Time Servers.
In monitoring direction the data stream coming from numerous RTUs increases from idle condition up to heavy load within a short time and also stay high for an unpredictably long time.

The data stream in the control direction is normally of lower volume; commands are issued less frequently and most require operator interaction.

2.8.2 Regular Data

2.8.2.1 Overview

Regular data is acquired in the RTU and transmitted only once, as soon as possible. The transmission from RTU to the Front-End is either spontaneous or within a fixed transmission period.

Renewal Check

For cyclically or periodically transmitted data it is possible to perform a Renewal Check. This check routine makes sure that the monitored data is refreshed within the preset time range. 2 different Renewal Check cycles are configured for each RTU. Each data point is assigned to one (or none) of the check cycles.

Data points assigned to a Renewal Check cycle are checked at the end of that cycle for being updated. Any missing data point results in the CFE Status Information **Renewal Check 1 or 2 Alarm** and the affected data point is tagged with the qualifier **Not Renewed**.

In contrast to regular data, other type of data is first collected in a buffer in the RTU waiting for being called up or to be transmitted twice (once without and once with time stamp).

For additional information, refer to [2.8.4 Sequence of Events \(SOE\) / Double Transmitted Information](#).

2.8.2.2 Digital Values

Old/New Comparison

Received information is subject to Old/New Comparison. The comparison can be configured for all indications, transformer step positions, and comparable digital information. The value and status attributes are compared with those stored in the CFE database. If a change in value or attributes is detected the new information point is stored in the CFE database and passed on to SCADA.

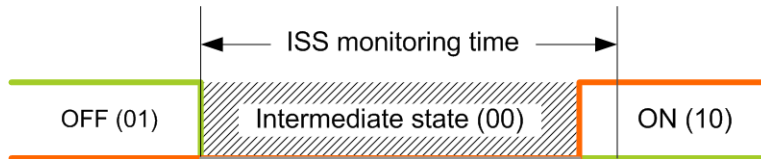
Information Inversion

If the Signal Inversion feature is enabled the transmitted signal is inverted from 0 to 1 or 1 to 0. Signal inversion is applicable for all type of single point and double point information as well as for single and double commands, configurable per data point.

For additional information, refer the entry **Monitor Info Conversion** in the **RTU-Datapoints** chapters of the protocol-specific sections in the Data Modeling Guide CFE.

Intermediate State Suppression ISS

For safety reasons, switch positions are represented by 2 data bits. In this case, the OFF-state is represented by **01**, the ON-state by **10**. When a slowly moving switch-gear, like an isolator, is switched from on to off or either ways its state is undefined or **intermediate** for a certain while and the 2 data bits are **00**.



[cfets032-150509-01.tif, 1, en_US]

Figure 2-13 Intermediate State Suppression (ISS)

As long as the intermediate state during a switch procedure is not abnormal long it is out of interest and is discarded by the system. This is done by the function **Intermediate State Suppression (ISS)** in the CFE. This option is activated by a data point related attribute in the IMM. The attribute can be set from 0 to any value in milliseconds. Value 0 means that the ISS is deactivated.

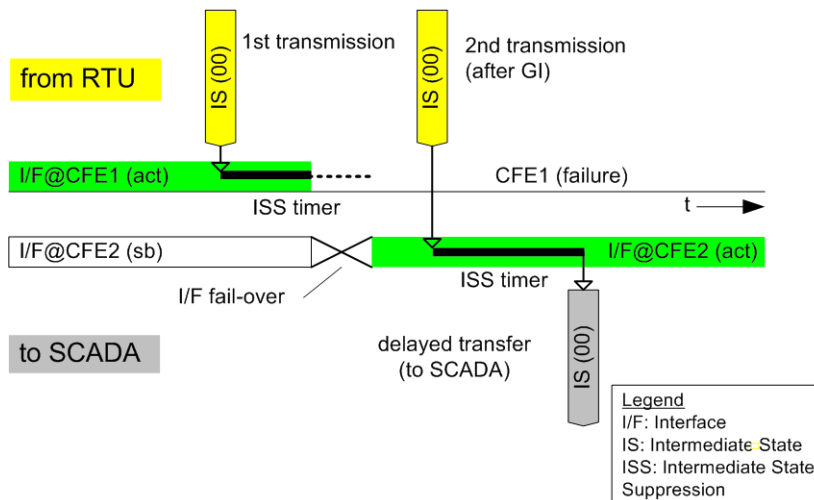
If ISS is activated the following happens in case of incoming intermediate states from a switching device:

- Regular Cases
 - Case 1 - intermediate state is shorter than the ISS timer setting (normal situation):
At the end of the intermediate state, the ISS timer is stopped and the new state (ON or OFF) is transmitted to SCADA.
 - Case 2 - intermediate state is longer than ISS timer setting (faulty situation):
After ISS monitoring time has expired the intermediate state (with time stamp and value 0) is transmitted to SCADA
- Abnormal Cases
 - Intermediate state is repeated for any reason from RTU while the ISS timer is still running:
In this case the ISS timer continues running and is not retriggered.
 - The undefined value 11 is received from RTU:
This value is considered as a major failure and not as an intermediate state. As such the value 11 is immediately passed on to SCADA for further processing.

CFE Redundancy

For a redundant pair of CFE (CFE1, CFE2) the behavior of the ISS depends on the implementation of the Line Level Redundancy. If both CFE are able to listen to the incoming data a failover from active to the standby interface has no affect to the ISS.

If the standby interface is not able to listen to incoming data there is a delay of data transmission to SCADA if a failover takes place during the ISS timer is running. In worst case, this delay is double as much as the normal ISS monitoring time.



[cfets033-150509-01.tif, 1, en_US]

Figure 2-14 Delay of Intermediate State Suppression at CFE Failover

2.8.2.3 Fleeting Information

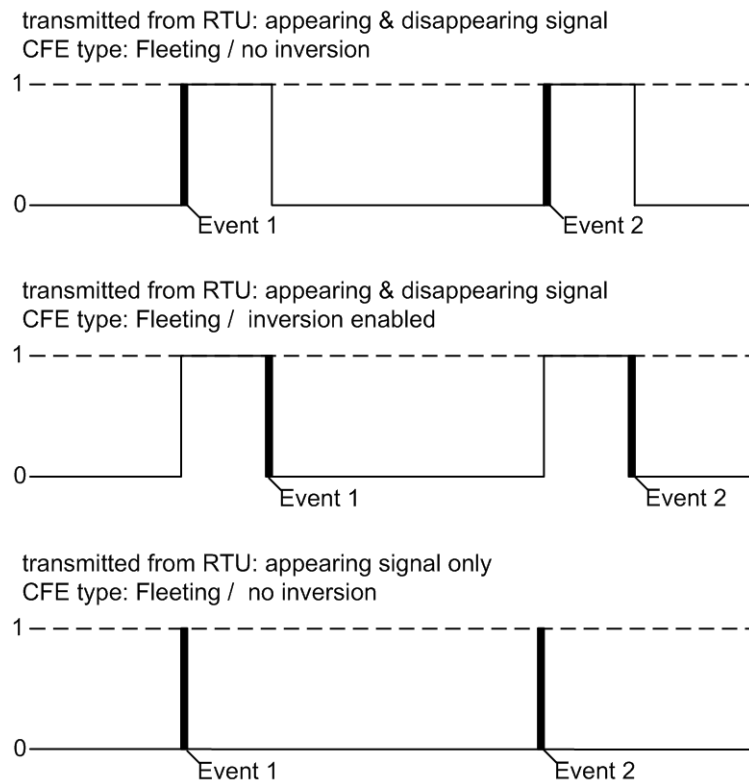
Some data points have an **event** or **fleeting** character. That is, only the appearance or, in some cases, the disappearance of the information is of interest.

Examples of fleeting information are: Circuit breaker trip, protection (trip) alarm and RTU service information, like **Information Loss** or **Command Not Executed**.

Fleeting points are specially processed in the CFE as follows:

- Old/New Comparison is not applied.
- Fleeting information is not stored in CFE database.
- Only the signal change from 0 to 1 is evaluated and passed on to SCADA.
- Depending on the type of RTU the transmission of the disappearing signal is suppressed (as demonstrated in the following 3rd figure)

The customer decides, whether fleeting information is included in the completeness check or not.



[cfets034-150509-01.tif, 1, en_US]

Figure 2-15 Processing of Fleeting Data (with and without inversion)

2.8.2.4 Analog Values

The following processing is carried out in the CFE for analog measurements:

- **Scaling**
Analog values are scaled on conversion from raw value to engineering value. The desired characteristic curve is taken from a list which contains all characteristics in the system. A characteristic curve is linear (factor) or defined by up to 8 pivot points with linear segments between the pivot points.
- **Zero-Point Suppression**
Inaccuracies in transducers or analog-digital converters sometimes result in analog values that do not show true zero. An example is a non-zero current or power flow through an isolated circuit.
To avoid possible operator confusion from this effect, a dead-band is defined for each analog value. Values falling within the dead-band are automatically set to zero by the CFE.

- Filtering (Smoothing)

Each analog value is filtered before passing on. The filtered analog value is calculated using the following formula:

$$\text{Filtered Value} = (\text{New Value} + \text{SF} * \text{Old Value}) / (\text{SF} + 1)$$

The intensity of the smoothing is customized by the Smoothing Factor SF (higher value makes stronger smoothing). If analog value filtering is used with threshold monitoring, then the values are first filtered before threshold check is performed.

- Threshold Check

Analog values are subject to a threshold value check. All new analog values arriving from the substation, which exceed this threshold, are passed on to SCADA. The threshold check is carried out by the integration method.

For slave protocols, there are several different forms of threshold calculation available.

For additional information, refer to Algorithms / Concepts.

The order in, which analog values are processed is: Scaling -> smoothing -> threshold check. Processing only takes place if configured accordingly.

2.8.2.5 Transformer Step Positions

Transformer step positions received from RTUs are converted from protocol-specific format into the Power 5 representation.

The following processing functions are optionally enabled:

- Completeness Check
- Old/New Check

2.8.2.6 Accumulators

Accumulators (or Integrated Totals in IEC terminology) received from RTUs are converted from protocol-specific format into the Spectrum Power 5 representation. Further data processing or completeness check does not take place in the CFE.

2.8.3 Disabling Processing Routines for GI Data

In order to force the refreshing of the process image in SCADA some of the described processing routines are suppressed for data that is requested by a General Interrogation (GI).

The following processing routines are disabled for incoming GI data:

- **Old/New Comparison** for single-point and double-point information as well as for step positions.
- **Threshold Check** for analog measurements.

This behavior is the same for any GI such as Run-up GI, Periodical GI, manually released GI.



NOTE

Old-New comparison is also implemented in SCADA. Alarming are suppressed for data points received several times (for example, at cyclic scan) having the same value and quality flags but different time stamp. As such this feature does not cause unnecessary alarms.

This feature is implemented in the **Device Master** module and is applicable for all types of communication protocols.

Exception: For the RTUCS protocols (TG800, TG8979, CDC2, TG065), the feature is not implemented for periodic GIs, since the RTUCS performs periodic GIs autonomously.

For additional information, refer to [Figure 2-1](#).

The GI-messages received from RTU is **marked** by the Protocol Module in order to notify the Device Master on how to process the information.

Some protocols provide suitable qualifiers like **Cause of Transmission** others do not. In such cases, the GI marker is derived from status information **GI Running**.

2.8.4 Sequence of Events (SOE) / Double Transmitted Information

Introduction

Sequence of Events (SOE) is a kind of information that is used to analyze loss-incurring incidences in the network grid such as breaker trips, activities of protection devices or similar events.

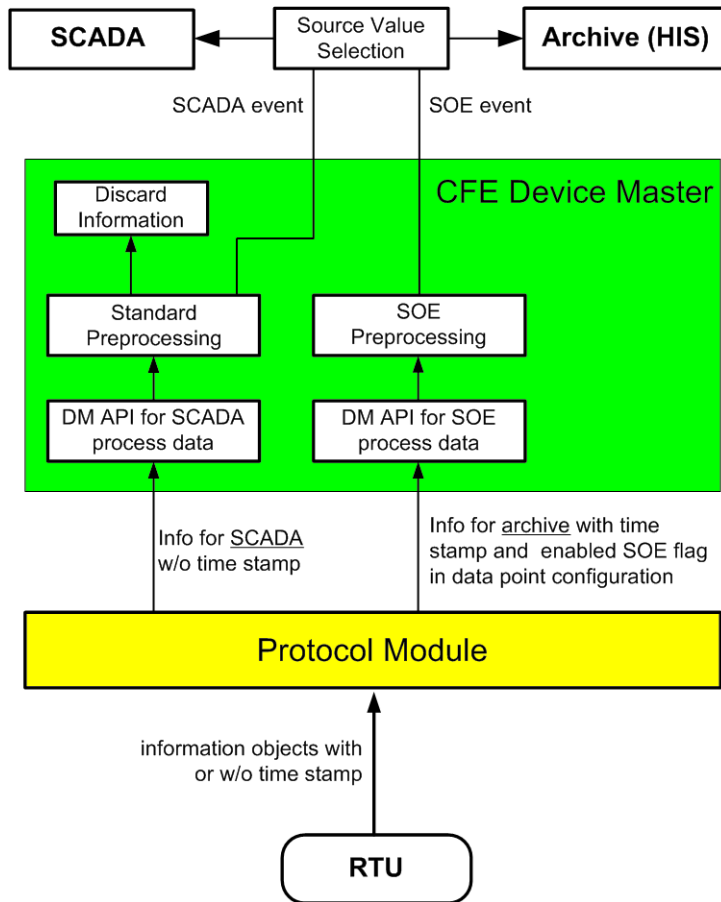
The nature of SOE is characterized through the following key properties:

- SOE messages received from RTU contain time information (normally at high accuracy)
- SOE information is exclusively used for archiving and never for alarming or displaying.
- SOE data objects are transmitted twice:
First, in a standard message format without time information like any regular indication and, in addition, as a time-tagged message that is transmitted later.

RTU protocols provide different representations of information objects for the same data point: static and change/event type. Event objects report current data point changes. Change types objects contain time stamps which show the time of occurrence. Such objects are used to transmit SOE information. However, they are handled differently from regular process information. The Protocol Module separates SOE data from other data and forwards them to the Device Master module for further processing.

Since these information objects are also used for transmission of online process data the Protocol Module requires a criterion to select the appropriate information objects destination.

In order to specify SOE data points the IMM provides an **SOE/Double Transmission** flag per data point. Only data points with set SOE flag and time stamp are forwarded (using **Source Value Selection** component) as SOE objects to the archive. Regular process information (no SOE / Double Transmitted Information) is directed to SCADA.



[cfets035-150509-01.tif, 1, en_US]

Figure 2-16 Handling of SOE and Messages with and without time stamp

Protocol CDC 2

RTU protocol CDC Type 2 permits the transmission of special SOE data. The SOE information is kept in the SOE-Buffer and requested from master station when ever required. Regular messages and SOE messages are processed accordingly in CFE in a different way as already described.

Protocol Sinaut 8FW

RTUs of the type Sinaut 8FW allow transmitting the same data object twice in different message formats:

- As a regular message without time information.
Regular messages without time information are transmitted by the RTU at a high priority
- As a time-tagged message
The same data object is then transmitted as a time-tagged message at a lower priority.

The SOE message follows the regular message automatically with more or less time delay.

Protocols IEC61870-5-101 and IEC61870-5-104

The Protocols IEC61870-5-101 and IEC61870-5-104 allow the transmission of data objects with or without time stamp. Special SOE messages as well as double transmission are not features of these protocols. However, all message types with time stamp (For example, type ID 2, 4, 6, 8, 10 etc.) is configured in IMM as a **SOE** information. As a result, this information is directly forwarded to the archive without old or new comparison.

2.9 Data Preprocessing - Outgoing Data Processing

2.9.1 Overview

The CFE converts normalized information from the Real-time server (RTS) into either process commands or organizational commands for the subordinate level (RTUs). Command information objects and organizational data are passed on to the communication section for the channel assigned to the RTU. The control direction has a higher processing priority than the monitoring direction.

In general, the following types of commands are issued:

- Process commands such as
 - Single commands
 - Double commands
 - Regulating step command
 - Set point commands
- Organizational commands including
 - General Interrogation command
 - Clock synchronization command

For additional information, refer to Engineering

2.9.2 Type of Commands (SC, DC)

Single Commands

Single commands are used infrequently. They serve to set / reset or to activate something in the substation. Some of them do not have return messages.

Typical applications are:

- Clear Alarm (For example fire alarm)
- Reset latching relays
- Activate fire extinguisher

The physical output at the RTU is just one relay per single command

Double Commands

Double Commands (DC) are used to control switch gear devices, like breakers, isolators, tap changers. They always consist of a command pair.

For example:

- Open / close command
- Lower / higher command
- ON / OFF command

The physical outputs at the RTU are 2 relays per double command.

2.9.3 Type of Command Execution

Direct Process Commands

Process commands are transmitted from the CFE to the RTU and executed immediately (direct commands). The monitoring of the feedback is carried out in the SCADA part of the Spectrum Power 5 System

Select-Before-Operate (SBO) Commands

Process commands are performed in 3 steps:

- The CFE transmits the select command to the RTU
- RTU returns the confirmation to the CFE (positive or negative)
- If confirmation is positive, the CFE transmits the release command to the RTU (operator is not involved)
If confirmation is negative, the CFE transmits an error message to the RTS to notify the operator.

The type of command execution, Direct Command or Select-Before-Operate, is configured for every data point individually. Both methods are applicable for switching commands as well as for analog set points.

2.9.4 Scaling of Set point Commands

Set points are analog values in command direction for controlling equipment in the substation. For example power generators. The value coming from SCADA is sent unmodified as raw value (case 1 in following table) or converted from raw value into the engineering value by applying a characteristic curve. It is possible to use a common characteristic curve both for monitoring and command direction (case 2 in following table) or to specify individual characteristics for both directions (cases 3 and 4 in following table).

If set points are passed on to RTS without conversion a dummy characteristic (Input value equal output value) is configured, refer case 5 in the following table.

Table 2-1 Characteristic Definition

Case	Characteristic for Monitoring Direction	Characteristic for Control Direction	Analog Measurement Converted	Analog Set point Converted
1	Not configured	Not configured	No (raw value)	No (raw value)
2	Configured	Not configured	As configured for monitoring direction	As configured for monitoring direction
3	Not configured	Configured	No (raw value)	As configured for control direction
4	Configured	Configured	As configured for monitoring direction	As configured for control direction
5	Configured	Dummy type configured (1:1)	As configured for monitoring direction	Raw value (1:1)



NOTE

If no characteristic is configured for set points the characteristic for monitoring direction is automatically applied for set points in control direction. (case 2 in preceding table)

2.9.5 Command Buffering and Aging Monitoring

Buffering RTU

Transmitting a command message to an RTU occurs at a time when the communication path is otherwise busy. To prevent loss of commands they are queued for sequential transmission. For this reason CFE provides a command buffer for each RTU.

The size of the Command Buffer is configurable for each RTU by 2 settings:

- Buffer disabled
Command Buffer Time Maximum set to zero.
- Buffer enabled
If Command Buffer time is greater than zero, it is possible to buffer up to 512 commands.

Aging Monitoring

To prevent commands from getting obsolete it is monitored how long a command stays in the buffer. For this purpose, a time stamp is added to each (internal) transmit order entering the command buffer. Before the transmit order is converted to a command message and passed to the RTU, this time stamp is checked against the current system time. If the maximum allowed life time of the command has expired the command is discarded.

If the communication link from CFE to the RTU is interrupted all queued commands of the affected RTU are immediately canceled even if their live time has not yet expired.

Summary of features:

- Size of the Command Buffer: min. 0 (no buffering) max. 512 commands.
- Settings are configured for each RTU individually.
- Command Buffering is applied for binary commands and analog set points.
- After disconnection of CFE from RTU all Command Buffer entries for the affected RTU are cleared.



NOTE

The settings for command confirmation monitoring in SCADA account for the delay caused by CFE command buffering. In particular for party lines or dial-out connections a considerable delay occurs before the communication link to the RTU is established and the command is carried out by the RTU. The timer in SCADA is set at higher value than the command aging time in the CFE.

2.9.6 Counter Interrogation and Freeze Commands

Depending on protocol, it is possible to transmit counter interrogation or freeze commands to handle accumulators. Refer to the Interoperability Lists in Appendixes C through J to ensure that the relevant RTU protocol supports the Counter Interrogation and Freeze command.

2.10 Time Processing

2.10.1 Overview

Time processing in the CFE comprises the following tasks:

- Time tagging of messages that do not have time information
- RTU clock synchronization by synchronization messages
- Formation of time information

As a member of the Spectrum Power 5 control center system LAN, the CFE periodically receives the local time from a central master clock.

2.10.2 Time Tagging

With respect to time information, in monitoring direction the following message types are distinguished:

- Messages without time information: In the CFE, each information item gets a system time stamp (UTC format). The time range is from 1 ms up to years.
- Messages with real-time stamp: In the CFE, the real-time information is converted to UTC time and passed on to SCADA.

In the Spectrum Power 5 system logs both message types have identical presentation.

If the time information carries a valid quality flag the time is converted into the internal time format. If the quality flag of the time stamp is set to **invalid**, the time information is replaced with the running local time of the CFE.

The time stamp presentation in the operating logs is as follows:

- Message received with valid time stamp:
28.01.2004 14:25.125 TT (the identifier **TT** in the column CAUSE stands for **true time** and freely is selected)
- Message received with invalid time stamp or message received without time stamp:
28.01.2004 14:25.125

On invalid or missing time stamp, time information is added by CFE.

2.10.3 Time Management and Time Synchronization

In general time synchronization of the RTU clock is done using either:

- Time receiver in the unit (for example, DCF77, GPS)
- Synchronization pulses from master clock on the sync input at the unit.
- Synchronization messages from CFE.

In case of using synchronization messages the generation and time-accurate transmission of these messages in configurable time intervals is performed by the CFE.

For additional information, refer to Real Time Communication Application Data Sheet.



NOTE

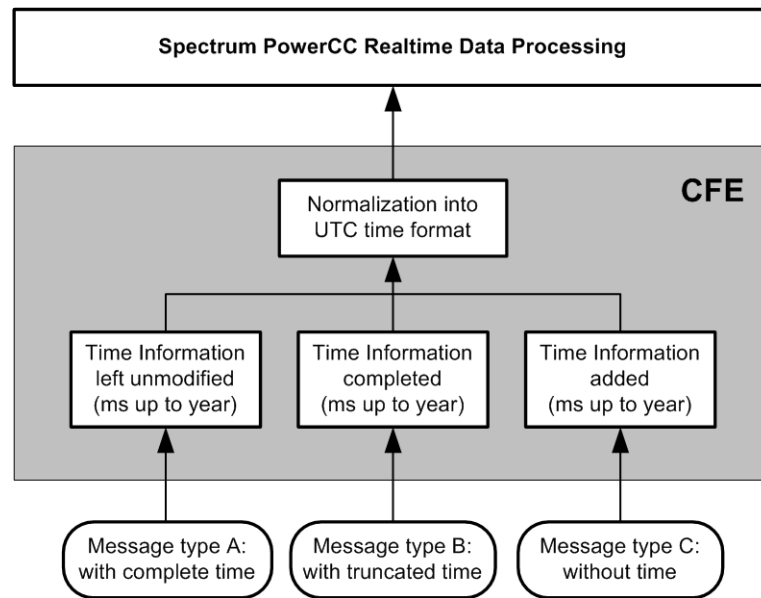
Time synchronization is implemented in a protocol-specific way.

2.10.4 Time Formation

RTU time information is normally based on local time. The CFE internal time format is UTC. As such, at least one time transformation, from RTU time to UTC takes place in the CFE.

Time tagged messages from RTUs carry complete time information (time range from ms up to year) or truncated time information (time range from ms up to minutes or hours, depending on protocol).

Truncated time information represents only a limited time span. For example 60 min (IEC 60870-5-101) and is completed in the CFE. Both truncated and complete time information in time-tagged messages from RTUs are converted to the CFE-internal UTC time format.



[cfets036-180509-01.tif, 1, en_US]

Figure 2-17 Processing of Message Time Information in CFE

2.11 Error Messages and Quality Flags

2.11.1 Error Messages

Error messages are issued when (for example) a faulty configuration or hardware problem is detected. In general, only the occurrence of the error is noted. These notifications are useful for logging as well as providing information useful in troubleshooting. Error messages are visible in the **Windows Event Viewer**.

2.11.2 Log Files

Error messages are categorized by device and maintained chronologically in log files. These log files can be readout during test and diagnosis phases and support the commissioning engineer and service personnel in removal of faults.

For additional information, refer to Volume Test & Diagnosis of the Communication Front End Data Modeling Guide.

2.11.3 Quality Flags

The CFE provides each data point in Spectrum Power 5 with value, quality, and time stamp. The Spectrum Power 5 quality codes specify the quality of the value, the transmission cause, and the source. Some of the quality codes are built locally in the CFE others are transmitted from the RTU and usually are specific for the telecontrol protocol.

2.12 CFE Interfaces

2.12.1 Serial Interfaces

Data exchange from/ to RTUs is carried out through serial communication interfaces of the PC.

The type of interface card used for a particular RTU depends upon the protocol supported by the RTU:

- **Standard serial COM-Boards (For example DIGI **AccelePort XEM**)**
This type of interface is used for byte-oriented protocols (for example, IEC 101, DNP3.0). The (usually 2) standard COM ports of the PC are also used for byte-oriented protocols
- Currently, no special hardware is readily available for use with bit-oriented protocols (for example, CDC II, SINAUT 8FW). Therefore, these kinds of protocols are not supported in Power 5.

Per CFE, up to 48 COM ports can be handled. Each COM port can be assigned to different protocols and protocol parameters.

For additional information, refer to CFE Application Sheet.

2.12.2 LAN/WAN Interface

For LAN-based protocols (for example IEC 104, DNP 3i), a standard LAN adapter is used. Also for connecting a CFE server to other Spectrum Power 5 servers a LAN adapter is used. Siemens recommends that a separate LAN card is used for RTU communications.

2.13 Startup Procedure

2.13.1 Startup of the CFE

The startup of the CFE server is carried out by the Spectrum Power 5 startup Service. The startup is performed in steps according to the configuration of the CFE software package in System Management Information Data Set (SMI).

2.13.2 Startup of a Connected Component (RTU)

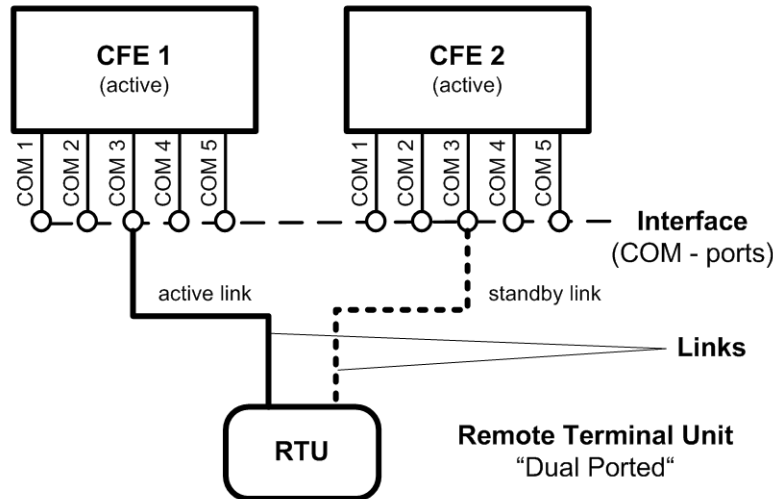
If a RTU runs-up or a communication link is reestablished the following procedure takes place:

- Link check is performed (protocol specific)
- The link layer reset procedure is carried out (protocol specific)
- Run-up messages are received from RTU and evaluated (protocol and RTU specific)
- The CFE sends a General Interrogation (GI) command to the RTU.
- The RTU responds to the GI command.
- After the GI monitoring time has expired or the GI termination message is received:
The CFE checks the completeness of the returned GI messages.
- If the result of the GI completeness check is negative, a **GI incomplete** Status message is generated and reported to the operator through standard alarming mechanism.

2.14 Line Level Redundancy

2.14.1 General

It is possible to define redundant links from several interfaces on the same or different CFE servers to one RTU.



[cfets037-180509-01.tif, 1, en_US]

Figure 2-18 Redundant CFE, Redundant RTU Link

In Line Level Redundancy all servers of a redundant set of CFE are in active state. However, it is possible to select one of the redundant lines to a remote terminal unit to become active.

The switchover on line level is triggered (manually or automatically) as follows:

- Manually:
 - Switchover of a line by locking (disabling) an active interface.
 - Switchover of a line by setting a preference on a standby interface.
 - Switchover of an RTU by locking an active link (not possible for single-port RTUs)
 - Switchover of an RTU by setting a preference on a standby interface or a standby link (not possible for single ported RTUs)
- Automatically:
 - Switchover of a single RTU after detection of a link failure (not possible for single ported RTUs)
 - Switchover of a whole line after detection of an interface failure
 - Switchover upon a server failure

For most protocols, the standby channel (2nd redundant channel) is passive. There is no data transmission, no polling, and no link test performed on this line as long it is not activated. For other protocols, the availability of redundant standby channels is being monitored. The monitoring of standby channels is implemented in a protocol-specific way.

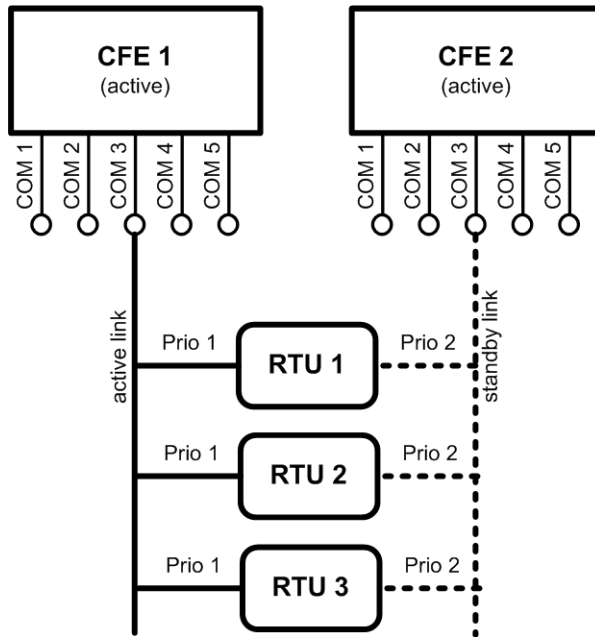
The priority attribute of the CFE Interface Link is used to decide, which link or interface is activated during run up and automatic failover.

For priority the following rules are relevant:

- The priority of each link to the same remote terminal unit is unique
- For single-port RTUs only, all links of a party-line have the same priority.

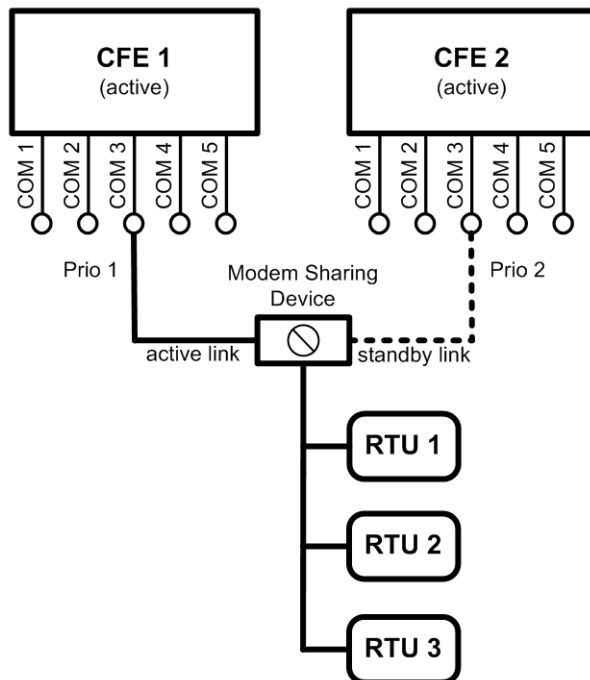
2.14.2 Redundant Connections

There are basically 2 different kinds of redundant connections between a Front-End interface and a Remote Terminal Unit. Redundant links to a single-port RTU are called dependent; links to an RTU with more than one port are called independent.



[cfets039-180509-01.tif, 1, en_US]

Figure 2-19 Independent Interfaces Connected to Dual Ported RTUs



[cfets040-180509-01.tif, 1, en_US]

Figure 2-20 Dependent Interfaces (with single-port RTUs)

In a dependent interface only a complete group of links is able to become active. Each of the other links run in standby mode. It is not possible to switchover a single RTU.

2.14.3 Line Level Redundancy and Device Status

Status information is provided for individual devices and subdevices. It describes the state of an individual device (for example, CFE interface) or a connected device (RTU). Status information is used to indicate the **state of health** of a device and whether it is enabled or disabled.

State changes in status information during operation are transferred as spontaneous events and is processed in the same manner as process data.

The following device states (assigned to both CFE Interface and CFE Interface Link) are relevant:

- Device Active
- Device Failure
- Device Locked
- Device Preferred

The possible CFE status assignments are shown in the following table:

DEVICE	STATUS INFORMATION
CFE interface link to the connected component (RTU)	Active
CFE interface link to the connected component (RTU)	Failure ¹⁾
CFE interface link to the connected component (RTU)	Locked
CFE interface link to the connected component (RTU)	Preferred (only if independent)
CFE interface	Active
CFE interface	Failure
CFE interface	Locked
CFE interface	Preferred (only if dependent)

¹⁾ specifies Link Failure prevails when communication is not possible. This also happens when the RTU is down. The status information **RTU Failure**, which is set ON when all available links have failed. The failure disappears as soon as one link to that RTU is able to establish a connection (the corresponding status information **Link Failure** has disappeared).



NOTE

More details on CFE switchover behavior in different configurations, situations, transmission modes etc. are obtained from Appendix [4.5.1 Switch-Over Behavior](#) of this document.

2.14.4 Device Status - Non-controllable States

The following states are non-controllable:

Device Active

The internal device state Device Active is used to signal that an interface or link is either active or standby. This status information marks that interface or link which is enabled for communication with RTUs (independent from the fact whether the establishment of link succeeded or not). Just one CFE Interface can be active at the same time.

If the CFE Interface does not belong to a CFE Dependency Group its status will always be active after run up. In case of dependent CFE Interfaces (single ported RTUs) only one CFE Interface of a CFE Dependency Group is active at the same time. In case of non-redundant connections that status is always active for CFE Interface and CFE Interface Link.

Device Failure

The status Device Failure indicates a problem opening a communication port or establishing a link to an RTU. If the protocol does not support standby-monitoring, the value of a standby-interface or a standby link has the qualifier Local Not Renewed.

The status disappears in following cases:

- The corresponding CFE Interface is active and is successfully opened.
- The link is successfully established.

2.14.5 Device Status - Controllable States

There are 2 controllable states associated with CFE Interface and (CFE Interface) Link, which allow to switch-over manually. They are designated as Device Locked and Device Preferred and are controllable using User Interface and/ or using CFE Equipment Status Display in the CFE view.

Device Locked

The status information **Device Locked** is used to disable an interface or link and to prevent it from becoming active. When a currently active CFE Interface is locked, an automatic switchover of all concerned links is forced (if possible). For a link that belongs to a party-line the following is considered: Locking one of the active links does not cause switchover as long as there is any other link (of that party-line) active.

Device Preferred

It is possible to specify a **preferred** interface or link through use of the Device Preferred state. This state is only considered at startup time of a link process and at the time the state is initially set by operator command. It is not considered during automatic failover. Only one of a link or interface group is preferred. If the preferred link or interface fails, an automatic switchover occurs.

2.14.6 Standby Monitoring and Buffering - General Description

Purpose

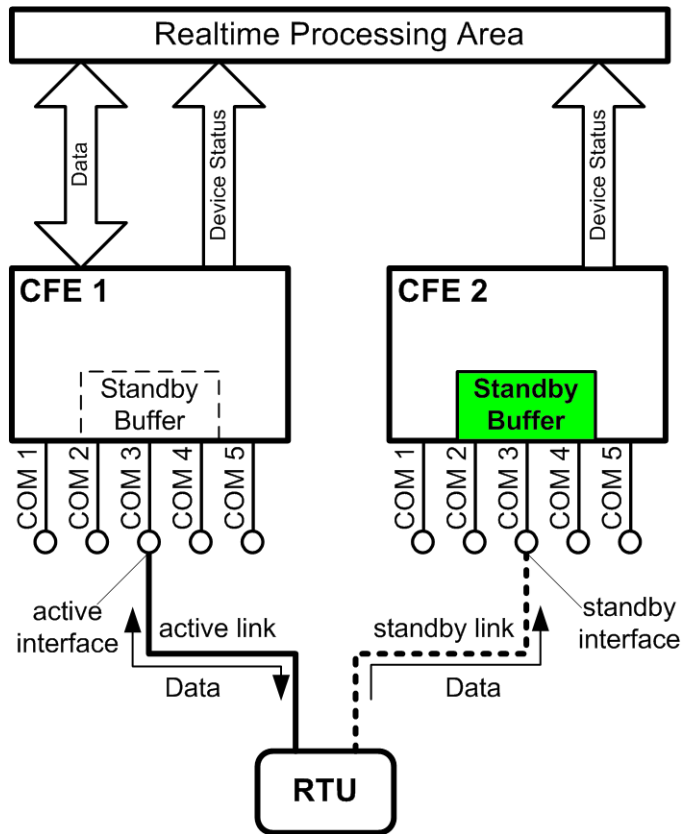
The main purpose of the Standby Monitoring and Buffering function is to ensure the availability of the inactive standby communication links and to prevent information losses during switchover of communication lines.

Features

The Standby Monitoring and Buffering features are as follows:

- All standby connections are monitored and their unavailability is reported.
- Incoming telegrams are stored in a standby buffer. When a standby connection becomes active, the content of the standby buffer is transmitted to the Real time Processing Domain.
- The behavior in standby mode differs for dependent and independent connections (refer further section). In addition, with each connection type it differs individually for each protocol.
- The device states are monitored (the acknowledge time-outs are monitored in order to detect connection failures)
- Protocols (dependent connection and unbalanced) not containing the RTU address in the monitoring messages cannot be supervised on the standby connection.
- Information of protocols not containing the info address in the monitoring messages cannot be buffered on the standby connection.

The following figure shows the CFE behavior during standby monitoring and buffering in different operating modes (active / standby).



[cfets041-190509-01.tif, 1, en_US]

Figure 2-21 Data Flow and Buffering for Active and Standby Links

Dependent/ Independent Connections

The monitoring of standby connections depends whether the connections to the RTU are:

- Dependent (connections to single-ported RTUs)
- Independent (connections to dual-ported [or multi-ported] RTUs)

[Figure 2-19](#) and [Figure 2-20](#) depict the basic kinds of redundant connections between an interface and an RTU.

Dependent Connection Features

With dependent connections, standby monitoring is possible in monitoring direction only (potential problems in control direction are not detected). Passive listening takes place, only monitoring telegrams are received by CFE but no control telegrams or any other signals are sent.

Independent Connection Features

With independent connections, the standby monitoring is possible in Monitoring and Control direction. Active listening takes place. Telegrams are sent via standby connections from CFE to RTU.

For monitoring IEC 60870-5-104 standby connections the following procedures are used:

The transport connection is established (same as for active connections). However, no application data is transmitted over standby connections. For monitoring the availability of the standby connections, test frames are transmitted in regular check routines (standby buffer is not used for such a protocol).

Standby Buffer

When an active link fails, it takes a while until the system recognizes the problem and the switchover procedure completes. In order to avoid data loss when a switchover occurs, spontaneously transmitted data is temporarily stored with time stamp in the standby buffer. Upon fail over, the messages stored in the standby

buffer is transmitted in chronological order to the real-time server. That avoids data loss but data is duplicated. Data from the standby buffer are not specially designated.

Configuring the Standby Buffer

In IMM, the operation of the standby buffer is configured according to requirements.

Using the relevant CFE Server attributes, configure the following:

- Time span
Specify how long (in seconds) incoming information is buffered for a standby connection. This is configurable per CFE Server. Default is 10 s
- Storing
This is a Boolean value that specifies, whether the following measurement types are buffered in the standby buffer
 - Accumulators
 - Analogs
 - Digitals
 - Positions
 - Fleetings

2.14.7 Redundancy Management

Overview

For distinct protocols standby channels are monitored so that a fail over is done if the backup link is not in failure state.

States of Redundancy Management

The above elements can assume one of the following states:

- Device Active ON:
When in this state, CFE Interface or CFE Interface Link is selected by redundancy management for process control.
- Device Active OFF:
When in this state, CFE Interface or CFE Interface Link is selected by redundancy management for standby.

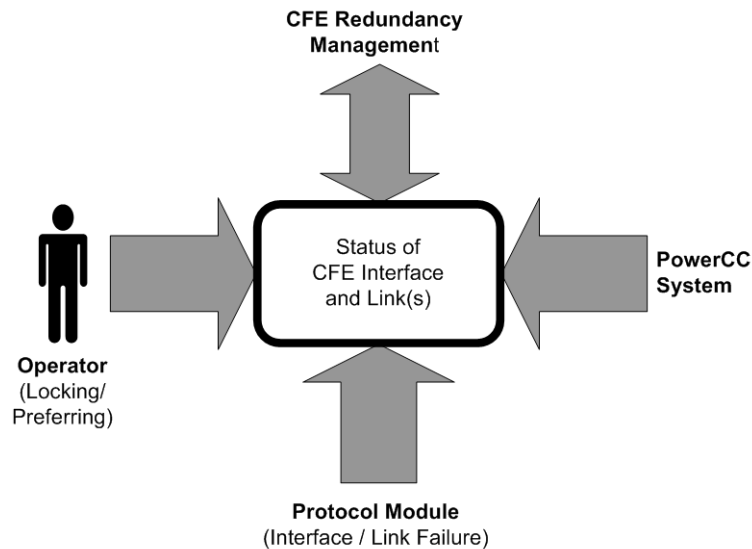
The following table explains the expected behavior for the **Protocol Module** in different states.

-	CFE Interface	CFE Interface Link
Device Active ON	The communication port is open in both directions.	Communication in both directions is allowed. Commands are processed.
Device Active OFF	Communication port is open in both directions. If the CFE Interface is a dependent one, the Control direction is blocked completely.	The link status is monitored and reported so that the redundancy management gets informed about the current failure status of the link. Incoming messages are interpreted and processed like in active mode, and they are stored in the Standby buffer. Command output is disabled (probably existing command queues are cleared when the CFE Interface Link changes from Device Active = ON to OFF). In case of an independent CFE Interface it is necessary to establish a link to the RTU.

(Other) Sources Affecting Redundancy Management States

Commonly, the CFE redundancy management controls the above state. However, it is affected by the user, Spectrum Power 5 system and the Protocol module as follows:

- The user can lock or unlock interfaces or links or increase the priority of a link or interface by setting preferences.
- If the Real time Processing Domain is down, the CFE redundancy management stops communicating with RTUs.
- Details of link and interface status reporting vary depending upon the particular protocol.



[cfets042-190509-01.tif, 1, en_US]

Figure 2-22 Sources Influencing the Status of CFE Interface and Links

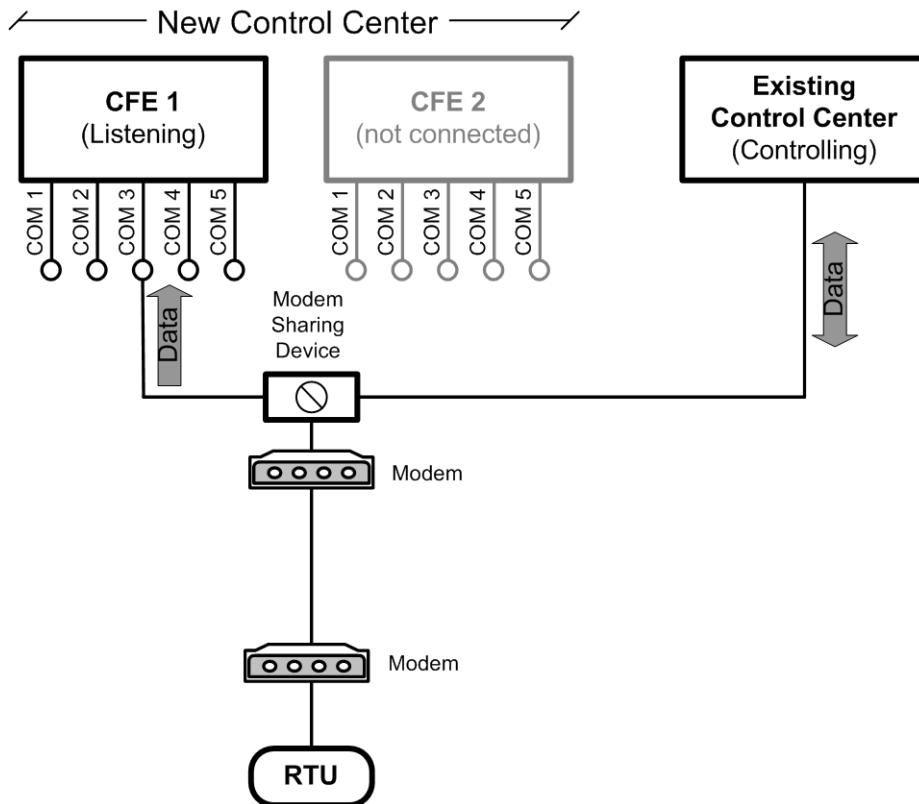
2.15 Listening Mode

Purpose

During commissioning, a new control center (such as Spectrum Power 5) is running in parallel to an existing system in order to monitor the communications with RTUs as well as processing of incoming messages and alarms.

The Listening Mode enables the transmission of process information from RTUs to SCADA and disables the transmission of control information from SCADA to RTUs.

For a typical example of a Listening Mode configuration using a modem sharing device (modem splitter).



[cfets043-190509-01.tif, 1, en_US]

Figure 2-23 Use case for Listening Mode



NOTE

Whether listening mode is possible or not depends on the protocol implementation.
For additional information, refer to Engineering.

Features

The features of the Listening Mode are as follows:

- It enables the transmission of the process information to SCADA.
- The control direction is always locked (commands coming from SCADA are negatively confirmed and never transmitted to an RTU).
- A listening communication port is not allowed to send any telegram to an RTU.
- In order to avoid automatic fail over actions, the link status Device Failure is always off.
- For standby connections, in Listening Mode there is no monitoring or buffering of incoming data.

- Completeness checks and time-out handling are turned off.
- Listening Mode is indicated by the CFE Status Information **Listening Mode**. This status can be checked through the Runtime Explorer in **Substation View**.
- Switching the Listening Mode on or off is done by changing the **Listen Mode** attribute of the corresponding interface instance in IMM.

2.16 Data Concentrator for IEC 60870-5-104 Interface

2.16.1 Requirements

Data Acquisition

The data concentrator keeps an up-to-date image of all process data of the connected RTUs with scans and / or general checks. If the data concentrator loses the connection to an RTU, it marks the information of this RTU as Not Topical (NT).

The data concentrator is able to answer a general check from the control center to each RTU. Usually the data concentrator generates this answer from its own process image. The data concentrator transmits any of the information in the image to the control center when it has changed. Possible changes include: change of value or of quality flags (Not Topical, Invalid).

All process information is transmitted from the data concentrator to the control center with the Common Address of the related RTU - not of the data concentrator.

When CFE demands a general interrogation, the data concentrator sets the Cause Of Transmission to **Interrogated by general interrogation** in the answer. (The CFE also accepts spontaneous data) When the data concentrator demands a general interrogation from the Sub-RTU, receives the data, and detects changes, it sends these changes to the Control Center with Cause Of Transmission **spontaneous**. If required, the data concentrator forwards the general checks to the Sub-RTUs instead of using its local process image. In either case, the information sent to the CFE is the same.

Commands

The data concentrator allows direct commands as well as select-before-operate commands. Select before-operate procedure is controlled from the control center.

Time Processing

Time synchronization using IEC 60870-5-104 is possible. However it is checked whether this method is accurate enough. In particular, in networks where routers or other time delaying devices are involved, time synchronization using IEC 60870-5-104 becomes unacceptably inaccurate. In this case the data concentrator is synchronized by an external clock (GPS or DCF 77).

Status Indications

The data concentrator sends status indications for the communication links between the data concentrator and the Sub-RTU. The status indications either have the Common Address of the related Sub-RTU or of the data concentrator. The Equipment Status Display (ESD) does not show the Sub-RTUs.

Redundancy

The data concentrator must be able to establish more than one transport connection to the control center (usually 2 connections). The control center initiates a **start data transfer** on the highest prior connection and sends a **stop data transfer** on the other connections.

A redundant pair of data concentrators must use block redundancy and must decide autonomously which part becomes **master** (active) and which becomes **slave** (standby). Therefore a cross coupling is required. Only the master data concentrator communicates with the Sub-RTUs. All Sub-RTUs is connected to both parts of the data concentrator.

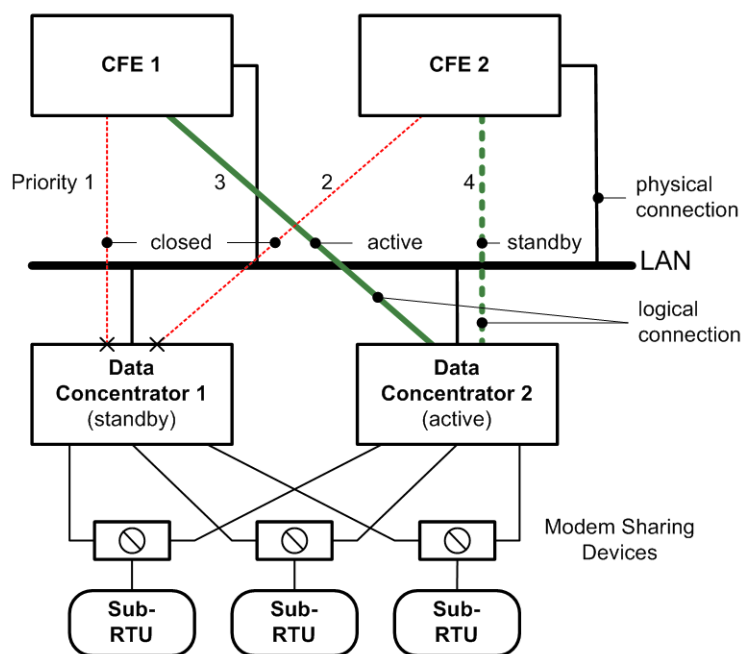
Redundant data concentrators have separate IP addresses. The standby does not communicate with the CFE. Thus, only the active data concentrator sends a **start process data transmission** signal to the master data concentrator.

In order to get the state of the standby data concentrator even though it is not connected to the CFE, the 2 data concentrators exchange their status information. The master data concentrator reports the state of both data concentrators to the CFE.

The data concentrator has a master Common Address. If there is a switchover from one data concentrator to the redundant partner, the same Common Address is assigned to the new master. That means that status

information from the redundant pair of data concentrators is always transmitted with the same (master) common address (regardless of the current master).

In the following figure, DACO 2 is active (master). DACO 1 is standby (slave); its connections (1 and 2) to CFEs are closed. As a result, link 3 is active. Link 4 is standby and being monitored.



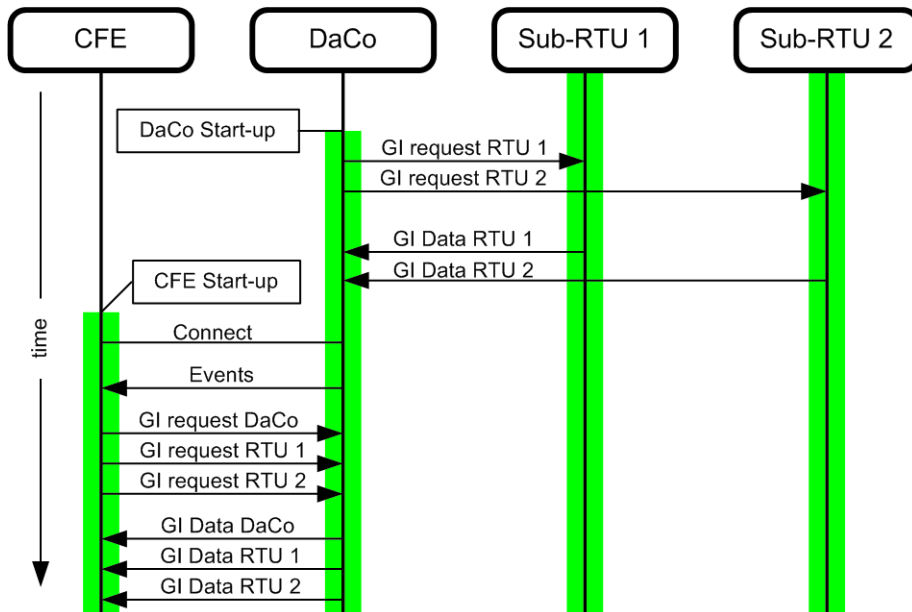
[cfets045-200509-01.tif, 1, en_US]

Figure 2-24 Example of a Redundant Data Concentrator Configuration

2.16.2 Communication CFE - Data Concentrator (DaCo)

CFE Startup

When the data concentrator starts up while the CFE is not yet running, it demands a general interrogation of all Sub-RTUs. When the CFE starts, it connects to the data concentrator and demands a general interrogation of the data concentrators' local data and of each Sub-RTU. The data concentrator answers with the data from its process image. It is possible that the data concentrator immediately sends stored events from local inputs and from Sub-RTUs as soon as the CFE connects and starts data transmission.

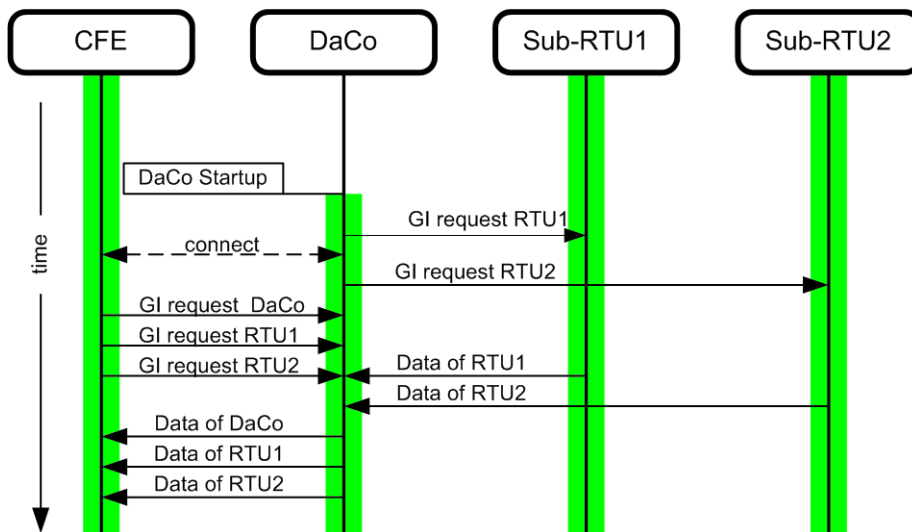


[cfets046-200509-01.tif, 1, en_US]

Figure 2-25 CFE Start-up

DACO Startup

When the data concentrator starts up while the CFE is already running, it requests a general interrogation of all Sub-RTUs. When the CFE successfully connects to the data concentrator, it performs a general interrogation. If the data concentrator's process image is up to date, it replies with data from the process image. If not, it performs a GI of the Sub-RTUs before responding.

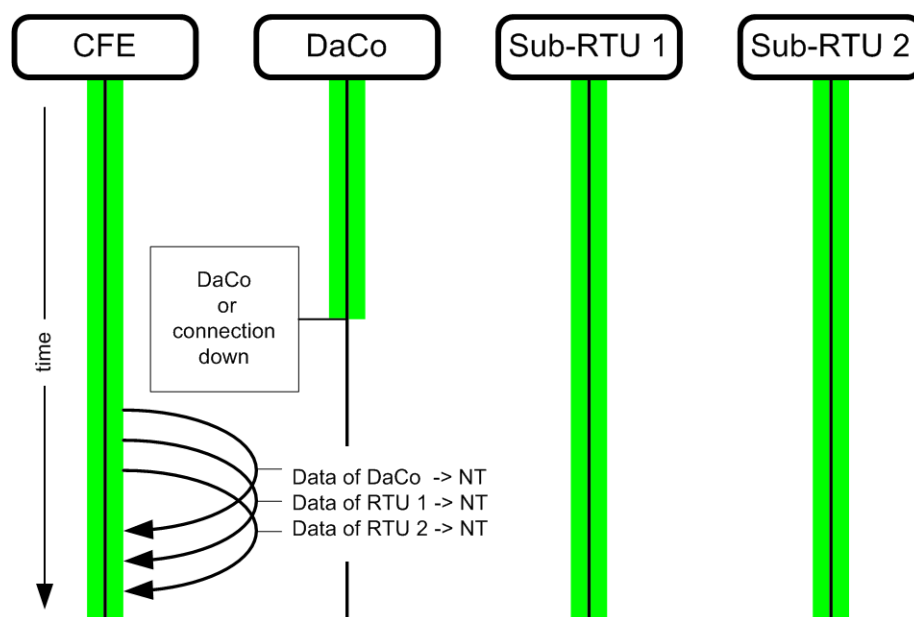


[cfets047-200509-01.tif, 1, en_US]

Figure 2-26 Data Concentrator Start

DACO Failure

If a data concentrator or connection fails, the CFE sets all data from that DACO to Not Topical (NT).

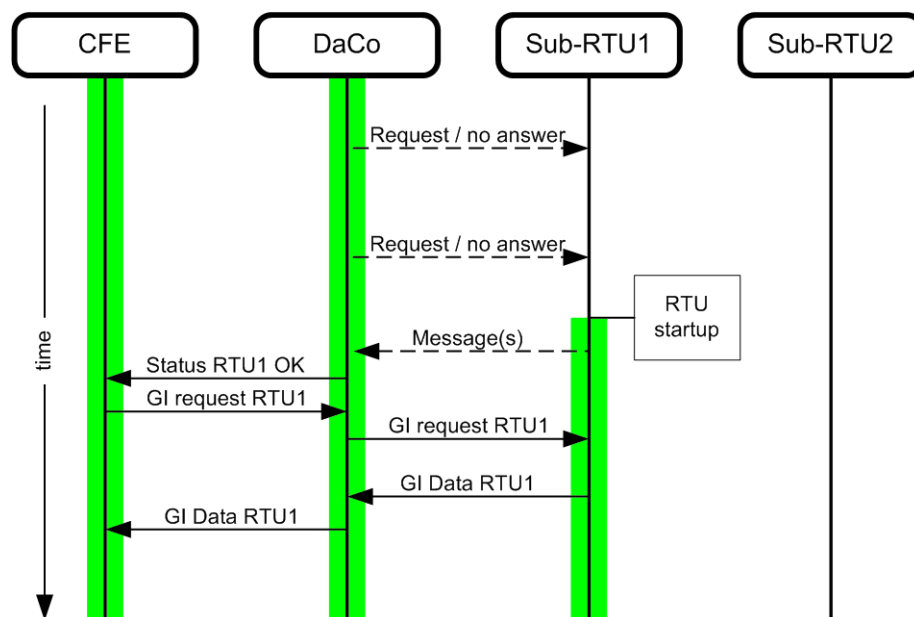


[cfets048-200509-01.tif, 1, en_US]

Figure 2-27 Data Concentrator Failure

Sub-RTU Startup

When a Sub-RTU starts up or its connection to the data concentrator is re-established, DACO resets the RTU Failure Status and requests a general interrogation from the Sub-RTU. The GI from the Sub-RTU is passed-on to the CFE.



[cfets049-200509-01.tif, 1, en_US]

Figure 2-28 Sub-RTU Startupt

Sub-RTU Failure

When a Sub-RTU does not answer, the data concentrator sets the RTU Failure Status and sets all related data to **Not Topical (NT)**.

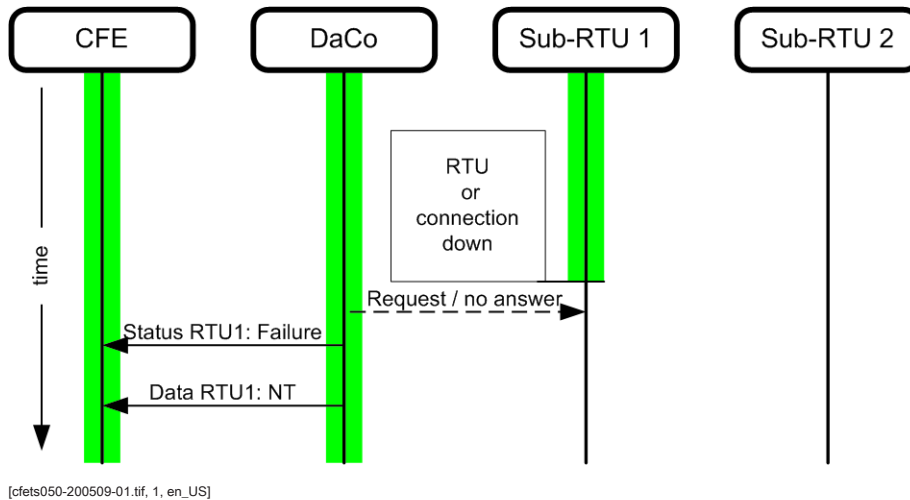


Figure 2-29 No Answer from Sub-RTU

Operator wants Data from DACO .

When an operator issues a general interrogation command (GI) to the data concentrator, DACO sends back its own dataset (its status information).

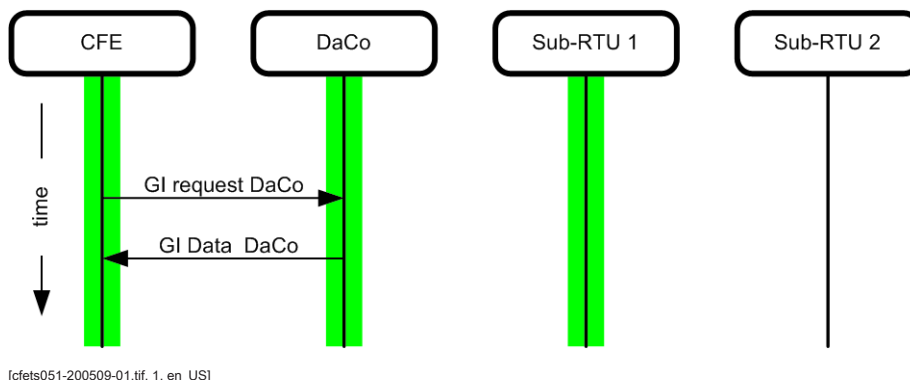


Figure 2-30 Operator GI Request to Data Concentrator

Operator wants Data from Sub-RTU

When an operator wants to check data of a particular Sub-RTU it issues a general interrogation command to the DACO for the RTU of interest. The data concentrator sends back the GI data of this Sub- RTU retrieved from its database.

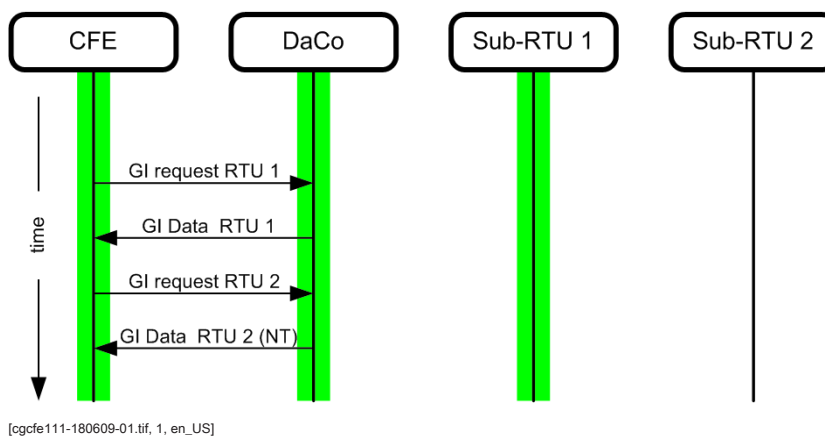


Figure 2-31 Operator GI Request of Sub-RTU Data

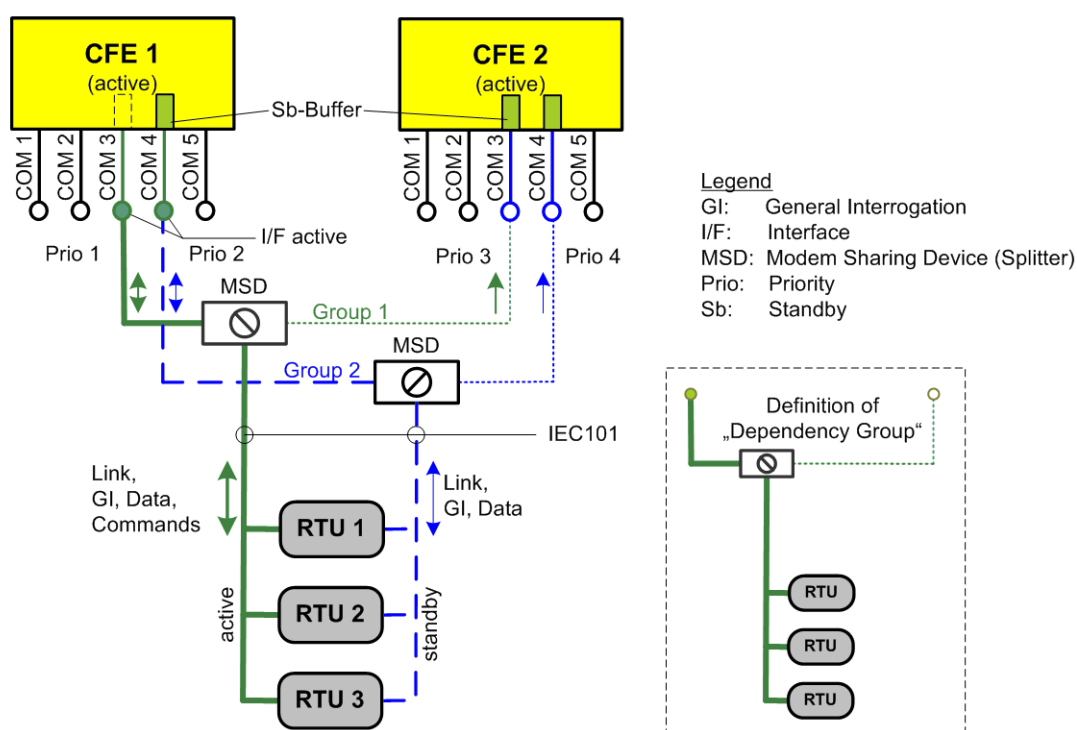
2.17 Enhanced Redundancy for IEC 101 IEC104

Introduction

The standard CFE redundancy allows interfacing single or dual ported RTUs either through serial lines or through TCP/IP connections. In the standard, CFE and communication links is redundant, however, redundancy of connected components is not taken into account by the CFE redundancy concept. In order to connect dual ported RTUs using 4 connection points or to connect intelligent redundant devices having their own redundancy control, CFE provides particular features to cope this requirement.

Enhanced Redundancy for Protocol IEC 101

In order to increase the reliability of (serial) IEC 101 communication, CFE supports the connection of dual ported RTUs in a party line arrangement with a total of 4 interface links as presented in the following figure.



[cfets054-200509-01.tif, 1, en_US]

Figure 2-32 Enhanced IEC 101 Redundancy

The Enhanced Serial Redundancy features as following:

- RTUs have 2 independent communication ports. (**Independent** means that the RTU is able to communicate with 2 different Control Centres at the same time).
- Each branch of the party line is connected using a Modem Sharing Device (MSD) to both units (CFE1 and CFE2) of the redundant front end. This makes 4 interface links for each RTU. One branch of the party line and associated interface points is called a **Dependency Group**, refer the preceding figure.
- Each interface link is allocated to a unique priority, ranging from 1 to 4.
- Depending on its priorities, one Dependency Group has a **fully active** connection and a standby connection. The other Dependency Group has a **half active** connection and a standby connection.
- RTU Data received through the active connection is passed on to SCADA.
- The **half active** interface operates like the full active interface, however, received data is stored in the Standby Buffer and not passed on to SCADA and commands from SCADA are not passed through to the RTU.

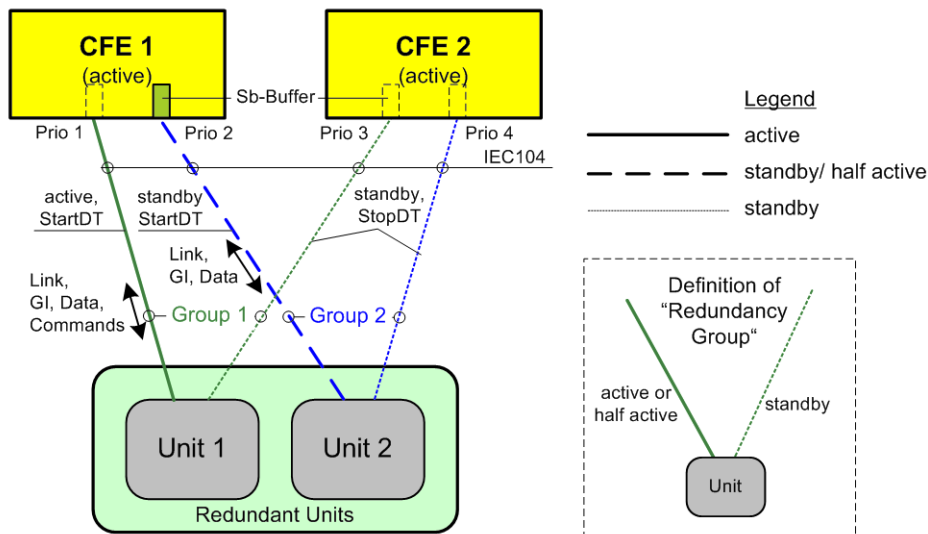
- Standby interfaces operate in listening mode. RTU data is also stored in their Standby Buffers.
- At all interfaces the activity of the link layer is monitored and corresponding status information generated. Status information of interfaces and links is summarized in the following table:

Table 2-2 Status of Link, Interface, and Data Handling for Protocol IEC 101

Group	Priority	Interface Status	Link Status	Data Handling
1	1	active	active	data acquired & processed, commands sent to RTU
1	3	standby	standby	data received and buffered by listening
2	2	active	standby	data acquired & buffered, no commands
2	4	standby	standby	data received and buffered by listening

Enhanced Redundancy for Protocol IEC 104

The enhanced IEC 104 redundancy concept allows to connect a redundant RTU or a redundant Data Concentrator through IEC 104 links to a redundant CFE block as illustrated in the following figure.



[cfets055-200509-01.tif, 1, en_US]

Figure 2-33 Enhanced IEC 104 Redundancy

The Enhanced Redundancy for protocol IEC 104 has the following characteristics:

- Connected units (RTU or Data Concentrator) are redundant and provide, if redundant, a total of 4 independent communication ports. (**Independent** means that the device is able to communicate with 2 different Control Centres at the same time)
- Both units (Unit 1, Unit 2) are active at the same time.
- The 2 Groups of redundant communication links are established. A pair of links, operating in (half) active / standby mode, is called a **Redundancy Group**, Refer the preceding figure.
- Each interface link is allocated to a unique priority, ranging from 1 to 4.
- Depending on priority, one Redundancy Group has one **fully active** connection and one standby connection. The other Dependency Group has one **half active** connection and one standby connection.
- RTU Data received through the active connection is passed on to SCADA.

- The **half** active interface operates like the full active interface, however, received data is stored in the Standby Buffer and not passed on to SCADA and commands from SCADA are not passed through to the RTU.
- Standby connections operate in test mode. Only test frames are exchanged to monitor the standby connections. For all Transport links and interfaces status information is generated as illustrated in the following table:

Table 2-3 Status of Link, Interface, and Data Handling for Protocol IEC 104

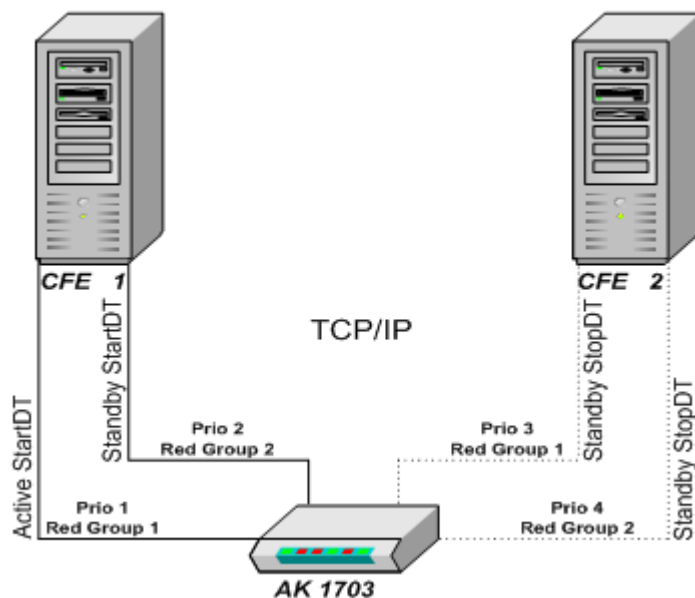
Group	Priority	Interface Status	Link Status	Data Handling
1	1	active	active	data acquired & processed, commands sent to RTU
1	3	standby	standby	no data acquisition
2	2	active	standby	data acquired & buffered, no commands
2	4	standby	standby	no data acquisition

2.18 Redundant AK1703 Configuration

Introduction - Redundancy and Dependency Groups

CFE provides a combination of two redundant pairs of connection for dual ported RTUs. In the following figure, the node AK 1703 represents the dual ported RTU.

For this requirement, the configuration of the interfaces are the same mechanism as for dependent serial interfaces with different dependency groups. The interfaces belonging to one redundancy group shall have a link to the same dependency group. This means for this document, whenever the term **dependency group** is used with IEC-104, it is equal to the term **redundancy group**. Also, the displayed state in Runtime Explorer is similar to dependent interfaces. This means that the visual behavior of IEC-104 will change slightly, if configured with dependency groups. The standby interfaces will be in state standby (**Active-Off**). Nevertheless, the standby interface which has sent a StartDT will work like an independent standby interface and be in state standby (**Active-On**).



[sc_enhanced redundancy_2014-02-14, 1, en_US]

Figure 2-34 Enhanced Redundancy - Dependency Groups

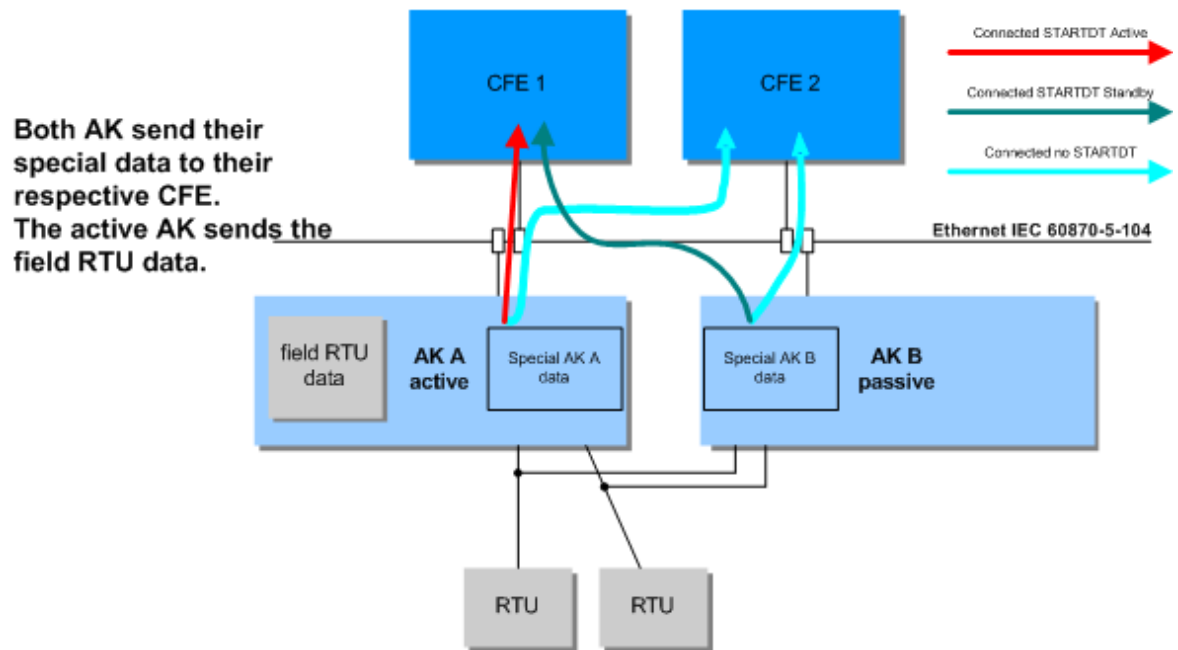
The active connection of the first redundancy group will send a StartDT to the RTU. On this connection, GIs and commands from SCADA are allowed and data from the RTU is received and transmitted to SCADA. The standby connection of the first redundancy group sends and receives test frames only.

The first standby connection of the second redundancy group will also send a StartDT and an initial GI but the received data from the RTU will be put into a standby buffer. It will also send manually triggered system commands (for example, GIs) and commands to node AK 1703 to update the process image. The link is in the new state dependent active (**Active-On**).

The standby connections of both redundancy groups will only send and receive test frames (**Active-Off**).

AK 1703 Redundancy

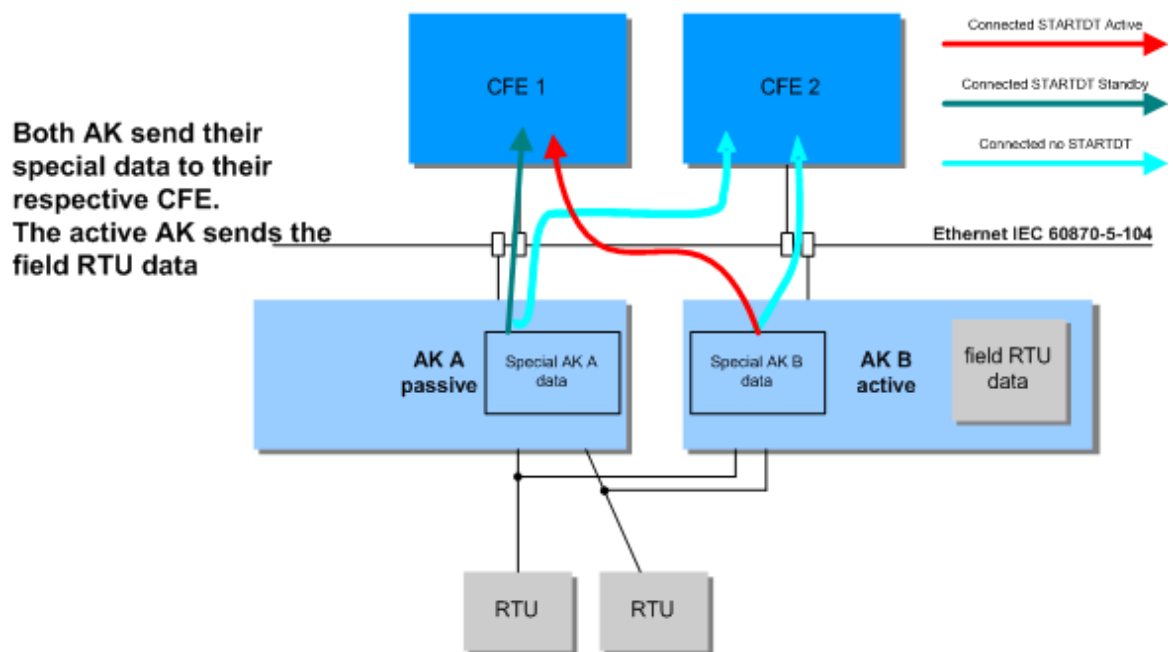
AK A sends special data from AK A only, and AK B sends special data from AK B only.



[sc_AK 1703 Redundancy _2014-02-14, 1, en_US]

Figure 2-35 AK 1703 Redundancy - Normal Situation

Situation after the switch over of the AK.



[sc_AK 1703 Redundancy after switchover_2014-02-14, 1, en_US]

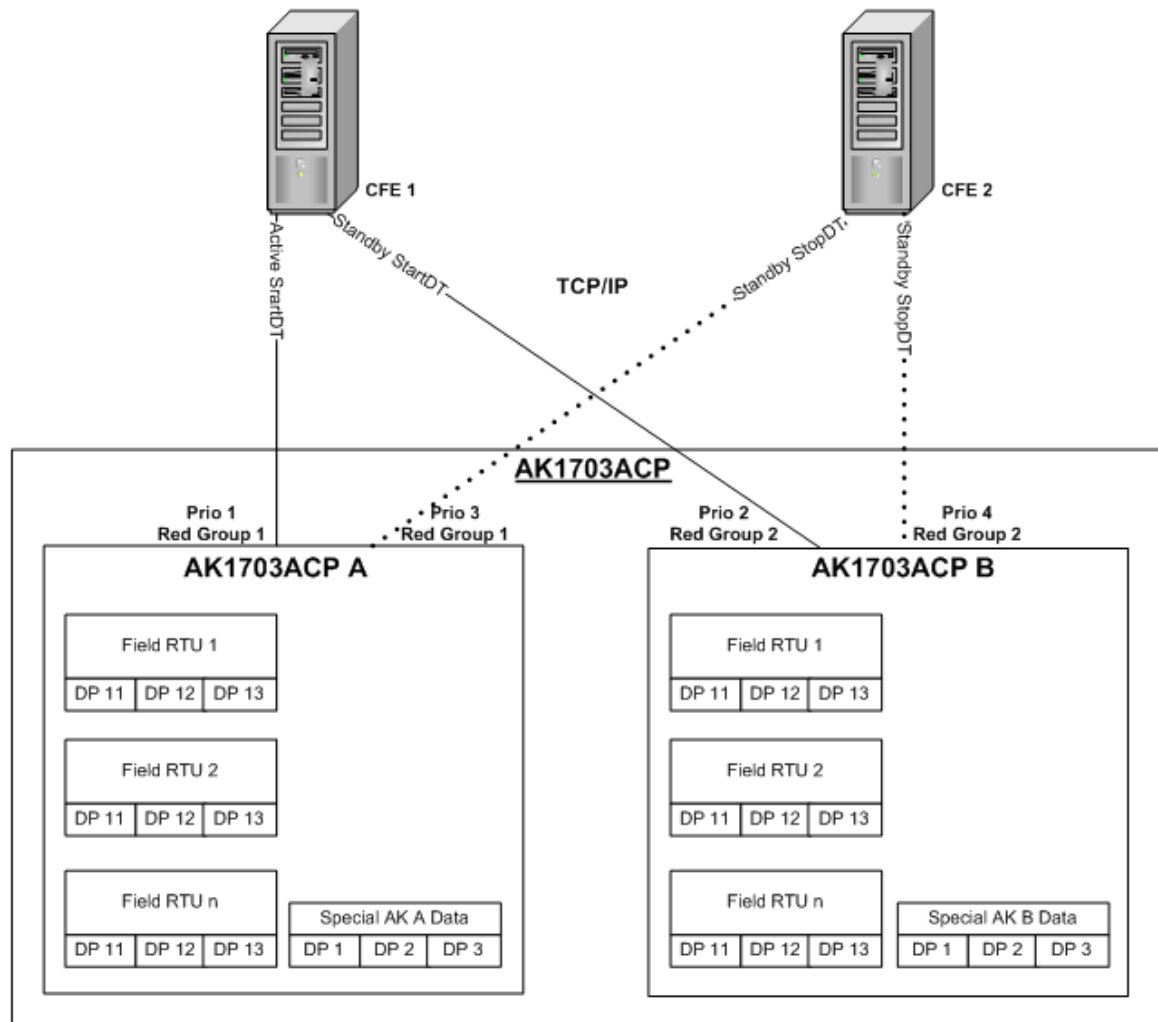
Figure 2-36 AK 1703 Redundancy - Situation after Switch Over

The CFE will process the field RTU data information from the active AK only and CFE will process the special AK A/B data from the active and passive AK.

Configuration with/without Redundancy Groups

Every AK 1703 ACP unit will be connected to one interface each on two CFE. If the AK 1703 ACP has a redundant setup, each AK 1703 ACP Unit consists of one special data RTU and one or more field RTU. If the

AK 1703 ACP is non-redundant, the special data RTU is not available. All RTU belonging to the AK 1703 ACP must be configured as CfeSubRtu, this applies also for the **Special AK A Data** and **Special AK B Data**.



[sc_Redundant AK 1703 ACP Configuration _2014-02-14, 1, en_US]

Figure 2-37 Redundant AK 1703 ACP Configuration

When having a redundant AK 1703 ACP, the redundancy groups have to be set up correctly. When having a non-redundant AK 1703 ACP, no redundancy groups need to be configured.

For more information on how to configure this type of redundancy, refer to DM-CFE-EN.

2.19 Communication Statistics

General

Communication Statistics are used to get information about the quality of communication interfaces and communication links connected to RTUs. For this purpose, CFE provides different communication statistic counters.

On detection of any communication anomaly a numerically encoded message is generated. This message shows the Error No of the detected error. The meaning of the **Error No** and the possible cause for the error is derived from Interoperability List of the respective protocol.

Statistic counters are configured in IMM to transfer them to SCADA for processing, displaying or for archiving. Each statistic counter can be read through the incorporated Test & Diagnosis tool no matter whether it is configured or not.



NOTE

Communication error detection and statistic counters are embedded in the protocol software and is missing or differently be implemented for the various protocols. Refer check the corresponding Interoperability List for details (see Appendix).

Communication Statistic Counters

Most of the protocol modules provide counters for communication statistics for the interface and the link. These counters are helpful in handling communication issues. That way weak communication lines are detected before they eventually break down.

Typical counters for communication statistics are:

- Counter for sent messages.
- Counter for received intact messages.
- Counter for received erroneous messages.
- Counter for time-outs.

Communication Errors

Due to poor communication links, switchover actions, configuration errors, device failures etc. communication errors occurs in the data traffic between CFE and RTUs.

Here are some typical (but not all) communication errors that occur:

- Invalid message
- Checksum / parity error
- Time-out

If the protocol software detects any communication anomaly or a configuration error it generates an error message. This error message is numerically encoded and represents a protocol-specific Error No. It is transported with a counter value to SCADA for displaying, logging and/or archiving. This special type of counter is called **Last Error** and has always the Info ID 900.

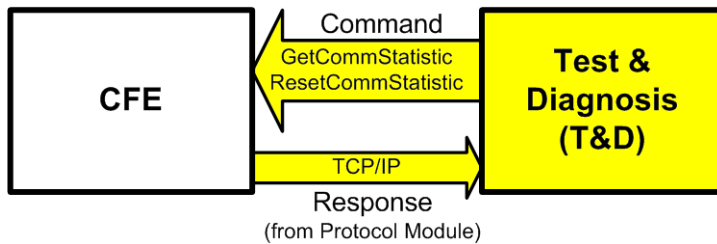
Counter Last Error (Info ID 900)

The special counter **Last Error** is used to transmit the last communication error that was detected. The counter **Last Error** behaves like an event message: it only appears; there is no corresponding **going** message when the error condition disappears. Typical for event information, there is no old-new comparison. That means, the same error is repeated several times in a sequence. Its value represents the number of the error (Error No), specified in Interoperability List of the respective protocol.

Accessing Statistic Counters through T&D Tool

Communication statistic counters and Last Error information of the relevant interfaces and links can be read or reset by simple commands using the Test and Diagnosis Tool (T&D) that is embedded in the CFE soft-

ware. The T&D is started on each Spectrum Power 5 server and communicates using TCP/IP over the system LAN.



[cfets058-210509-01.tif, 1, en_US]

Figure 2-38 Accessing Communications Statistics using T&D Tool

Regarding communication statistics the T&D tool features as following:

- Reading the current state of the counters.
- Resetting the counters (to zero).
- Identical T&D output scheme for all protocols.

Further Characteristics of Errors and Communication Statistic Counters

Further characteristics of communication errors and statistic counters are:

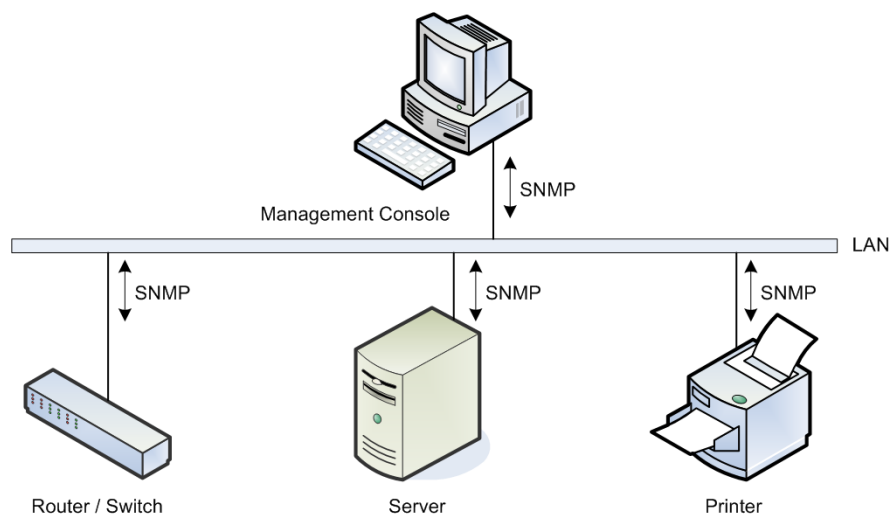
The value for not implemented counters is zero and the attribute is **NOT VALID**.

- The response to the T&D command **getcommunicationstatistics** only contains implemented counters.
- If no counter is implemented or statistic counters are not supported the answer is **Command is not supported**.
- When the interface process starts or when the counters are reset by T&D command, all counters are set to 0 and their attribute to **NOT RENEWED**.
- The range of the counters is from 0 to 2 147 483 647.
- Value 0 (zero) of **Last Error** counter always means: no error occurred since last start-up or reset.

2.20 Monitoring of Network Components by SNMP

2.20.1 Introduction

It is possible to monitor and control network components such as printers, routers, switches, servers with the Simple Network Management Protocol (SNMP) along with a suitable application program for handling incoming and outgoing data.



[cgcf112-180609-01.tif, 1, en_US]

Figure 2-39 Monitoring Network Devices through SNMP protocol

The SNMP protocol is a member of the internet protocol (IP) family and comprises 5 layers of the standardized ISO/OSI Layer Model.

Application layer	SNMP
Transport layer	UDP
Network/internet layer	IP
Data link layer	Ethernet
Physical layer	Ethernet physical layer

[cgcf113-180609-01.tif, 1, en_US]

Figure 2-40 SNMP Protocol Structure (According to OSI Layer Model)

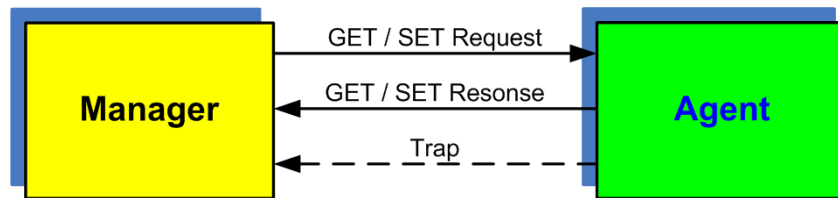
SNMP consists of a set of standards for network management, including an Application Layer protocol, a database schema, and a set of data objects. It presents management data in the form of variables on the managed devices, which describe the system configuration. These variables are requested (and sometimes set) by managing applications.

SNMP comprises the following tasks:

- Monitoring of network components.
- Telecontrol and tele-configuration of network components (presently not supported by CFE)
- Error detection and error information.

For detailed scope of implemented functionality, refer to the corresponding section in Interoperability Lists for Master Protocols.

In typical SNMP usage, there is a number of devices to be monitored, and one (or several) systems managing them. A software component called an **Agent** runs on each managed system and reports information via SNMP protocol to the managing systems.



[cgcfe114-180609-01.tif, 1, en_US]

Figure 2-41 SNMP Manager - Agent Communication

SNMP Agents present management data on the managed devices as variables (such as **free memory**, **system name**, **number of running processes**, **default route**). The protocol permits also active management tasks, as modifying and applying a new configuration.

The following message types are provided as per SNMP specification:

- **GET**, to request a management dataset
- **GETNEXT**, to request the subsequent data (in case of tables)
- **GETBULK**, to request several datasets at once for example, several rows of a table (available from version SNMPv2)
- **SET**, to modify one or several datasets of a network device
- **RESPONSE**, response from Agent on the preceding commands
- **TRAP**, data sent by the agent without being asked (also in case of failed modification)

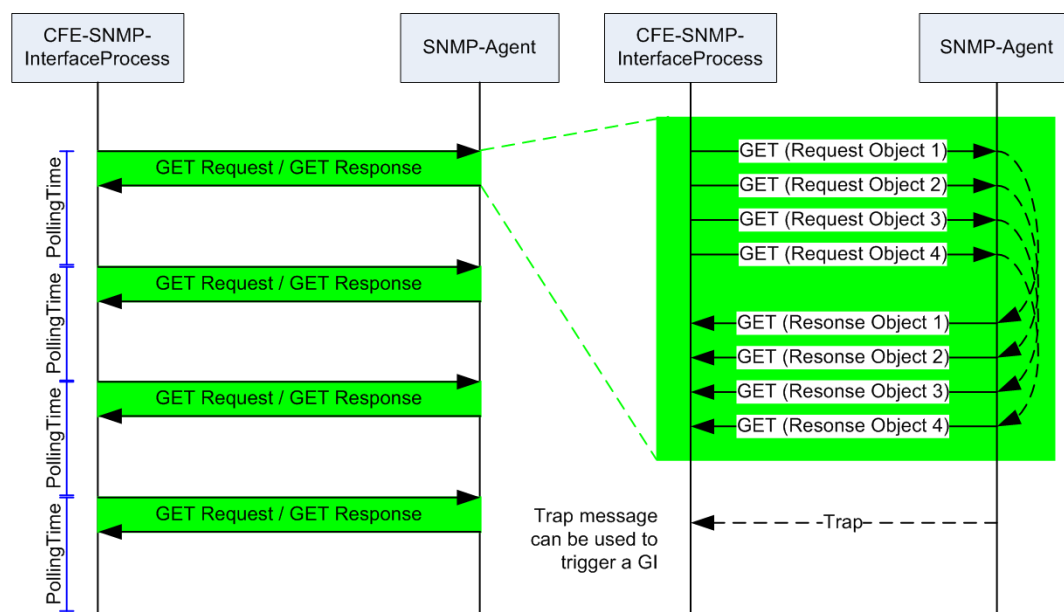
The following figure shows the SNMP-Message structure used for the **GetRequest / GetResponse** mechanism.

<u>Logical structur of an SNMP package:</u>				<u>Example / Explanation:</u>	
Package Header	Message Length in Bytes			total size of SNMP-package	
	Version number			SNMPv1, SNMPv2, SNMPv3	
	Community name			public, private,	
PDU (Protocol Data Unit)	PDU-Control Fields	Package type		Get, GetNext, GetBulk, Set,...	
		Request ID		for Request/Response correlation	
		Error status		Error type: noError(0), tooBig(1), noSuchName(2), badValue(3), ...	
		Error index		referece to object (the error occurs)	
	Variable Bindings	Variable Binding 1	Object name 1	OID: e.g. 1.3.6.1.2.1.1.1.0	
			Object value 1	Value of the specific object	
		Variable Binding 2	Object name 2		
			Object value 2		
					
		Variable Binding n	Object name n		
			Object value n		

[cgcfe115-180609-01.tif, 1, en_US]

Figure 2-42 SNMP Message Structure

The monitoring operations are performed on a periodic basis as shown in the following figure.



[cgcfe116-180609-01.tif, 1, en_US]

Figure 2-43 Monitoring Process, Principle, and Details

2.20.2 SNMP Implementation for CFE

The SNMP protocol runs as a process within the CFE server package. The CFE interface is configured as a type `CfeTransportInterface`, same as for the protocol IEC 104. The devices monitored are treated as regular Remote Terminal Units (RTU).

The following data types can be configured for monitoring devices connected to Siemens Spectrum Power 5

- Telemetered Digital
- Telemetered Analog
- Telemetered Accumulators
- Telemetered Step Positions

The following mapping of SNMP types is allowed:

Table 2-4 Mapping of SNMP Types to Siemens Spectrum Power 5 Data Point Types

SNMP Simple Type	Allowed Cfe Datapoint Types	Additional Info
Counter/Counter32	CfeTelemeteredAccumulator	-
Counter64	CfeTelemeteredAccumulator	-
Gauge	CfeTelemeteredAccumulator	-
TimeTicks	CfeTelemeteredAccumulator	-
Integer	CfeTelemeteredAnalog	MeasuredValue normalized
	CfeTelemeteredDigital	Single Point, value masked with 0x00000001 SCADA value = 1 or 2
	CfeTelemeteredDigital	Double Point, value masked with 0x00000003 SCADA value = 0 to 3
	CfeTelemeteredPosition	Step Position

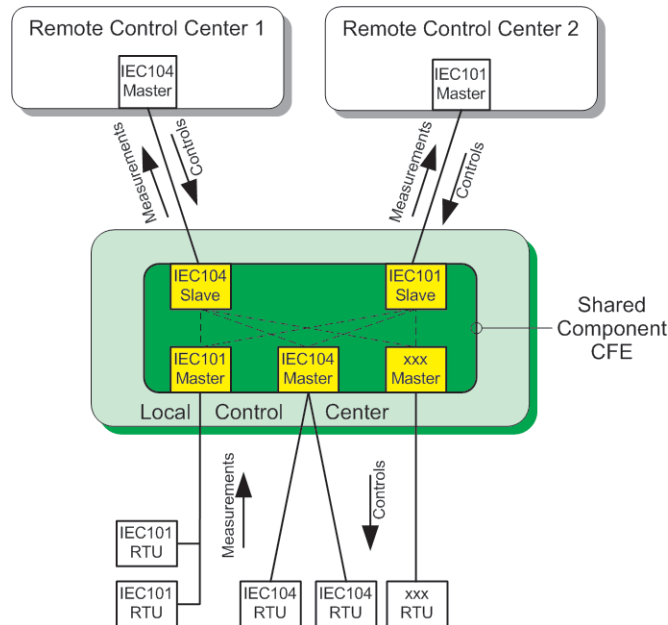
The following protocol functionality is implemented and available for usage:

- Message type:
 - GET Request
 - GET Response
 - TRAP: Optionally be used for triggering a General Interrogation on receipt.
- Polling: Polling Periodicity can be set to any value. Under normal circumstances, a value of 60 s is adequate.

2.21 CFE Slave Functionality

2.21.1 Introduction

The Spectrum Power 5 Front-End system CFE normally acts as a Master for the connected RTUs. The CFE Slave functionality is the opposite way: It provides the ability to forward process data to external Remote Control Centers (RCC) and to receive commands from them for passing on to connected RTUs.



[cgcf117-180609-01.tif, 1, en_US]

Figure 2-44 Master/Slave Communication, Overview

2.21.2 Scope of Communication

Protocols

For easy interfacing external Control Centers supplied by any manufacturer to Spectrum Power 5 Front-End system the following standard protocols are available:

- IEC 60870-5-101 (short: IEC 101) for UART-based serial connections.
- IEC 60870-5-104 (short: IEC 104) for TCP/IP based network connections.
- Others: The system is prepared to deploy other protocols instead of IEC101/IEC104 as a project specific add-on.

Starting the Communication

As soon as the connection to the RCC is established, the RCC usually issues a General Interrogation and the CFE as a Slave component, sends all (GI relevant) data. To indicate its readiness for communication the CFE sends an **End-of-Initialization** message to each RCC after running-up.

Forwarding Process Data

The primary goal of a **Slave** communication is to provide the connected Remote Control Centers with process data. This data can be any measurements such as analog values, digital values, accumulators, or step positions of the local Control Center, obtained from any communication way like ICCP, OPC, RTU CFE links as well as manual updates or internally generated.

Further, changed values are sent spontaneously with or without time tags. In case of analog values, only most recently updated analog values are sent. Analog values can also be sent periodically or in a subordinate transmission circle (background transmission). Data requests from RCC for updating accumulator values are responded accordingly by the CFE Slave functionality.

Receiving Commands

All commands received from Remote Control Center are sent to the supervisory control interface (SUPY) of the local Control Center. These commands are routed and supervised by SCADA. The target can be any substation reached using ICCP, OPC, CFE, or an internal status information (for example **CFE Interface Lock** or **CFE Interface Preference**). For these commands the Slave CFE sends a response to the Remote Control Center. If a digital command is executed, a termination message is sent; if monitor information is available the termination of the command is sent after the status change. For **Select-Before-Operate** commands (SBO), the CFE autonomously handles the select command. Only the execute command is forwarded.

2.21.3 Processing in Monitor Direction

Processing Analog Values

Analog values are transmitted to the RCC as IEC101/104 type **Normalized**, **Scaled** or **Short Floating** values. For converting the internal values into the protocol-specific values, pre-defined characteristic curves are applied. The most recently updated values are also transmitted in a General Interrogation response as well as in cyclic transmissions. If the internal value is beyond the characteristic curve, the maximum physical value is sent with qualifier OVERFLOW set.

Processing Accumulator Values

Accumulator values are transmitted to the RCC as IEC101/104 Integrated Totals without conversion. It is considered that accumulator values are transmitted as an integer number to RCC. As a consequence, the value is rounded down if in SCADA a floating point format was used.

Processing Digital Values

For digital values the only potential processing at RCC interface is the inversion: If the attribute Inverting of monitor direction is set, a single point value or both bits of a double point value is/are inverted.

Processing Fleeting Data (Events)

A fleeting is an event and as such it has no value. For the transmission to an RCC, it is converted into a single point information. When an event is received from subordinate RTU, 2 single point messages are generated and sent to the RCC: one for the transition from zero to one and then one back to zero.

Processing Step Positions

Transformer step positions are sent to the RCC as IEC101/104 type **Step Position** values ranging from - 64 to 63. There is no further processing or conversion in CFE.

2.21.4 Processing in Command Direction

Processing Switch Commands

All 3 types of IEC101/104 digital commands issued from RCC are accepted by the IEC101/104 Slave: Single Commands, Double Commands, and Regulation Step Commands. The routing of an RCC command towards target RTU are executed by way of a direct command or by **Select-Before- Operate** procedure. CFE checks and confirms the Select Command: If the object has accordingly been configured in the database, CFE confirms positive if not, negative. When SCADA detects a status change in monitor direction of the switch is controlled or when command time-out occurs, CFE sends a Command Termination message to the RCC.

Command Inversion

If the IMM attribute Inverting of Control Direction is set, the information of the command is inverted, ON to OFF command becomes OFF to ON command and either ways.

Processing Set-Point Commands

All 3 types of IEC101/104 set-point command such as Normalized, Scaled, and Short Floating Point are accepted by the IEC101/104 Slave. Digital switch commands, the routing of an RCC set-point command towards target RTU is executed by way of a direct command or by **Select-Before- Operate** procedure. Like for digital commands, CFE checks and confirms the Select Command either positive or negative. A characteristic curve is defined for the control direction. When SCADA has accepted or rejected the set-point command, the CFE sends a corresponding Command Termination message to the RCC.

General Interrogation

After reception of a General Interrogation (GI) request from RCC, CFE-Slave sends all GI-relevant data points to the RCC. This data represents the most recent state deposited in the CFE process image.

Time Synchronization from RCC

Depending on attribute settings, time synchronization commands from RCC is rejected or accepted by CFE. However, in case of accepting, the time information of the synchronization command is not evaluated.

2.22 Master/Slave Configurations

2.22.1 Overview

The protocol independent part of CFE is designed to handle not only the IEC101/104 Slave protocol, but also other serial or network Slave protocols.

The following is considered:

- Network interfaces (TCP/IP) are always independent
- Serial interfaces can be independent or dependent
- For availability reason, both external Master and CFE Slave are set up in a redundant configuration as demonstrated in the following figures.

2.22.2 Network Interfaces

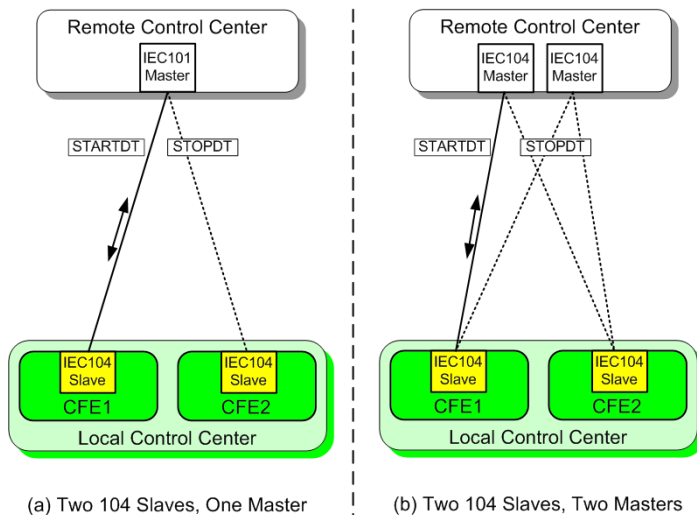
IEC 104 Network interfaces are always independent and controlled by the Master. Only the Master is in a position to establish connections and to decide which one (of a redundant pair) is to be used for data transmission. The following figures show typical IEC 104 Master/Slave configurations using TCP/ IP based network connections.

One Master, One Slave (No Redundancy)

The simplest configuration without redundancy is just one Master and one Slave without any redundancy. The Slave opens a listening protocol socket; the Master establishes an IP-connection and sends the **STARTDT** control frame to the Slave for starting the data transmission.

Redundant Master, Non-Redundant Slave

Figure 2-45 (b) shows an RCC with redundant front end and thus it is able to connect using 2 connections. It is allowed to start data transmission only over one connection, called active connection (or started connection) in the following. On the other (stopped) connection only test frames are exchanged for monitoring the standby channel by the Master. When the active connection fails, the Master can activate the standby connection by sending the **STARTDT** control frame over the alternative connection.



[cgcf118-180609-01.tif, 1, en_US]

Figure 2-45 Master/Slave Configurations, Redundant CFE, Independent Network Links

Non-Redundant Master, Redundant Slave

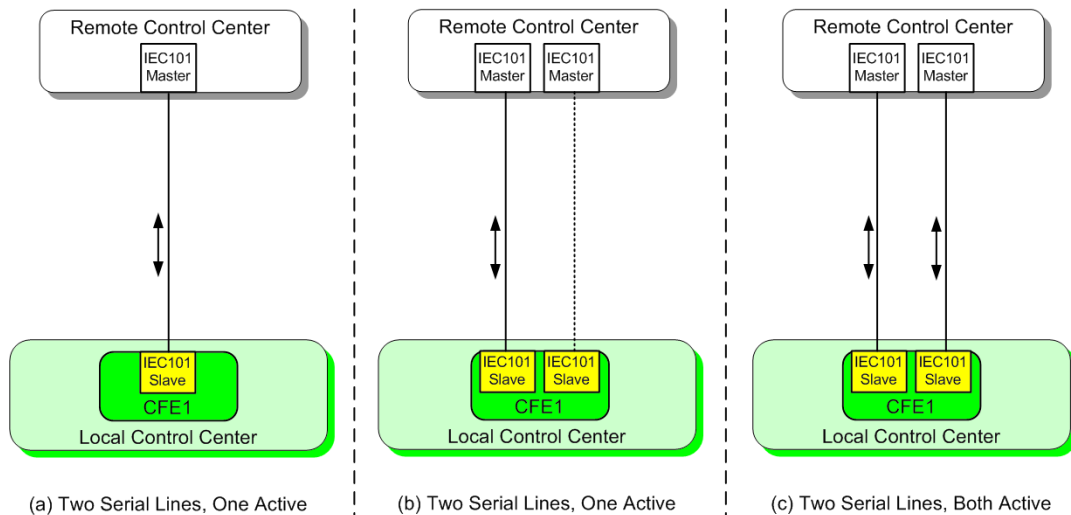
Figure 2-45 (a) shows a redundant Slave CFE. Each Slave opens a listening socket. The Master connects both, but it is only allowed to start data transmission over one connection. Again, test frames are exchanged over the standby connection. If the active connection fails, the Master sends a STARTDT control frame over the standby connection in order to use this connection from now on for data transmission. Stopping data transmission is done by the Master by sending a STOPDT control frame over the active connection.

Redundant Master, Redundant Slave

Figure 2-45 (b) shows both Master and Slave are redundant, with a total of 4 connections. Again, the Master connects all 4, but is only allowed to start data transmission over one connection. Test frames are exchanged over the standby connections, and when the active connection fails, the Master sends a STARTDT control frame over one of the standby connections to the Slave system.

2.22.3 Serial Interfaces

The following figures show Master/Slave configurations with serial connections.



[cgcf119-180609-01.tif, 1, en_US]

Figure 2-46 Master/Slave Configurations, Non Redundant CFE, Independent Serial Links



NOTE

The Master additionally use modem splitters and duplicate the serial ports internally to improve availability; this does not affect the Slave and thus is not shown in the figure.

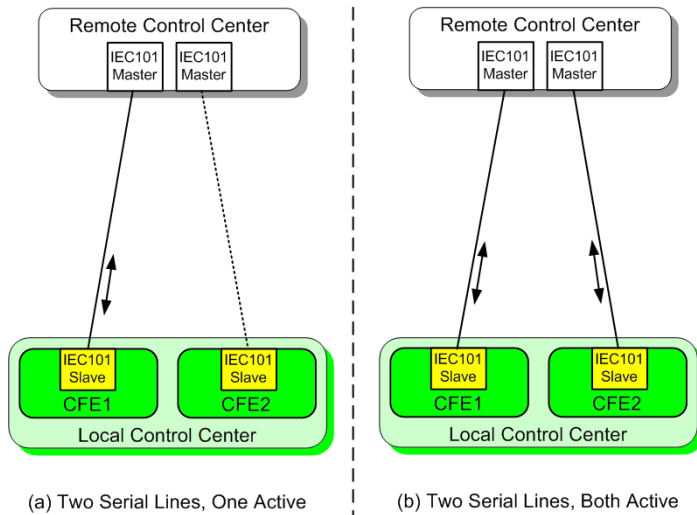
Independent Links, Non-redundant CFE

Figure 2-46 (a) shows the simplest case without any redundancy. In case of a line failure or a CFE failure the Master/Slave communication is interrupted. Both balanced and unbalanced transmission mode (polling from Master) is possible.

In *Figure 2-46* (b) the serial line is duplicated for redundancy. In unbalanced transmission mode, the Master decides, which line is used to poll the Slave; this line is called **active (polling) line** in the following. Both Slaves are ready to answer possible polls. If the active (polling) line fails, the Master uses the standby line.

In balanced mode, the Slave permanently sends **Status of Link** requests and - if the Master reacts to these requests - data as well.

In *Figure 2-46* (c), the situation is nearly the same, but the Master uses both lines at the same time. Both Slaves answer the polls, accept commands, and report the same events. As such, the Master receives all data from Slaves twice and must discard one half. If a line fails, the Master uses the other line (while still sending polls on both). The quality of redundancy is the same, but the Master behaves differently.



[cgcf120-180609-01.tif, 1, en_US]

Figure 2-47 Master/Slave Configurations, Redundant CFE, Independent Serial Links

**NOTE**

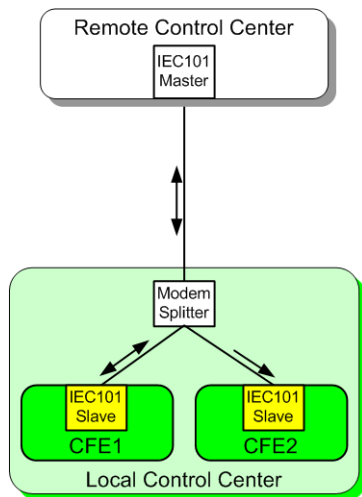
In balanced mode, there is no difference between case (b) and (c) for CFE.

Independent Links, Redundant CFE

In [Figure 2-47](#) (a) and [Figure 2-47](#) (b) also, the Slave CFE is doubled for redundancy. If the active (polling) line or the connected CFE fails, the Master activates the standby line.

Dependent Interfaces

The following figure shows a configuration with redundant Slave CFE. The serial line is connected to 2 communication ports on different CFE servers using a modem splitter.



[cgcf121-180609-01.tif, 1, en_US]

Figure 2-48 Master/Slave Configurations, Redundant CFE, Dependent Serial Links

In case of a single serial link as shown above, both Slaves receive the polls and commands from Master. However, only one of them answers. When it does not hear (or understand) the polls, the other Slave answers.

Monitoring of Standby Links

With dependent interfaces, also the standby communication links are monitored. A status information indicates if they are no more able to take over the communication should the active link fail.

2.22.4 Data Buffering

Temporary data storing (buffering) is used to prevent communication links from losing data. Due to the data buffering the system behavior in different situations is as following:

Failover

No data is lost on failover. In rare cases, duplication of events occurs after failover.

Burst

When at an extreme data inrush more events occur than can be sent to the Remote Control Center the buffer of an active link becomes full. In this case, the oldest entries in the affected buffer are cleared to gain space for new data. Further, a CFE status information **Buffer Overflow** is generated that can be forwarded to the Remote Control Center to signal the data loss and finally, a General Interrogation is used to update all static data.

All Links Down / Reconnection

As long as the RCC Master does not start data transmission on any link, all events and accumulator values are buffered for all links. When a connection is (re-)established, buffered data is transmitted to the Remote Control Center. Then, all GI data is sent (without interrogation command) to update the image of the RCC.

One Link Active

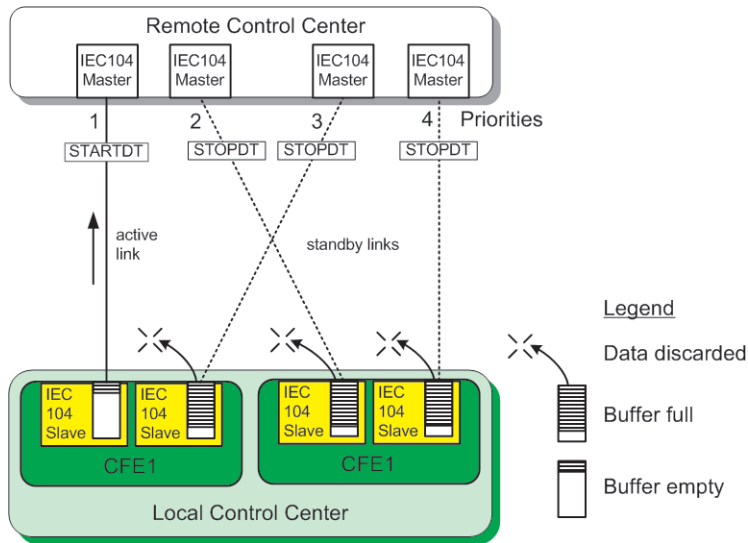
Until event data is transmitted on (at least) one active link to an RCC, it remains buffered on the alternative (standby) links to the same RCC. That way data is kept ready for a possible retransmission at failover. In order not to transmit obsolete data at failover **older** data is cleared after Buffer Time Span. This principle is applied for both dependent and independent interfaces.

Analog Value Transmission

Analog measurements are stored in the buffer only once per data point, even if they change. That way no buffer overflow or data loss can happen. On transmission at any time, the most recent values are sent.

Example

The following figure shows a simplified example with a Remote Control Center having 4 IEC 104 Masters. The RCC maintains 4 alternative connections. Only the connection with the highest priority is made active by the RCC. The other 3 links are standby. In this case the IEC 104 Slaves keep their local database updated and save events in the buffers. When the retention time of these events has expired, that is assumed that the data is safely transmitted the events in the buffers are discarded. When a buffer of an inactive (standby) link becomes full, the oldest items are **silently** replaced with newer ones.



[cgcf122-180609-01.tif, 1, en_US]

Figure 2-49 Buffering Principle for Network Connections

2.22.5 Granular Activation

A Slave process can handle only one Remote Control Center. Whenever configuration changes regarding interfacing this Remote Control Center or its data points are carried out, only this process is restarted and all GI-relevant data is sent to the Remote Control Center. The other interface processes are not affected.

2.22.6 IEC 60870-5-101 / ...-104 Protocol Specifications

CFE Slave Startup

The behavior of Slave CFE after run-up can be configured for each RCC as follows: Either the system message **End of Initialization** is sent or all GI relevant data points are transmitted automatically (as a **Background GI**) after startup time-out.

If the protocol type IEC 60870-5-101 is used, for each logical RTU a synchronization message (Type ID 103) is sent to RCC after CFE startup.

The synchronization messages has/have the following settings:

Common Address:	same as for the corresponding LRTU
Object Address:	0
Cause of Transmission:	3 (spontaneous)

This message is used for interpreting the time information of later transmitted time tagged messages (if any). CFE does not expect any response from RCC to this command.

Transmission Priorities

The processing priorities of the IEC-101/104 Slave are shown in following table. Highest priority comes first, lowest is at the end.

Prio	Process Response
1	Command confirmation
2	End of Initialization
3	Event processing
4	Confirmation of Test Command
5	Single Read command

Prio	Process Response
6	General Interrogation response
7	Cyclic or periodic transmission

Multiple RTUs

The IEC101/104 Slave supports **Multi-RTU** functionality. With this feature, it is possible to configure several logical RTUs on one Link Address (IEC 101) or one IP-address (IEC 104) but with different Common Addresses. From Remote Control Center it looks like a data concentrator with connected sub- RTUs. The assignment of measurements to these logical RTUs is permissive.



NOTE

For performance reasons the number of logical RTUs should not exceed 20 per RCC.

2.22.7 Device Status Information

Non-Controllable Status Information

For Slave protocols, the CFE provides the non-controllable CFE status information **Device Active** and **Device Failure** as per the following table.

Table 2-5 Non-Controllable Status Information

Item	Device Status	Independent Interface (IEC101SL/ IEC104SL)	Dependent Interface (IEC101SL)
Interface	Device Active	Ready for data exchange (not locked)	Ready for data exchange (not locked or standby)
Interface	Device Failure	Interface error	Interface error
Interface Link	Device Active	Exchanging data	Exchanging data
Interface Link	Device Failure	No connection established, no test frames received	No connection established, no test frames received

Controllable Status Information

The controllable status information **Device Locked** and **Device Preferred** have the following meaning for Slave protocols as mentioned in the following table.

Table 2-6 Controllable Status Information

Item	Device Status	Independent Interface (IEC101SL/ IEC104SL)	Dependent Interface (IEC101SL)
Interface	Device Locked	Interface deactivated	Interface deactivated
Interface	Device Preferred	Interface becomes active over-writing pre-defined priority	N/A
Interface Link	Device Locked	N/A	N/A
Interface Link	Device Preferred	N/A	N/A



NOTE

They do not all exist, as only one RCC link can be assigned to one interface.

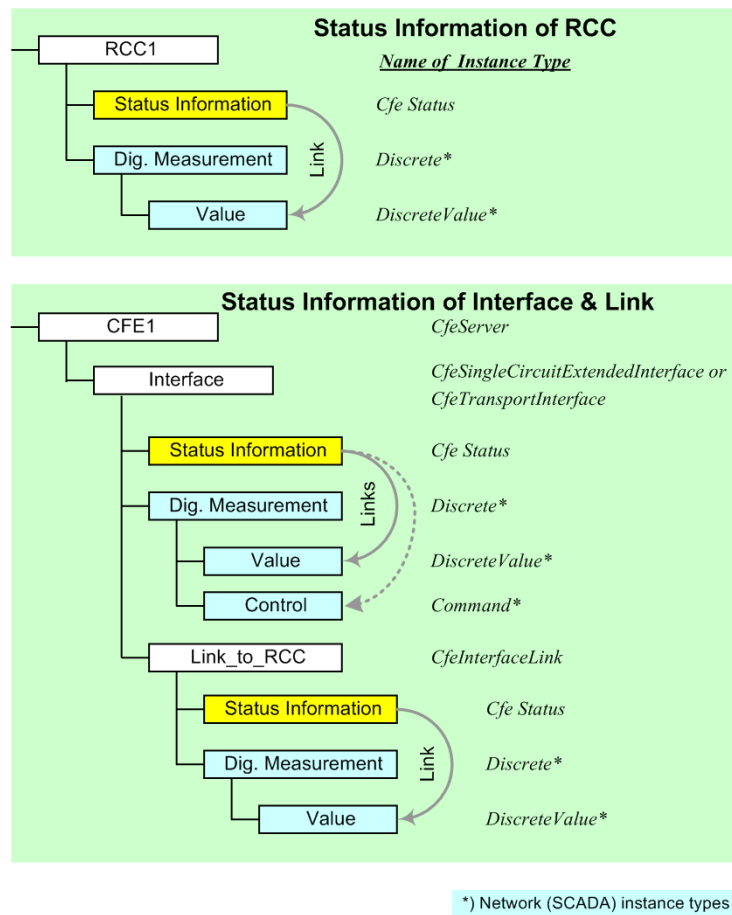
2.22.8 Status Information for Communication with RCCs

In order to monitor and control the communication with Remote Control Centers, appropriate status information is available in the system.

The following status information is configured in IMM:

- Non-controllable status information:
 - Device Active (for all types of interfaces and links)
 - Device Failure (for all types of interfaces and links)
 - Buffer Overflow (for each link)
 - RCC Failure (if all links are down)
- Controllable status information:
 - Device Locked (applicable only for interfaces, not for links)
 - Device Preferred (applicable only for interfaces, not for links)

Status information for RCC and RCC-CFE communication is arranged in hierarchical order as shown in the following figure.



[cgcf135-230609-01.tif, 1, en_US]

Figure 2-50 Arrangement of Status Information for RCC, Interface and Link



NOTE

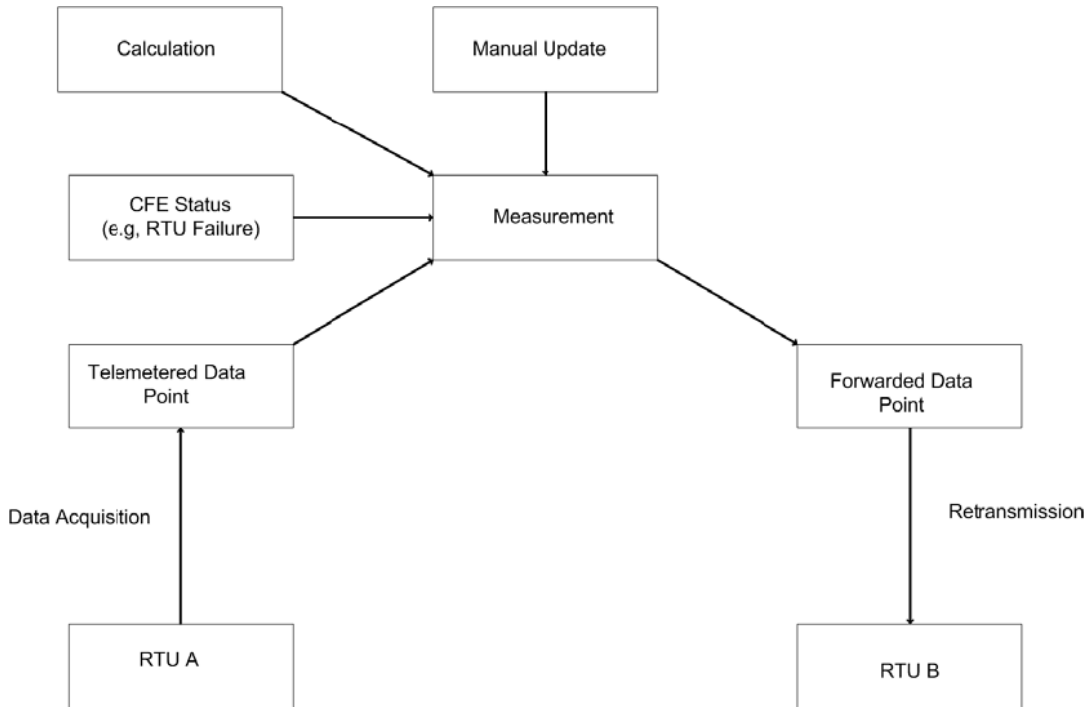
System counters for communication statistics are not available for RCC connections.

Configuring status information for RCC communication is done the same way as for that of RTU communication:

- A SCADA instance of the type **Discrete** is created.
- A child instance of the type **DiscreteValue** is created.
In case of controllable status information: Also a child instance of the type **Command** is created.
- The desired Status Information (type CFE Status) is created.
- Status Information and **Value** instance are interlinked.
In case of controllable status information also a link to the **Command** instance is established.

2.23 Retransmission

Retransmission basically means automatically forwarding data point values received from one RTU to another RTU, where they may be used for local displays or checking interlocking conditions. The source and target RTUs can have different protocols. Also calculation results or CFE Status Information like "RTU Failure" stored in a measurement can be forwarded to an RTU.



[cfets001-180111-01.tif, 1, en_US]

Figure 2-51 Principle of Retransmission

Manually Entered Values

If an operator manually overrides a value, this manually entered value is retransmitted. When the operator unblocks a value, the current original value is sent to the RTU.

General Interrogation

When the connection to the target RTU is established or when the target RTU sends a general interrogation request, all data points defined for retransmission to that target are sent once.

Time Stamp

Spontaneous retransmitted values can be sent with the time stamp.

Preprocessing

Preprocessing is usually done during data acquisition. Forwarded data points can only be inverted or scaled. For additional information, refer to CFE Data Modeling Guide, Forwarded Data Points.

Buffering

When analog measurement values change faster than they can be forwarded, only the most recent values are sent. Unlike for slave protocols, events of forwarded data points are not buffered while an RTU fails.



NOTE

Only selected protocols, CFE protocol implementations and RTUs support the transmission of values in control direction. Please check the Interoperability List in the Appendix of this document and of the RTU vendor.

3 Technology

3.1	User Interface	90
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3.1 User Interface

3.1.1 CFE Parameter Definition Interface

Engineering through IMM

All CFE-related parameters (attributes), links and settings are present in the Spectrum Power 5 system. These values are entered via the Information Model Management (IMM). IMM runs on the PSOS server and is accessible from each Spectrum Power 5 machine.

Data entry is done using various editors of the IMM. It is a web-based user interface for entering all system and customer data into the configuration database DOR.

The following screenshot illustrates a typical IMM editor for modifying attributes of an analog data point.

General Attributes Links IEC60870-5-101	
Attributes/Associations	Value
Command operation mode	Direct Operation
Control Direction Type	Not used
Control-TC-Address	?
Description	Voltage of 110kV-Busbar A
GI Relevant	False
Monitor Direction Type	Measured Value (Normalized)
Monitor-TC-Address	8010
Renewal check	Cycle 1 of renewal check enabled
Smooth Factor	0
Threshold Value	6400
Zero Suppress Lower Border	0
Zero Suppress Upper Border	3200
(Has A Cntrl) CfeCharacteristicCurve	
(HasA) CfeCharacteristicCurve	
(ReceivesFrom) MeasurementControl	
(SendsTo) MeasurementValue	

[cfets021-140509-01.tif, 1, en_US]

Figure 3-1 Attributes of the Instance Type-CFE Telemetered Analog



NOTE

For additional information on CFE attributes, refer to CFE Data Modeling Guide. All editors relevant to CFE and their use is described in this document.

3.1.2 CFE Operation Interface

Runtime Explorer

The Runtime Explorer that represents the operation user interface is used to monitor the status of system components.

To check whether the system or components are in operation, the following means are available:

- System Management Display
It shows the status of all major program modules
- Customized optional user interface diagram
It shows the relevant status of the communications system at a glance

CFE Equipment Status Display

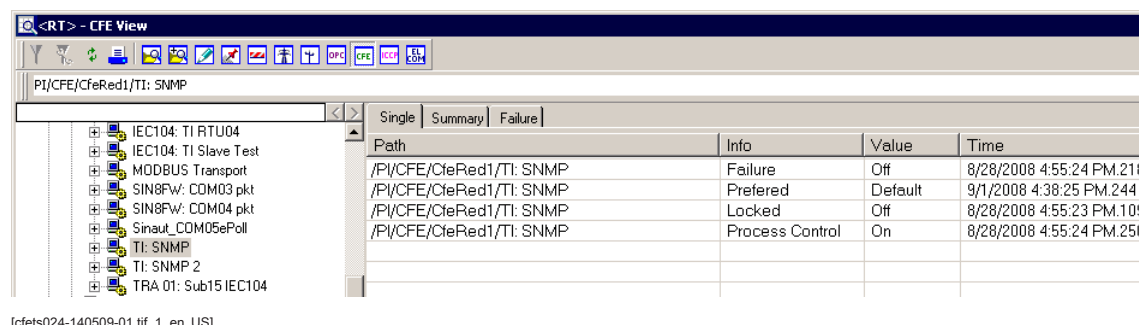
The Communication Front End has different status information. All this information is displayed in the CFE Equipment Status Display (CFE View), provided by the Runtime Explorer.

The CFE Equipment Status Display:

- Displays the status information for:
 - CFE Interfaces (Process Control, Locked, Failure, Preferred)
 - CFE Interface Links (Process Control, Locked, Failure, Preferred)
- Supports online refreshing
- Provides the following views:
 - Single View
 - Summary View
 - Failure View

Single View

Single View displays the status information for the selected device.

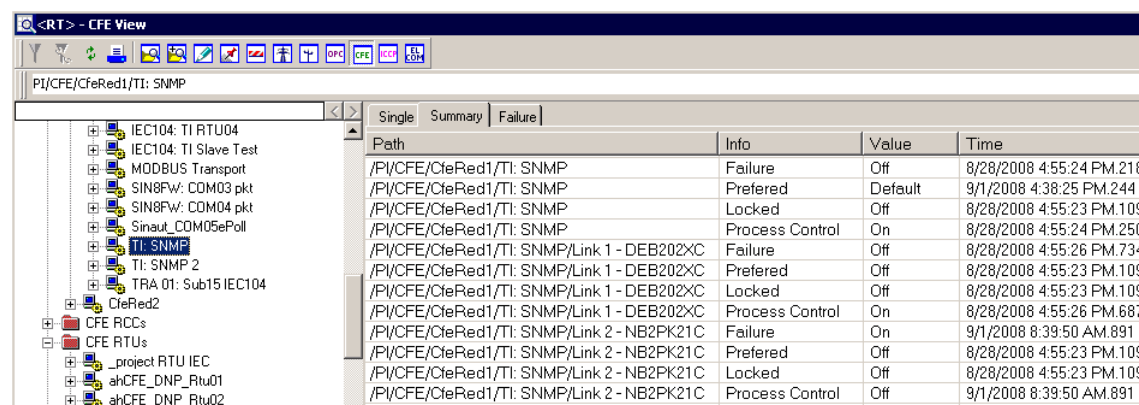


[cfets024-140509-01.tif, 1, en_US]

Figure 3-2 Runtime Explorer: CFE Status Information, Single View

Summary View

Summary View displays the device status information for the selected device as well as for all subdevices.



[cfets025-140509-01.tif, 1, en_US]

Figure 3-3 Runtime Explorer: CFE Status Information, Summary View

Failure View

Failure View displays the device status information Failure for the selected device and all subdevices. Only the failed devices are reported.

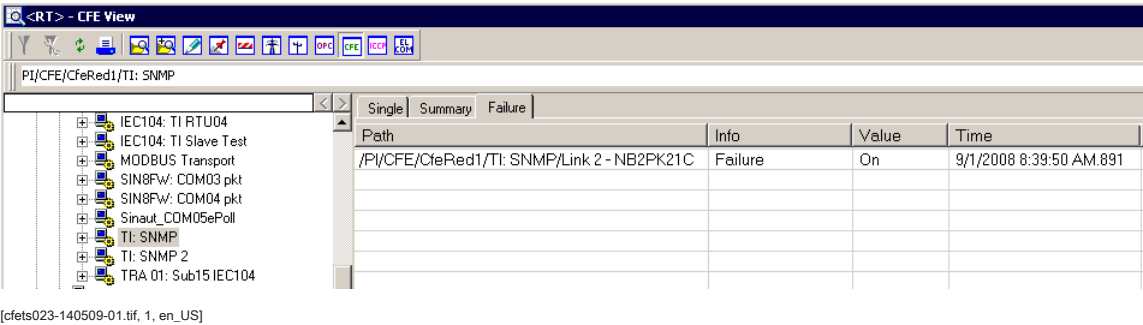


Figure 3-4 Runtime Explorer: CFE Status Information, Failure View

3.2 CFE Presented in Runtime Explorer

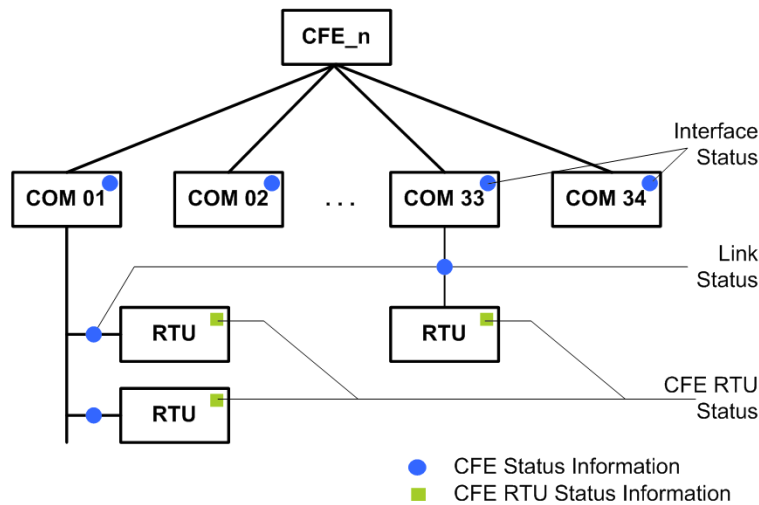
3.2.1 CFE /RTU Status Information - Introduction

Status data provide information about the availability of CFE components such as CFE Interfaces, links to RTUs as well as for RTUs. CFE status information is derived in the CFE through periodical device monitoring routines, like Link Test.

The following standard routines in CFE are applied to obtain the RTU Status Information:

- General Interrogation
- Group Interrogation
- Completeness Check
- Renewal Check

For additional information, refer to [2.4 CFE / RTU Status Information and Communication Statistics](#). The following diagram shows an overview of CFE components and RTUs providing Status Information.



[cgcf123-230609-01.tif, 1, en_US]

Figure 3-5 Overview of CFE and RTU Status Information

The Run-Time Explorer is a user interface for displaying status information including its current value of system components. To start the Runtime Explorer, select **Tools** in the **Power Application Tool Bar (PAT)** and then **Runtime Explorer** in the sub menu.

For displaying CFE or RTU Status Information there are 2 Views available:

- Substation View, showing a comprehensive overview including RTU Status Information.
- CFE View, showing selective CFE Status Information with tabs **Single**, **Summary**, and **Failure**.

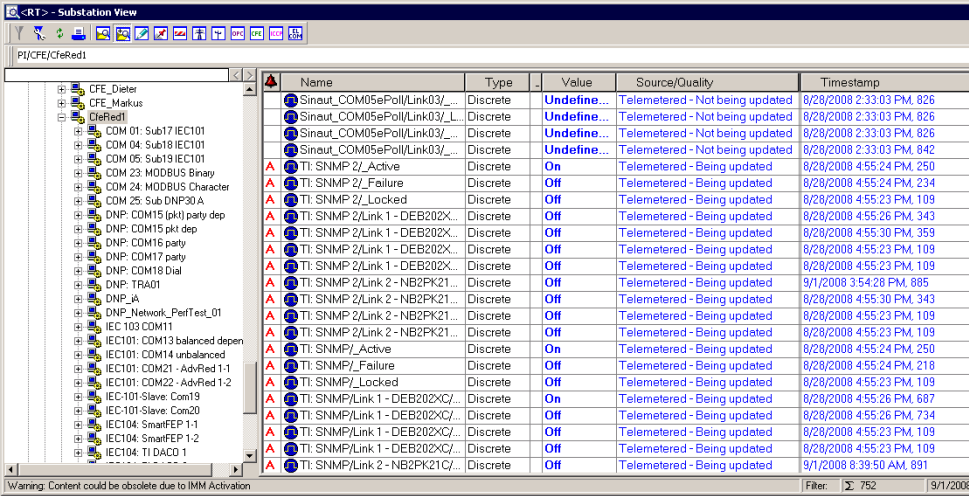
It is possible to toggle between Substation View and CFE View by means of icons in the tool bar.

3.2.2 CFE Status Information, Presented Through Runtime Explorer

Select a particular CFE, for example CFE1, in the Overview Pane, the status of all CFE1 components are shown in the table on the right side. The meaning of the columns is as follows:

1st Column (Bell symbol):	A Indicates a not yet acknowledged status change
Name:	Name of the Status Info Instances
Type:	Type of the Instance (for Status Information: all are of the type Digital Measurement).

Value:	The current value of the Information for example on, off or undefined (if not updated yet)
Source/Quality:	Indicates whether the instance is telemetered and updated.
Last change:	The instant when the displayed value was most recently updated.



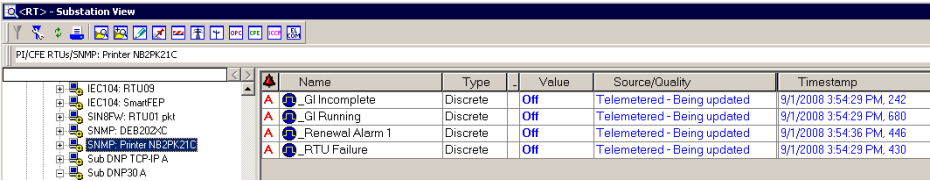
Name	Type	Value	Source/Quality	Timestamp
Sineut_COM05ePoll/Link03/...	Discrete	Undefined...	Telemetered - Not being updated	8/28/2008 2:33:03 PM, 826
Sineut_COM05ePoll/Link03/...	Discrete	Undefined...	Telemetered - Not being updated	8/28/2008 2:33:03 PM, 826
Sineut_COM05ePoll/Link03/...	Discrete	Undefined...	Telemetered - Not being updated	8/28/2008 2:33:03 PM, 826
Sineut_COM05ePoll/Link03/...	Discrete	Undefined...	Telemetered - Not being updated	8/28/2008 2:33:03 PM, 842
TI: SNMP 2/ Active	Discrete	On	Telemetered - Being updated	8/28/2008 4:55:24 PM, 250
TI: SNMP 2/ Failure	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:24 PM, 234
TI: SNMP 2/ Locked	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 PM, 109
TI: SNMP 2/Link 1 - DEB202X...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:26 PM, 343
TI: SNMP 2/Link 1 - DEB202X...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:30 PM, 359
TI: SNMP 2/Link 1 - DEB202X...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 PM, 109
TI: SNMP 2/Link 1 - DEB202X...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 PM, 109
TI: SNMP 2/Link 2 - NB2PK21...	Discrete	Off	Telemetered - Being updated	9/1/2008 3:54:28 PM, 885
TI: SNMP 2/Link 2 - NB2PK21...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:30 PM, 343
TI: SNMP 2/Link 2 - NB2PK21...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 PM, 109
TI: SNMP 2/Link 2 - NB2PK21...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 PM, 109
TI: SNMP/ Active	Discrete	On	Telemetered - Being updated	8/28/2008 4:55:24 PM, 250
TI: SNMP/ Failure	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:24 PM, 218
TI: SNMP/ Locked	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 PM, 109
TI: SNMP/Link 1 - DEB202X/...	Discrete	On	Telemetered - Being updated	8/28/2008 4:55:26 PM, 687
TI: SNMP/Link 1 - DEB202X/...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:26 PM, 734
TI: SNMP/Link 1 - DEB202X/...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 PM, 109
TI: SNMP/Link 1 - DEB202X/...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 PM, 109
TI: SNMP/Link 2 - NB2PK21C/...	Discrete	Off	Telemetered - Being updated	9/1/2008 8:39:50 AM, 891

[cgcf124-230609-01.tif, 1, en_US]

Figure 3-6 Runtime Explorer: CFE Status Information

3.2.3 RTU Status Information Presented Through Runtime Explorer

To display the RTU Status Infos select the substation instance of interest in the Overview Pane. The substation is a child instance of the instance /PI/CFE RTUs.



Name	Type	Value	Source/Quality	Timestamp
GI Incomplete	Discrete	Off	Telemetered - Being updated	9/1/2008 3:54:29 PM, 242
GI Running	Discrete	Off	Telemetered - Being updated	9/1/2008 3:54:29 PM, 680
Renewal Alarm 1	Discrete	Off	Telemetered - Being updated	9/1/2008 3:54:36 PM, 446
RTU Failure	Discrete	Off	Telemetered - Being updated	9/1/2008 3:54:29 PM, 430

[cgcf125-230609-01.tif, 1, en_US]

Figure 3-7 Runtime Explorer Showing RTU Status Information

Instances are displayed in alphabetical order. To arrange Status Information items all together in top position special characters, for example underscore (_) or dots (...), are often used in front of the instance name.

3.3 Sizing

The total number of CFE processes that runs on a server has to be less than 100. This number is calculated as follows:

- Two general CFE processes run on each server (CfeRti.exe and CfeGateway.exe).
- For each type of protocol, 1 master CFE process runs on each server:
For example, if IEC101 and IEC104 are handled by a server, 2 such master processes will run on that server.
- For each CFE interface assigned to the server, 1 CFE process runs on the server.

A single CFE interface process can handle the communication to several RTUs (**SubRTUs**). In case of IP-based protocols, these can also be standard RTUs. IP-based interface processes create one TCP/IP connection per linked RTU. It has to be taken into account that depending on the type of protocol, the communication to the RTUs is sequential in this case.

4 Algorithms / Concepts

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4.3	Mapping of SCADA Qualifiers to CFE Quality Flags - Slave Protocols	101
4.4	RTU Connection Diagrams	102
4.5	Switch-Over Behavior	105

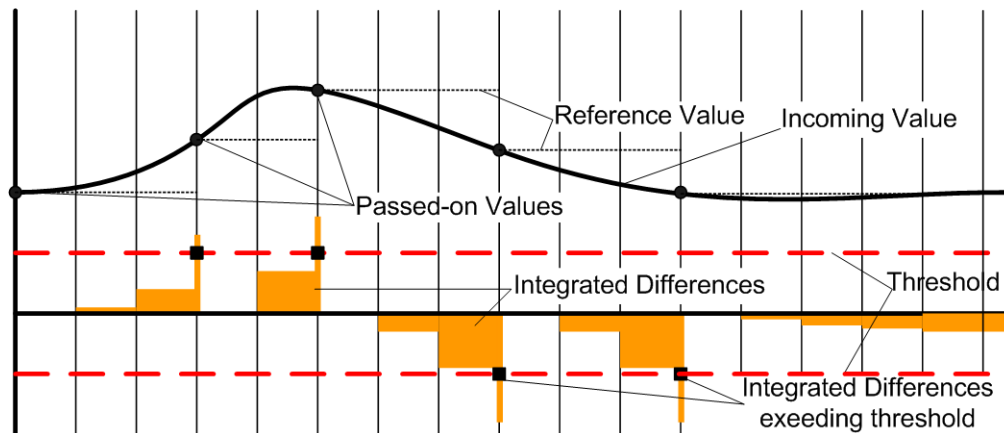
4.1 Methods of Threshold Processing

Analog values can be subject to a threshold value check. All the new analog values arriving from the substation which, exceed this threshold are passed on to SCADA. The Integration Method is used to carry out the threshold check.

For slave protocols, four different threshold calculation types can be chosen. Two of them are chosen based on a direct comparison and two of them are chosen by using a method of integration.

Principle 1 – Integration

If the integration of the value, as compared to the most recently transmitted value, is less than the threshold, it is not passed-on. When the sum of the individual differences exceeds the threshold, the new value is passed-on. Thus, minor variations also eventually result in transferring a new value.

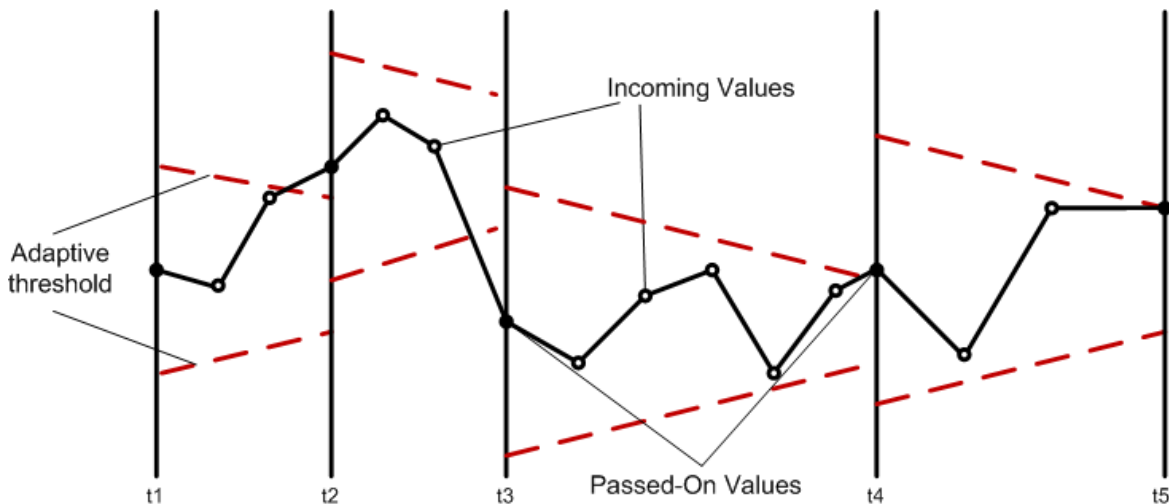


[cfets018-140509-01.tif, 1, en_US]

Figure 4-1 Threshold Check by Integration of Differences

Principle 2 – Adaptive Threshold

The threshold is determined by the highest and lowest value received and set automatically as a percentage of their difference. The threshold boundary decreases over time while being recalculated every second ensuring that small changes in value also are transmitted eventually.



[cfets019-200115-01, 1, en_US]

Figure 4-2 Threshold Check by Direct Adaptive Threshold

Master Protocols

For master protocols, principle 1 is used. A new measured value is transmitted as soon as the value of the integrated change has reached or exceeded the threshold value. To this end, the amount of the value change since the last transmission is added to an internal accumulator once a new value is received. A transmission is triggered as soon as the value of the accumulator has exceeded the threshold value. After the transmission of the measured value, the accumulator is deleted and the calculation starts again.

Forwarded Values in Master Protocols

For forwarded values in master protocols, only the direct comparison with the threshold value is available. See below explanation for **Absolute Direct** calculation method.

Slave Protocols

Absolute Direct: A new measured value is transmitted without any further delay as soon as there is a change in the amount of the measured value (absolute value) since the last transmission has reached or exceeded the threshold value.

Absolute Integrating: Refer principle 1. With the exception that for slave protocols only, the integration is done once per second with the last received value ensuring that the value is transmitted in a reasonable amount of time nonetheless.

Percentage Direct/Percentage Integrating with adjustment: If the value range for a measured value is not known, these two methods can also be used. The integration is done once per second.



NOTE

Outliers (sporadic extreme values) can block the transmission because they set the thresholds at a correspondingly high value. If the dynamics of the measured value are known, it is generally better not to use this functionality.

4.2 Mapping of SCADA Qualifiers to CFE Quality Flags - Master Protocols

The following table shows the mapping of SCADA quality bits, ranging from 0 through 24 to corresponding CFE Quality flags for Master protocols. It depends on the protocol which subset hereof is used.

Bit Number	SCADA Qualifiers	CFE Quality Flags from Device Master to Gateway															CoT						
		cfeTimeStampValid	cfeTimeStampTrue Time	cfeTimeStampInvalid	cfeValid	cfeHeld	cfeSuspect	cfeNotValid	cfeNotRenewed	cfeNotUpdated	cfeChatterFiltered ¹	cfeDouble Transmit ¹	cfeNotFrozen	cfeTelemetered	cfeCalculated ³	cfeEntered	cfeEstimated ³	cfeNormal	cfeAbnormal ²	Spontaneous	Requested/Interrogated	Local Command Return	Default
0	Telemetry blocked local																						
1	Telemetry blocked remote																						
2	Alarm blocked																						
3	Control blocked local																						
4	Control blocked remote																						
5	Not updated																						
6	Invalid local																						
7	Invalid remote																						
8	Suspect local																						
9	Suspect remote																						
10	Selection manually adjusted																						
11, 12	Source selection	Irrelevant for CFE																					
13, 14	<u>Value Source</u>																						
	00: Telemetered																						
	01: Entered																						
	10: Calculated																						
	11: Estimated																						
15, 16	<u>Value Source Telemetered:</u>																						
	00: Normal																						
	01: Spontaneous																						
	10: Gen. Interrogation																						
	01: Rem. Control Response																						
15, 16	<u>Value Source Entered:</u>																						
	00: Local																						
	01: Remote																						
15, 16	<u>Value Source Calculated:</u>	Irrelevant for CFE																					
15, 16	<u>Value Source Estimated:</u>	Irrelevant for CFE																					
17	True Time																						
18	Object in test																						
19	Clamped to zero																						
20	Initial data																						
21	Not in normal state																						
22, 23	RAP authorization source	Irrelevant for CFE																					
24	Incomplete aggregation																						

1) Quality flag is defined but not used by Master Device

2) Caused (e.g.) by IEC101/104 flag „Overflow“

3) Only used for CFE internal messages (communication statistic counters)

[cfets019-140509-01.tif, 1, en_US]

Figure 4-3 Mapping of SCADA Qualifiers to CFE Quality Flags: Device Master to Gateway

4.3 Mapping of SCADA Qualifiers to CFE Quality Flags - Slave Protocols

The following table shows the mapping of SCADA quality bits, ranging from 0 through 24 to corresponding CFE Quality flags for Slave protocols and for retransmission. It depends on the protocol which subset hereof is used.

Bit Number	SCADA Qualifiers	CFE Quality Flags from Gateway to Device Slave																	
		cfeTimeStampValid	cfeTimeStampTrueTime	cfeTimeStampInvalid	cfeValid	cfeHeld	cfeSuspect	cfeNotValid	cfeNotRenewed	cfeNotUpdated	cfeChatterFiltered ¹	cfeDoubleTransmit ¹	cfeNotFrozen	cfeTelemetered	cfeCalculated	cfeEntered	cfeEstimated	cfeNormal	cfeAbnormal
0	Telemetry blocked local																		
1	Telemetry blocked remote																		
2	Alarm blocked																		
3	Control blocked local																		
4	Control blocked remote																		
5	Not updated																		
6	Invalid local																		
7	Invalid remote																		
8	Suspect local																		
9	Suspect remote																		
10	Selection manually adjusted																		
11, 12	Source selection	Irrelevant for CFE																	
13, 14	<u>Value Source</u>																		
	00: Telemetered																		
	01: Entered																		
	10: Calculated																		
	11: Estimated																		
15, 16	<u>Value Source Telemetered:</u>																		
	00: Normal																		
	01: Spontaneous																		
	10: Gen. Interrogation																		
	01: Rem. Control Response																		
15, 16	<u>Value Source Entered:</u>	Irrelevant for CFE																	
15, 16	<u>Value Source Calculated:</u>																		
	00: Valid																		
	01: Manually entered																		
	10: Suspect																		
	01: Invalid																		
15, 16	<u>Value Source Estimated:</u>	Irrelevant for CFE																	
17	True Time																		
18	Object in test																		
19	Clamped to zero																		
20	Initial data																		
21	Not in normal state																		
22, 23	RAP authorization source	Irrelevant for CFE																	
24	Incomplete aggregation																		

1) not used any more

[cfets020-140509-01.tif, 1, en_US]

Figure 4-4 Mapping of SCADA Qualifiers to CFE Quality Flags: Gateway to Device Slave

4.4 RTU Connection Diagrams

There are different ways to connect RTUs to CFE.

The following characteristics of a RTU play a role in the decision regarding a connection to an RTU:

- Interface capability
Single ported or dual ported RTUs, that is, 2 independent communication ports at the RTU
- Communication mode
Balanced communication for point-to-point configuration or unbalanced for party line communication

Other aspects that are considered when a RTU connection diagram is designed are:

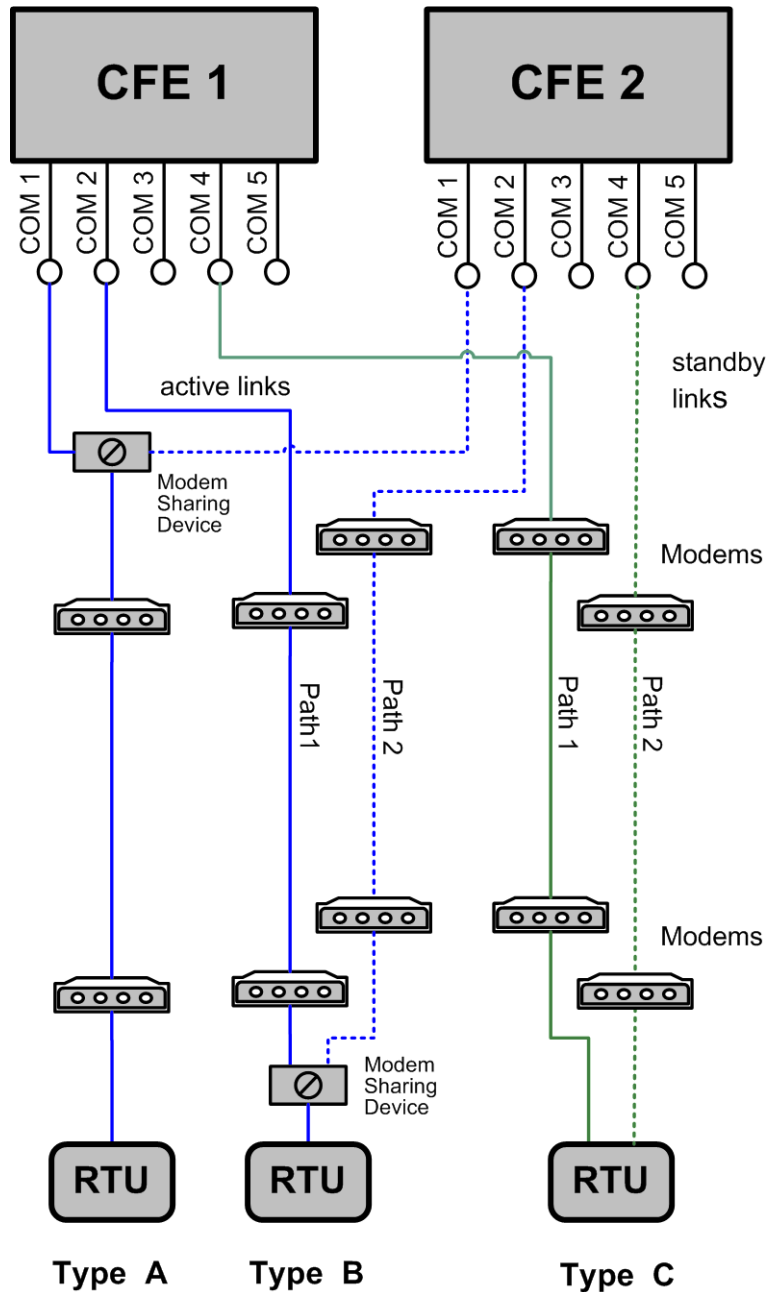
- Reliability
- Economy
- Availability of communication links
- Speed of communication links (baud rate)

The following interdependencies exist in general:

- Point-to-point connections are more reliable and effective than party-lines
- Redundant links operating with a single-port RTU offer a high degree of transmission communication reliability
- Redundant independent links operating with 2 independent RTU interfaces offer the highest degree of communication reliability
- Party-lines provide communication at low investment cost. If there is lack of communication links, party-lines are the only alternative available.

The following 2 diagrams depict the basic connection types for interfacing Remote Terminal Units to the CFE Front end System:

- Point-to-Point configurations as described in the following Interfacing of Point-to-Point Connections



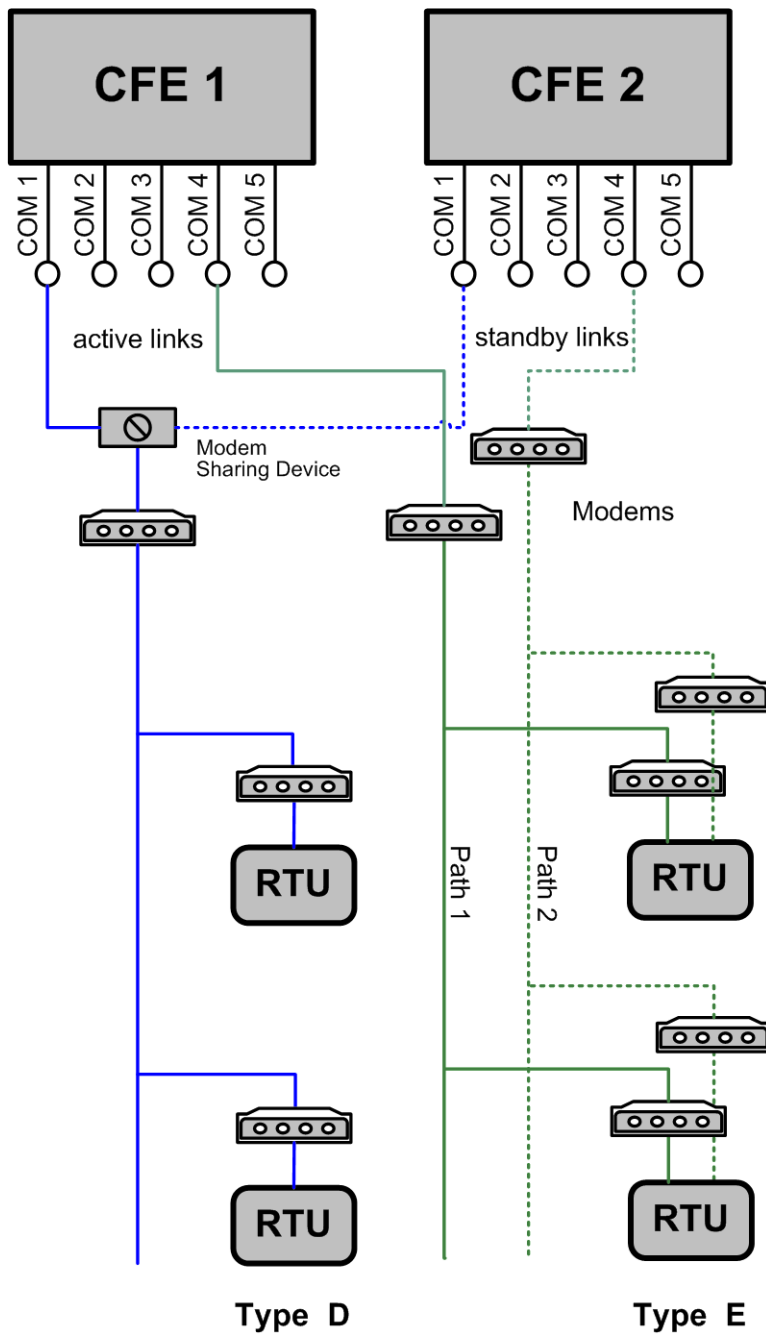
[cfets001-040509-01.tif, 1, en_US]

Figure 4-5 Interfacing of Point-to-Point Connections

As described in Interfacing of Point-to-Point Connections

- Type A: Non-redundant point-to-point communication at single-ported RTUs
- Type B: Redundant point-to-point communication at single-ported RTUs
- Type C: Redundant point-to-point communication at dual-ported RTUs

- Party-Line configurations as described in Interfacing of Multipoint Party Lines



[cfets002-040509-01.tif, 1, en_US]

Figure 4-6 Interfacing of Multipoint Party Lines

As described in Interfacing of Multipoint Party Lines

- Type D: Non-redundant party-line communication at single-ported RTUs
- Type E: Redundant party-line communication at dual-ported RTUs

4.5 Switch-Over Behavior

4.5.1 Switch-Over Behavior

The switch-over and switch-back behavior of CFE communication links depends on different parameters, such as:

- Balanced or unbalanced transmission mode
- Dependent or independent communication links
- Point-to-point or party-line configuration
- Whether standby link is monitored by CFE

The following section answers the following questions:

- What happens if status preference is set to a standby channel?
- What happens if an active channel is locked?
- What happens if settings like preference or locked are removed?
- What happens if an active communication link is disconnected from CFE interface and reconnected again?

The CFE switch-over behavior is investigated and summarized for the following 4 basic configurations:

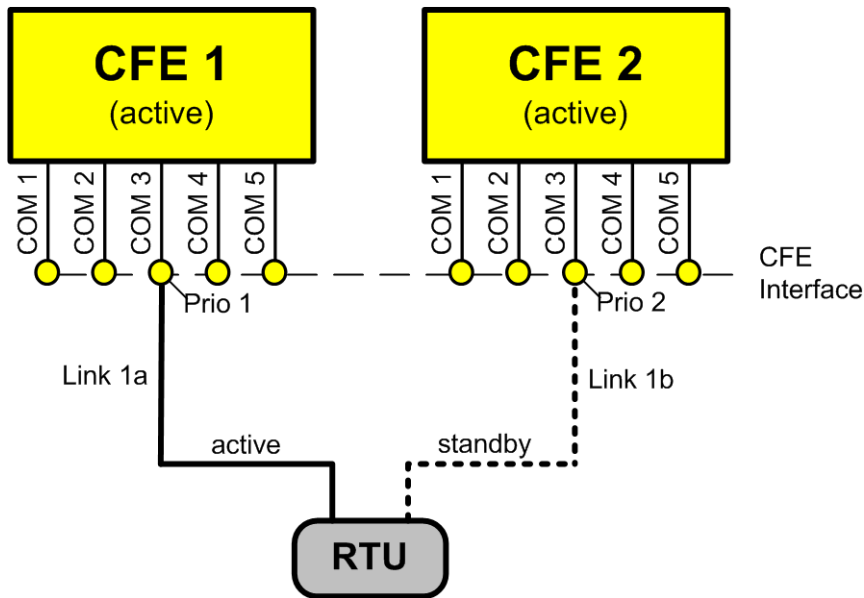
- Balanced transmission mode, independent communication links (1 RTU in end-end configuration)
- Balanced transmission mode, dependent communication links (1 RTU in end-end configuration)
- Unbalanced transmission mode, independent communication links (3 RTUs in party-line configuration)
- Unbalanced transmission mode, dependent communication links (3 RTUs in party-line configuration)

4.5.2 Configuration 1

Configuration 1 has the following properties:

Channel Configuration:	Independent
RTU arrangement:	RTU, end-end configuration
Transmission mode:	Balanced

Configuration1 is described in the following figure:



[cfets003-040509-01.tif, 1, en_US]

Figure 4-7 Independent Links / End-End Configuration

Switch-over behavior for protocol software with monitoring of standby links for example, IEC101 (balanced)



NOTE

For additional information on numbers in the columns **Start state** and **Result state**, refer to table [Table 2-2](#).

Table 4-1 Independent links, end-end, standby link monitored

Case No.	Start State	Action to Cause Switchover	Reaction of Links	Result State
1	1	Set locking at high priority link 1a	Switch over: link 1b becomes active link 1a becomes standby / no upd	3
2	3	Unlock high priority link 1a	Switch back: link 1a becomes active link 1b becomes standby	1
3	1	Set locking at high priority interface (CFE1/COM3)	Switch over: link 1b becomes active link 1a becomes standby / no upd	4
4	4	Unlock high priority interface	Switch back: link 1a becomes active link 1b becomes standby	1
5	1	Set preference at low priority link 1b	Switch over: link 1b becomes active link 1a becomes standby	2
6	2	Reset preference to high priority link 1a	Switch back: link 1a becomes active link 1b becomes standby	1

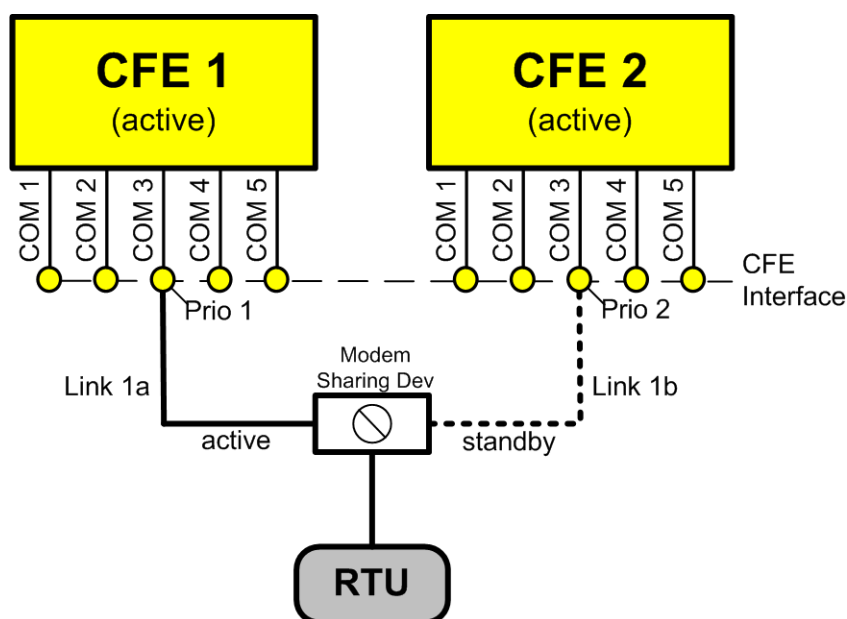
Case No.	Start State	Action to Cause Switchover	Reaction of Links	Result State
7	1	Disconnect link from high priority interface (CFE1/COM3)	Switch over: link 1b becomes active as soon as link 1a failure on link 1a was detected. link 1a becomes standby / failure	5
8	5	Reconnect link to high priority interface	Switch back: link 1a becomes active as soon as link test on link 1a is positive link 1b becomes standby	1

4.5.3 Configuration 2

Configuration 2 has the following properties:

Channel Configuration:	Dependent
RTU arrangement:	1 RTU, end-end configuration
Transmission mode:	Balanced

Configuration2 is described in the following figure:



[cfets004-040509-01.tif, 1, en_US]

Figure 4-8 Dependent Links / End-End Configuration

Switch over behavior for protocol software with monitoring of standby links for example, IEC101 (balanced)

Table 4-2 Dependent links, end-end, standby link monitored

Case no.	Start State	Action to Cause Switch-over	Reaction of Links	Result State
9	6	Set locking at high priority link 1a	Switch over: link 1b becomes active link 1a becomes standby / no upd	7

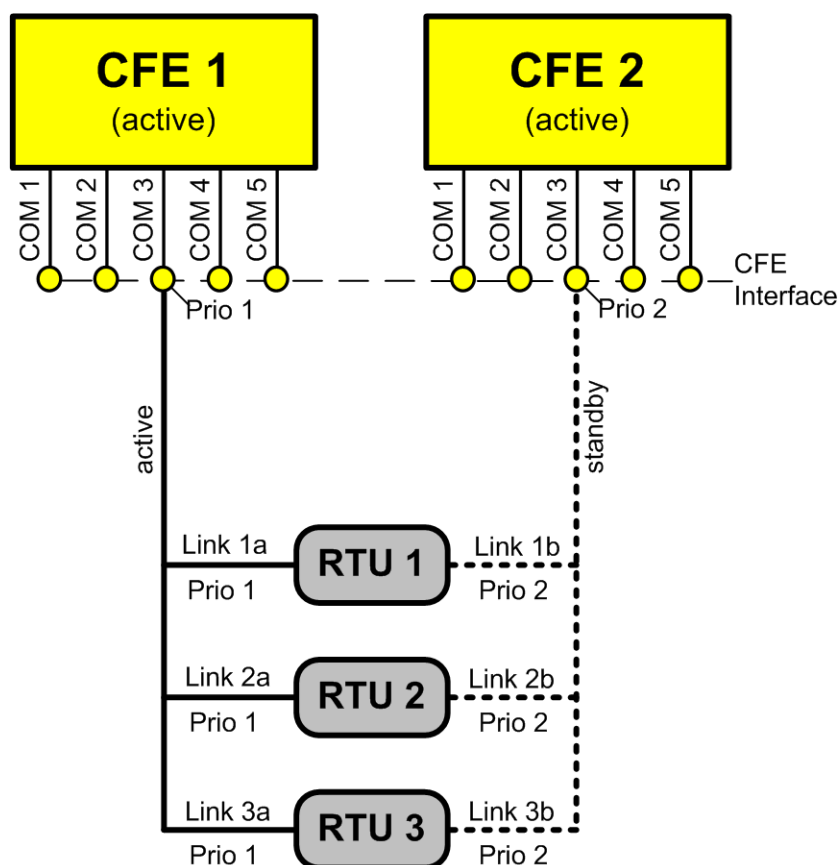
Case no.	Start State	Action to Cause Switch-over	Reaction of Links	Result State
10	7	Unlock high priority link 1a	Switch back: link 1a becomes active link 1b becomes standby	6
11	6	Set locking at high priority interface (CFE1/ COM3)	Switch over: link 1b becomes active link 1a becomes standby / no upd	8
12	8	Unlock high priority interface	Switch back: link 1a becomes active link 1b becomes standby	6
13	6	Set preference at low priority link 1b	Switch over: link 1b becomes active link 1a becomes standby	9
14	9	Reset preference to high priority link 1a	Switch back: link 1a becomes active link 1b becomes standby	6
15	6	Disconnect link 1a from high priority interface (CFE1/ COM3)	Switch over: link 1b becomes active as soon as link failure at link 1a was detected link 1a becomes standby / failure	10
16	10	Reconnect link 1a to high priority interface	Switch back: link 1a becomes active as soon as CFE1 receives messages from RTU at COM3 link 1b becomes standby	6

4.5.4 Configuration 3

Configuration 3 has the following properties:

Channel Configuration: Independent
RTU arrangement: 3 RTUs, party-line
Transmission mode: Unbalanced

Configuration3 is described in the following figure:



[cfets005-040509-01.tif, 1, en_US]

Figure 4-9 Party-Line with Independent Links

Switch over behavior for protocol software with monitoring of standby links for example, IEC101 (unbalanced)

Table 4-3 Independent links, party-line, standby links monitored

Case No.	Start State	Action to Cause Switch-over	Reaction of Links	Result State
17	1	Set locking at high priority link 1a	Switch over: Link 1b becomes active Link 1a becomes standby / no upd	11
18	11	Unlock high priority link 1a	Switch back: Link 1a becomes active Link 1b becomes standby	1
19	1	Set locking at high priority interface (CFE1/COM3)	Switch over: Links 1b...3b become active Links 1a...3a becomes standby / no upd	12
20	12	Unlock high priority interface	Switch back: Links 1a...3a become active Link 1b...3b become standby	1
21	1	Set preference at low priority link 1b	Switch over: Link 1b becomes active Link 1a becomes standby	13

Case No.	Start State	Action to Cause Switch-over	Reaction of Links	Result State
22	13	Reset preference to high priority link 1a	Switch back: Link 1a becomes active Link 1b becomes standby	1
23	1	Disconnect party-line from high priority interface (CFE1/ COM3)	Switch over: Links 1b...3b become active as soon as their link failure was detected Links 1a...3a become standby / failure	5
24	5	Reconnect party-line to high priority interface	Switch back: Link 1a...3a become active Link 1b...3b become standby	1

Switch over behavior for protocols without monitoring of standby links, for example, DNP3.0

Table 4-4 Independent links, party-line, standby links not monitored

Case No.	Start State	Action to Cause Switch-over	Reaction of Links	Result State
25	14	Set locking at high priority link 1a	Switch over: Link 1b becomes active Link 1a becomes standby / no upd	15
26	15	Unlock high priority link 1	Switch back: Link 1a becomes active Link 1b becomes standby / no upd	14
27	14	Set locking at high priority interface (CFE1/ COM3)	Switch over: Links 1b...3b become active Links 1a...3a become standby / no upd	16
28	16	Unlock high priority interface	Switch back: Links 1a...3a become active Links 1b...3b become standby / no upd	17
29	14	Set preference at low priority link 1b	Switch over: Link 1b becomes active Link 1a becomes standby / no upd	19
30	19	Reset preference to high priority link 1a	Switch back: Link 1a becomes active Link 1b becomes standby / no upd	14
31	14	Disconnect party line from high priority interface (CFE1/ COM3)	Switch over: Links 1b...3b become active (after link failure) Links 1a...3a become standby / no upd	11
32	11	Reconnect link to high priority interface	No switch back: Links 1b...3b stay active Links 1a...3a stay standby / no upd	11

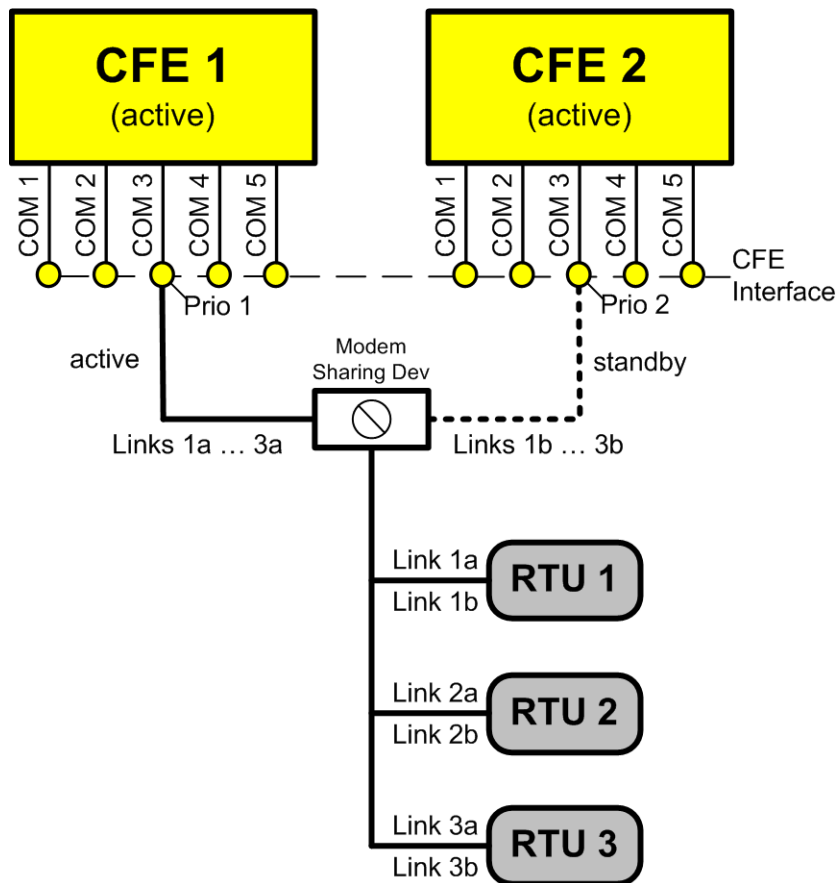
Case No.	Start State	Action to Cause Switch-over	Reaction of Links	Result State
33	11	Force switch-over: set locking at low priority interface	Switch back: Links 1a...3a become active Links 1b...3b become standby / no upd	18

4.5.5 Configuration 4

Configuration 4 has the following properties:

Channel Configuration: Dependent
RTU arrangement: 3 RTUs, party-line
Transmission mode: Unbalanced

Configuration4 is described in the following figure:



[cfets006-040509-01.tif, 1, en_US]

Figure 4-10 Party-Line with Dependent Links

Switch over behavior for protocol software with monitoring of standby links, for example, IEC101 (unbalanced)

Table 4-5 Dependent links, party-line, standby links monitored

Case No.	Start State	Action to Cause Switch-over	Reaction of Links	Result State
34	6	Set locking at all high priority links 1a...3a one by one	Switch over: Link 1b...3b become active after all links 1a...3a have been locked Link 1a...3a become standby / no upd	7
35	7	Unlock high priority link 1a...3a one by one	Switch back: As soon as the 1st link (for example, 1a) is unlocked it becomes active at CFE1/COM3 As long as links 2a & 3a are locked no communication is possible with RTU 2 and RTU 3. When all links are unlocked they become active at the highprior interface	6
36	6	Set locking at high priority interface (CFE1/COM3)	Switch over: Links 1b...3b become active Links 1a...3a becomes standby / no upd	8
37	8	Unlock high priority interface	Switch back: Links 1a...3a become active Link 1b...3b become standby	6
38	6	Set preference at low priority link 1a	Switch over: Link 1b becomes active Link 1a becomes standby	9
39	9	Reset preference to high priority link 1a	Switch back: Link 1a becomes active Link 1b becomes standby	6
40	6	Disconnect link from high priority interface (CFE1/COM3)	Switch over: Links 1b...3b become active Links 1a...3a become standby / failure	10
41	10	Reconnect link to high priority interface	Switch back: Link 1a...3a become active Link 1b...3b become standby	6

Switch over behavior for protocols without monitoring of standby links, for example, DNP3.0

Table 4-6 Dependent links, party-line, standby links not monitored

Case No.	Start State	Action to Cause Switch-over	Reaction of Links	Result State
42	20	Set locking at all high priority link sone by one	Switch over: Link 1b...3b become active after all links 1a...3a have been locked Link 1a...3a become standby / no upd	7

Case No.	Start State	Action to Cause Switch-over	Reaction of Links	Result State
43	7	Unlock high priority links 1a...3a one by one	Switch back: As soon as the 1st link (for example, 1a) is unlocked it becomes active at CFE1/COM3 As long as links 2a & 3a are locked no communication is possible with RTU 2 and RTU 3. When all links are unlocked they become active at the highprior interface	20
44	20	Set locking at high priority interface (CFE1/COM3)	Switch over: Links 1b...3b become active Links 1a...3a become standby / no upd	8
45	8	Unlock high priority interface	Switch back: Links 1a...3a become active Links 1b...3b become standby / no upd	20
46	20	Set preference at low priority interface	Switch over: Links 1b...3b becomes active Links 1a...3a becomes standby / no upd	21
47	21	Reset preference to high priority interface	Switch back: Links 1a...3a becomes active Links 1b...3b becomes standby / no upd	20
48	20	Disconnect party-line from high priority interface (CFE1/ COM3)	Switch over: Links 1b...3b become active Links 1a...3a become standby / no upd	22
49	22	Reconnect link to high priority interface	No switch back: Links 1b...3b stay active Links 1a...3a stay standby / no upd	22
50	22	Force switchover: set preference to high priority interface. Remove preference	Switch back: Links 1a...3a become active and stay active after removing preference	20

5 Engineering

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5.1 Interoperability Lists for Master Protocols

5.1.1 Protocol

Master protocols are standard protocols for CFE to communicate with RTUs. In this case, CFE is the **Controlling Station** or **Master** and the RTUs are the **Controlled Stations** or **Slaves**. If a protocol is used in the opposite direction, that is, CFE acts as a **Controlled Station**, this is called a **Slave Protocol**.

5.1.2 Interoperability Check List for Protocol IEC60870-5-101

5.1.2.1 Overview

The defined **Companion Standard** presents sets of parameters and alternatives. Subsets are selected from these parameters and alternatives to implement particular telecontrol systems. Certain parameter values, such as the number of octets in the COMMON ADDRESS of ASDUs represent mutually exclusive alternatives. This means that only one value of the defined parameters is admitted per system. Other parameters, such as the listed set of different process information in command and in monitor direction allow the specification of the complete set or subset as appropriate for given applications.

This section summarizes the parameters to facilitate a suitable selection for a specific application. If a system is composed of equipment that originate from different manufacturers, it is required that all partners agree on the selected parameters.



NOTE

In addition, the complete specification of a system requires individual selection of certain parameters for certain parts of the system. For example, individual selection of scaling factors for individually addressable measured values.

The symbols in the first column of the list have the following meanings:

☐	Implemented for Siemens Communications Frontend
n/i	Not implemented for Siemens Communications Frontend. If necessary, this feature is implemented as a project-specific add-on
☒	Selected for the defined project (empty check box are replaced with this symbol)
■	Mandatory, no alternative choice

5.1.2.2 Network Configuration

	Configuration	Remark
☐	Point-to Point	redundant communication links (yes no)
☐	Multiple Point-to Point	redundant communication links (yes no)
☐	Multipoint-Partyline	redundant communication links (yes no)
n/i	Multipoint-Star	
☐	Data Concentrator	
n/i	Multipoint-Ring	
n/i	Dial in	
n/i	Dial out	
n/i	Modem Bank	

5.1.2.3 Physical Layer

Transmission Speed (Common for Both Directions)

Unbalanced Interchange
Circuit V.24 / V.28 Standard

	Speed	Remark
☐	110 bit/s	
n/i	200 bit/s	
☐	300 bit/s	
☐	600 bit/s	
☐	1200 bit/s	
☐	2400 bit/s	
☐	4800 bit/s	
☐	9600 bit/s	
☐	19200 bit/s	
☐	38400 bit/s	
n/i	56 800 bit/s	
n/i	64 000 bit/s	

5.1.2.4 Link Layer

Link Transmission Procedure

Frame format FT 1.2, single character¹ (=E5) and the fixed time-out interval are used exclusively in this companion standard.

	Procedure	Remark
☐	Balanced Transmission	
☐	Send / No reply	
☐	Unbalanced Transmission	
n/i	max. length of the message (number of octets)	Monitoring direction: set up in the RTU Command direction: irrelevant (no blocking)
☐	Repetition of messages	default: 1
☐	ACD Bit Handling	
n/i	DFC Bit Handling in monitor direc- tion	

Address Field of the Link

	Description	Remark
☐	not present	only for balanced transmission the link address is omitted

	Description	Remark
☐	1 Octet	
☐	2 Octets	
n/i	Structured	
	Unstructured	

5.1.2.5 Application Layer

Transmission Mode for Application Data

Mode 1 (least significant octet first), as defined in clause 4.10 of IEC 870-5-4, is used exclusively in this standard.

Common Address of ASDU

	Description	Remark
☐	1 Octet	
☐	2 Octets	
n/i	Structured	
	Unstructured	

Information Object Address

	Description	Remark
☐	1 Octet	
☐	2 Octets	
☐	3 Octets	
☐	Structured	
☐	Unstructured	

Cause of Transmission

	Description	Remark
☐	1 Octet	CoT, P/N, Telegrams with Test bit T not evaluated!
☐	2 Octets	1 st octet: CoT, P/N, T Telegrams with Test bit T not evaluated 2 nd octet originator address (configurable in command direction, not evaluated in monitoring direction).

Selection of Standard ASDUs

Process Information in Monitor Direction

	Type Ident	ASDU	Description	Remark
☐	<1>	M_SP_NA_1	Single-point information	

	Type Ident	ASDU	Description	Remark
☐	<2>	M_SP_TA_1	Single-point information with time tag	
☐	<3>	M_DP_NA_1	Double-point information	
☐	<4>	M_DP_TA_1	Double-point information with time tag	
☐	<5>	M_ST_NA_1	Step position information	
☐	<6>	M_ST_TA_1	Step position information with time tag	
n/i	<7>	M_BO_NA_1	Bitstring of 32 bit	
n/i	<8>	M_BO_TA_1	Bitstring of 32 bit with time tag	
☐	<9>	M_ME_NA_1	Measured value, normalized value	recommended type
☐	<10>	M_ME_TA_1	Measured value, normalized value with time tag	
☐	<11>	M_ME_NB_1	Measured value, scaled value	
☐	<12>	M_ME_TB_1	Measured value, scaled value with time tag	
☐	<13>	M_ME_NC_1	Measured value, short floating point value	
☐	<14>	M_ME_TC_1	Measured value, short floating point value with time tag	
☐	<15>	M_IT_NA_1	Integrated totals	
☐	<16>	M_IT_TA_1	Integrated totals with time tag	
n/i	<17>	M_EP_TA_1	Event of protection equipment with time tag	
n/i	<18>	M_EP_TB_1	Packed start events of protection equipment with time tag	
n/i	<19>	M_EP_TC_1	Packed output circuit information of protection equipment with time tag	
n/i	<20>	M_PS_NA_1	Packed single-point information with status change detection	
n/i	<21>	M_ME_ND_1	Measured value, normalized value without quality descriptor	
☐	<30>	M_SP_TB_1	Single-point information with time tag CP56Time2a	
☐	<31>	M_DP_TB_1	Double-point information with time tag CP56Time2a	
☐	<32>	M_ST_TB_1	Step position information with time tag CP56Time2a	
n/i	<33>	M_BO_TB_1	Bitstring of 32 bit with time tag CP56Time2a	
☐	<34>	M_ME_TD_1	Measured value, normalized value with time tag CP56Time2a	
☐	<35>	M_ME_TE_1	Measured value, scaled value with time tag CP56Time2a	
☐	<36>	M_ME_TF_1	Measured value, short floating point value with time tag	

	Type Ident	ASDU	Description	Remark
☐	<37>	M_IT_TA_1	Integrated totals with time tag CP56Time2a	
n/i	<38>	M_EP_TD_1	Event of protection equipment with time tag CP56Time2a	
n/i	<39>	M_EP_TE_1	Event of protection equipment with time tag CP56Time2a	
n/i	<40>	M_EP_TF_1	Packed output circuit information of protection equipment with time tag CP56Time2a	

Process Information in Control Direction

	Type Ident	ASDU	Description	Remark
☐	<45>	C_SC_NA_1	Single command	
☐	<46>	C_DC_NA_1	Double command	
☐	<47>	C_RC_NA_1	Regulating step command	
☐	<48>	C_SE_NA_1	Set point command, normalized value	
☐	<49>	C_SE_NB_1	Set point command, scaled value	
☐	<50>	C_SE_NC_1	Set point command, short floating point value	
n/i	<51>	C_BO_NA_1	Bitstring of 32 bits	

Cause of Transmission

	CoT	Cause	Remark
☐	<1>	periodic, cyclic	
☐	<2>	background scan	
☐	<3>	spontaneous	
☐	<5>	requested	
☐	<6>	activation	
☐	<7>	activation confirmation	
n/i	<8>	deactivation	
n/i	<9>	deactivation confirmation	
☐	<10>	activation termination	
☐	<11>	return information caused by a remote command	
☐	<12>	return information caused by a local command	
☐	<20>	interrogated by general interrogation	
☐	<21> ... <36>	Interrogated by group interrogation	

Quality Descriptors Evaluated by the Spectrum Power 5

	QoI	Cause	Remark
☐	<NT>	not topical	applicable for all information
☐	<IV>	invalid	for measured values, integrated totals, digitals
☐	<OV>	overflow	for measured values, tap changer positions treated as invalid.
☐	<SB>	substituted	
☐	<BL	blocked	

System Information

System Information in Monitoring Direction

	Type Ident	ASDU	Description	Remark
☐	<70>	M_EI_NA_1	End of initialization COI parameters is not evaluated	After receiving TID70 from substation a General Interrogation command is sent

System Information in Control Direction

	Type Ident	ASDU	Description	Remark
☐	<100>	C_IC_NA_1	Interrogation command	
☐	<101>	C_CI_NA_1	Counter interrogation command	
n/i	<102>	C_RD_NA_1	Read command	not supported
☐	<103>	C_CS_NA_1	Clock synchronization command	Clock synchronization supported by RTU command only, not implemented as broadcast; accuracy ca. +/- 100 ms
☐	<104>	C_TS_NA_1	Test command	
n/i	<105>	C_RP_NA_1	Reset process command	not supported
n/i	<106>	C_CD_NA_1	Delay acquisition command	not supported

Parameter Transfer in Control Direction

	Type Ident	ASDU	Description	Remark
n/i	<110>	P_ME_NA_P q KPA=1 KPA=2 KPA=3/4	Parameter of measured value, normalized value - Threshold - Smoothing factor - Low / high limit	
n/i	<111>	P_ME_NB_1 KPA=1 KPA=2 KPA=3/4	Parameter of measured value, scaled value - Threshold - Smoothing factor - Low / high limit	

	Type Ident	ASDU	Description	Remark
n/i	<112>	P_ME_NC_1 KPA=1 KPA=2 KPA=3/4	Parameter of measured value, short floating point value - Threshold - Smoothing factor - Low / high limit	
n/i	<113>	P_AC_NA_1 PA=3	Parameter activation - Activation of cyclic transmission - deactivation of cyclic transmission	

File Transfer

	Type Ident	ASDU	Description	Remark
n/l	<120>	F_FR_NA_1	File ready	not supported
n/l	<121>	F_SR_AN_1	Section ready	not supported
n/l	<122>	F_SC_NA_1	Call directory, select file, call file, call selection	not supported
n/l	<123>	F_LS_NA_1	Last section	not supported
n/l	<124>	F_AF_NA_1	Ack file, ack section	not supported
n/l	<125>	F_SG_NA_1	Segment	not supported
n/l	<126>	F_DR_TA_1	Directory	not supported

Type Identifier and Cause of Transmission Assignments

The following chart is an extract from IEC60870-5 Companion Standard 101. It shows for each message Type ID the supported Cause of Transmission assignments (white boxes). Shaded boxes show combinations that are not supported.

[cfets011-050509-01.tif, 1, en, US]

Figure 5-1 Type Identifier and Cause of Transmission Assignment

5.1.2.6 Basic Application Functions

General Interrogation

	Type Ident	ASDU	Description	Remark
☐	<100>	C_IC_NA_1 (QOI=20)	global G I	the following COT parameters are applied in control direction: <6> activation (C_IC ACT) in monitoring direction: <7> activation confirmation (C_IC ACTCON) <10> activation termination (C_IC ACTTERM)

Group Interrogation

	Type Ident	ASDU	Description	Remark
☐	<100>	C_IC_NA_1 (QOI=21)	group 1	the following COT parameters are applied in control direction: <6> activation (C_IC ACT) in monitoring direction: <7> activation confirmation (C_IC ACTCON) <10> activation termination (C_IC ACTTERM)
☐	<100>	C_IC_NA_1 (QOI=22) ... C_IC_NA_1 (QOI=36)	group 2 ... group 36	the following COT parameters are applied in control direction: <6> activation (C_IC ACT) in monitoring direction: <7> activation confirmation (C_IC ACTCON) <10> activation termination (C_IC ACTTERM)

Command Transmission

	Description	Remark
☐	Direct command transmission	
☐	Direct set-point command transmission	
n/i	Command termination	Not supported
☐	Select and execute command	EXECUTE command is generated automatically after reception of the positive SELECT confirmation
☐	Select and execute set-point command	
☐	C_SE ACTTERM used	

Qualifier for Command Transmission

	QOC	Description	Remark
☐	<0>	no additional definition	individual command output pulses are to be parameterized in the RTU

	QOC	Description	Remark
☐	<1>	Short pulse duration (duration determined by a system parameter in the outstation)	
☐	<2>	Long pulse duration (duration determined by a system parameter in the outstation)	
☐	<3>	Persistent output	
n/i		(others to be defined)	

Counter Interrogation Commands

	Type Ident	ASDU	Description	Remark
☐	<101>	C_CI_NA_1 ...(FRZ=0)	no freeze or reset	
☐		...(FRZ=1)	counter freeze without reset	
☐		...(FRZ=2)	counter freeze with reset	
☐		...(FRZ=3)	counter reset	
n/i		...(RQT=0)	no counter request	
☐		...(RQT=1-4)	request counter group 1...4	
☐		...(RQT=5)	general request counter	



NOTE

Counter acquisition commands (freeze, reset) are preferably to be carried out by the RTU. The transmission of the counter values should be initialized by the RTU spontaneously after each freezing.

5.1.2.7 Special Communication Features

Communication Statistics

Info ID	Counter Designation	Remark
901	Messages Sent Counter	Link & Interface
902	Messages Received Counter	Link & Interface
903	Messages Receive Error Counter	Interface only
904	Message Timeout Counter	Link & Interface

Listening Mode

	Description	Remark
☐	Listening Mode	User parameters

Monitoring of Standby Channels and Information Buffering

	Description	Remark
☒	Monitoring	
☐	Buffering	Buffer configurations with user parameters

5.1.3 Interoperability Check List for Protocol IEC60870-5-104

5.1.3.1 Network Configurations

	Configuration	Remark
☐	Point-to Point	Redundant communication links (yes no)
n/i	Multiple Point-to Point	
n/i	Multipoint-Partyline	Redundant communication links (yes no)
n/i	Multipoint-Star	
☐	Data Concentrator	

5.1.3.2 Device Function

Indicates the stations function

	Description	Remark
☒	Controlling station (Master)	
☐	Controlled station (Slave)	

5.1.3.3 Protocol Structure

The following table shows the structure of the end system

Selection of Application Functions of IEC 60870-5-5 according to IEC 60870-5-101	Initialization	User Process
Selection of Application Service Data Units of IEC 60870-5-101 and IEC 60870-5-104		Application (layer 7)
APCI (Application Protocol Control Information) Transport Interface (User to TCP Interface)		
Selection of TCP/IP Protocol Suite (RFC 2200)		Transport (layer 4)
		Network (layer 3)
		Link (layer 2)
		Physical (layer 1)

5.1.3.4 Application Layer

Transmission Mode for Application Data

Mode 1 (least significant octet first), as defined in clause 4.10 of IEC 870-5-4, is used exclusively in this standard.

Common Address of ASDU

	Description	Remark
☐	2 Octets	

Information Object Address

	Description	Remark
☒	3 Octets	
☐	structured	
☐	unstructured	

Cause of Transmission

	Description	Remark
☒	2 Octets	1 st octet: CoT, P/N, T Telegrams with Test bit T not evaluated 2 nd octet originator address (always zero in command direction, not evaluated in monitoring direction).

Length of APDU

The maximum length of the ADDU is 253 (default). The maximum length may be reduced per system (to be setup in the RTU).

Selection of Standard ASDUs

Process Information in Monitor Direction

	Type Ident	ASDU	Description	Remark
☐	<1>	M_SP_NA_1	Single-point information	
☐	<3>	M_DP_NA_1	Double-point information	
☐	<5>	M_ST_NA_1	Step position information	positive values only!
n/i	<7>	M_BO_NA_1	Bitstring of 32 bit	
☐	<9>	M_ME_NA_1	Measured value, normalized value	recommended type
☐	<11>	M_ME_NB_1	Measured value, scaled value	
☐	<13>	M_ME_NC_1	Measured value, short floating point value	
☐	<15>	M_IT_NA_1	Integrated totals	
n/i	<20>	M_PS_NA_1	Packed single-point information with status change detection	
n/i	<21>	M_ME_ND_1	Measured value, normalized value without quality descriptor	
☐	<30>	M_SP_TB_1	Single-point information with time tag CP56Time2a	
☐	<31>	M_DP_TB_1	Double-point information with time tag CP56Time2a	
☐	<32>	M_ST_TB_1	Step position information with time tag CP56Time2a	
n/i	<33>	M_BO_TB_1	Bitstring of 32 bit with time tag CP56Time2a	

	Type Ident	ASDU	Description	Remark
☐	<34>	M_ME_TD_1	Measured value, normalized value with time tag CP56Time2a	
☐	<35>	M_ME_TE_1	Measured value, scaled value with time tag CP56Time2a	
☐	<36>	M_ME_TF_1	Measured value, short floating point value with time tag	
☐	<37>	M_IT_TA_1	Integrated totals with time tag CP56Time2a	
n/i	<38>	M_EP_TD_1	Event of protection equipment with time tag CP56Time2a	
n/i	<39>	M_EP_TE_1	Packed start events of protection equipment with time tag CP56Time2a	
n/i	<40>	M_EP_TF_1	Packed output circuit information of protection equipment with time tag CP56Time2a	

Process Information in Control Direction

	Type Ident	ASDU	Description	Remark
☐	<45>	C_SC_NA_1	Single command	
☐	<46>	C_DC_NA_1	Double command	
☐	<47>	C_RC_NA_1	Regulating step command	
☐	<48>	C_SE_NA_1	Set point command, normalized value	
☐	<49>	C_SE_NB_1	Set point command, scaled value	
☐	<50>	C_SE_NC_1	Set point command, short floating point value	
n/i	<51>	C_BO_NA_1	Bitstring of 32 bits	
n/i	<58>	C_SC_TA_1	Single command with time tag CP56Time2a	
n/i	<59>	C_DC_TA_1	Double command with time tag CP56Time2a	
n/i	<60>	C_RC_TA_1	Regulating step command with time tag CP56Time2a	
n/i	<61>	C_SE_TA_1	Set point command, normalized value with time tag CP56Time2a	
n/i	<62>	C_SE_TB_1	Set point command, scaled value with time tag CP56Time2a	
n/i	<63>	C_SE_TC_1	Set point command, short floating point value with time tag CP56Time2a	

	Type Ident	ASDU	Description	Remark
n/i	<64>.	C_BO_TC_1	Bitstring of 32 bit with time tag CP56Time2a	

Monitor Process Information in Control Direction

	Type Ident	ASDU	Description	Remark
☐	<1>	M_SP_NA_1	Single-point information	
☐	<11>	M_ME_NB_1	Measured value, scaled value	
☐	<13>	M_ME_NC_1	Measured value, short floating point value	
☐	<15>	M_IT_NA_1	Integrated totals	

System Information

System Information in Monitoring Direction

	Type Ident	ASDU	Description	Remark
☐	<70>	M_EI_NA_1	End of initialization COI parameters is not evaluated	After receiving TID70 from substation a General Interrogation command is sent

System Information in Control Direction

	Type Ident	ASDU	Description	Remark
☒	<100>	C_IC_NA_1	Interrogation command	
☐	<101>	C_CI_NA_1	Counter interrogation command	
n/i	<102>.	C_RD_NA_1	Read command	
☐	<103>	C_CS_NA_1	Clock synchronization command	
n/i	<105>	C_RP_NA_1	Reset process command	

Parameter Information in Control Direction

	Type Ident	ASDU	Description	Remark
n/i	<110>	P_ME_NA_1	Parameter of measured value, normalized value	
n/i	<112>	P_ME_NB_1	Parameter of measured value, scaled value	
n/i	<113>	P_ME_NC_1	Parameter of measured value, short floating point value	
n/i	<114>	P_AC_NA_1	Parameter activation	

File Transfer

	Type Ident	ASDU	Description	Remark
n/i	<120>	F_FR_NA_1	File ready	
n/i	<121>	F_SR_AN_1	Section ready	
n/i	<122>	F_SC_NA_1	Call directory, select file, call file, call selection	
n/i	<123>	F_LS_NA_1	Last section	
n/i	<124>	F_AF_NA_1	Ack file, ack section	
n/i	<125>	F_SG_NA_1	Segment	
n/i	<126>	F_DR_TA_1	Directory	

Selection of Non-Standard private ASDUs

Process Information in Monitor Direction

	Type Ident	ASDU	Description	Remark
☐	<135>	undefined	SICAM1703 error telegram	Only component summary diagnostics set is evaluated regarding the following eight error bits: Internal Error External Error Communication Error Test Warning ComponentOutage AEOutage Runup

Type Identifier and Cause of Transmission Assignments

The following chart is an extract from IEC60870-5 Companion Standard 104. It shows for each message Type ID the supported Cause of Transmission assignments (white boxes). Shaded boxes show combinations that are not supported.

[illegible]

[cfets012-050509-01.tif, 1, en US]

Figure 5-2 Type Identifier and Cause of Transmission Assignment

5.1.3.5 Basic Application Functions

Overview

	Description	Remark
n/i	Remote station initialization	
☐	Cyclic data transmission	
n/i	Read procedure	
☐	Spontaneous transmission	
☐	Select and execute set-point command	
☐	C_SE ACTTERM used	
☐	Clock synchronization	

Double Transmission of Information Objects

The following type identifications are transmitted in succession caused by a single status change of an information object with CoT **Spontaneous**. The particular information object addresses for which double transmission is enabled are defined in a project-specific list.

	Description	Remark
n/i	Single-point information	M_SP_NA_1, M_SP_TA_1, M_SP_TB_1 and M_PS_NA_1
n/i	Double-point information	M_DP_NA_1, M_DP_TA_1 and M_DP_TB_1
n/i	Step position information	M_ST_NA_1, M_ST_TA_1 and M_ST_TB_1

	Description	Remark
n/i	Bitstring of 32 bit3	M_BO_NA_1, M_BO_TA_1 and M_BO_TB_1 (if defined for a specific project)
n/i	Measured value, normalized value	M_ME_NA_1, M_ME_TA_1, M_ME_ND_1 and M_ME_TD_1
n/i	Measured value, scaled value	M_ME_NB_1, M_ME_TB_1 and M_ME_TE_1
n/i	Measured value, short floating point number	M_ME_NC_1, M_ME_TC_1 and M_ME_TF_1

General Interrogation

	Type Ident	ASDU	Description	Remark
☒	<100>	C_IC_NA_1 (QOI=20)	global G I	the following COT parameters are applied in control direction: <6>activation (C_IC ACT) in monitoring direction: <7> activation confirmation (C_IC ACTCON) <10> activation termination (C_IC ACTTERM)

Group Interrogation

	Type Ident	ASDU	Description	Remark
☐	<100>	C_IC_NA_1 (QOI=21)	group 1	the following COT parameters are applied in control direction: <6> activation (C_IC ACT) in monitoring direction: <7> activation confirmation (C_IC ACTCON) <10> activation termination (C_IC ACTTERM)
☐	<100>	C_IC_NA_1 (QOI=22) ... etc. ... C_IC_NA_1 (QOI=36)	group 2 ... group 36	the following COT parameters are applied in control direction: <6> activation (C_IC ACT) in monitoring direction: <7> activation confirmation (C_IC ACTCON) <10> activation termination (C_IC ACTTERM)

Command Transmission

	Description	Remark
☐	Direct command transmission	
☐	Direct set-point command transmission	
☐	Select and execute command	
☐	Select and execute set point command	
n/i	C_SE ACTTERM used	
n/i	Supervision of maximum delay in command direction	commands and set point commands

Qualifier for Command Transmission

	QOC	Description	Remark
☐	<0>	no additional definition	individual command output pulses are to be parameterized in the RTU
☐	<1>	Short pulse duration (duration determined by a system parameter in the outstation)	
☐	<2>	Long pulse duration (duration determined by a system parameter in the outstation)	
☐	<3>	Persistent output	

Transmission of Integrated Totals

	Description	Remark
☐	Mode A	Local freeze with spontaneous transmission
☐	Mode B	Local freeze with counter interrogation
☐	Mode C	Freeze and transmit by counter interrogation commands
☐	Mode D	Freeze by counter interrogation command, frozen values reported

	Description	Remark
☐	Counter read	
☐	Counter freeze without reset	
☐	Counter freeze with reset	
☐	Counter reset	

	Description	Remark
☐	General request counter	
☐	Request counter group 1	
☐	Request counter group 2	
☐	Request counter group 3	
☐	Request counter group 4	

Parameter Loading

	Description	Remark
n/i	Threshold value	
n/i	Smoothing factor	
n/i	Low limit for transmission of measured values	
n/i	High limit for transmission of measured values	

	Description	Remark
n/i	Parameter activation	Act/deact of persistent cyclic or periodic transmission of the addressed object

Parameter Activation

	Description	Remark
n/i	Parameter activation	Act/deact of persistent cyclic or periodic transmission of the addressed object

Test Procedure

	Description	Remark
n/i	Test procedure	

File Transfer**File Transfer in Monitor Direction**

	Description	Remark
n/i	Transparent file	
n/i	Transmission of disturbance data of protection	
n/i	Transmission of sequences of events	
n/i	Transmission of sequences of recorded analogue values	

File Transfer in Control Direction

	Description	Remark
n/i	Transparent file	

Background Scan

	Description	Remark
n/i	Background scan	

Definitions of Time-Outs

Parameter	Default Value	Meaning	User parameter of Cfe Interface Link
t_0	30s	Time out of connection establishment	30s
t_1	15s	Time out of send or test ADPUs	15s
t_2	10s	Time out for acknowledges in case of no data messages $t_2 < t_1$	10s
t_3	20s	Time out for sending test frames in case of a long idle state	20s

Maximum range of values for all timeouts: 1 to 255 s, accuracy 1 s

Maximum number of Outstanding I Format ADPUs k and Latest Acknowledge

Parameter	Default Value	Meaning	User parameter of Cfe Interface Link
k	12 ADPUs	Maximum difference receive sequence number to send state variable	12 ADPUs
w	8 ADPUs	Latest acknowledge after receiving w I-format ADPUs	8 ADPUs

- Maximum range of values for k: 1 to 32767 ADPUs
Accuracy: 1 APDU
- Maximum range of values for w: 1 to 32767 ADPUs
Accuracy: 1 APDU (Recommendation: w should not exceed 2/3 of k).

Port Number

Parameter	Value	Remarks
Portnumber	2404	Any port number can be configured. 2404 is default value and specified in IEC

RFC 2200 Suite

RFC 2200 is an official Internet Standard. It describes the state of standardization of protocols used in internet as determined by the Internet Architecture Board (IAB). It offers a broad spectrum of actual standards used in the internet. The suitable selection of documents from RFC 2200 defined in this standard for given projects is chosen by the user of this standard.

- ☐ Ethernet 802.3
- ☐ Serial X.21
- ☐

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

[cfets013-060509-01.tif, 1, en_US]

Figure 5-3 List of valid documents from RFC 2200

5.1.3.6 Special Communication Features

Communication Statistics

Not implemented.

Listening Mode

Not implemented.

Monitoring of Standby Channels / Standby Channel Information Buffering

	Description	Remark
☒	Monitoring	
n/i	Buffering	Buffer configurations with user parameters

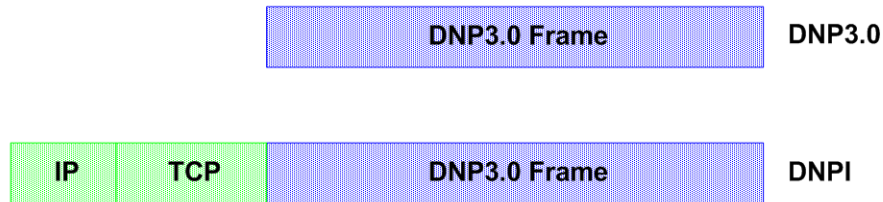
5.1.4 Interoperability Check List for Protocol DNP3.0 / DNP3i

5.1.4.1 General

Release 4.0 on the DNP3.0 protocol implementation offers the following 2 different ways of communication:

- Serial links through COM ports according to RS232 standard either via leased lines or dial-in / dial-up connections
- LAN / WAN communication over TCP/IP transport protocol. This version of DNP3.0 is called in this context **DNPI**

The following figure shows the LAN/WAN communication over the TCP/IP protocol



[tscfe004-290411-01.tif, 1, en_US]

Figure 5-4 DNP3.0: In TCP/IP Embedded DNP3.0 Frame

Since all DNP3.0 Layers - except Physical Layer are also applicable for the DNPI version, both versions are commonly specified in this Interoperability document.

If there is any difference or restriction in functionality, it is indicated in the corresponding table by a remark.

The following figure describes the DNPI Presentation in ISO/OSI Model

DNP3.0 Application		Application (Layer 7)
DNP3.0 Pseudo Transport		
DNP3.0 Link		
		Presentation (Layer 6)
		Session (Layer 5)
TCP		Transport (Layer 4)
IP		Network (Layer 3)
PPP	LLC/MAC	Link (Layer 2)
WAN	LAN	Physical (Layer 1)

[tscfe005-290411-01.tif, 1, en_US]

Figure 5-5 DNPI, Presentation in ISO/OSI Model

Restrictions:

- Only the master function is supported
- RTU configuration is an offline process
- CFE does not write configuration data to RTUs at system start-up time

5.1.4.2 Network Configurations

Network Configuration for Serial Links

	Configuration	Remark
☐	Point-to Point	Redundant communication links (yes no)
☐	Multiple Point-to Point	Redundant communication links (yes no)
☐	Multipoint-Partyline	Redundant communication links (yes no)
n/i	Multipoint-Star	
n/i	Data Concentrator	
n/i	Multipoint-Ring	
☐	Dial out	
☐	Dial in	Dial-out to RTUs is used to perform scans Dial-in from RTU is used to support unsolicited messages

Network Configuration for TCP/IP Links

	Configuration	Remark
☐	Point-to Point	Redundant communication links (yes no)
☐	Multiple Point-to Point	Redundant communication links (yes no)
n/i	Data Concentrator	

5.1.4.3 Physical Layer for Serial Links

Transmission Speed (common for both directions)

Unbalanced Interchange

Circuit V.24 / V.28 Standard

	Speed	Remark
n/i	100 bit/s	
n/i	200 bit/s	
☐	300 bit/s	
☐	600 bit/s	
☐	1200 bit/s	
☐	2400 bit/s	
☐	4800 bit/s	
☐	9600 bit/s	
☐	19200 bit/s	
n/i	38400 bit/s	
n/i	56000 bit/s	
n/i	64000 bit/s	

5.1.4.4 Data Link Layer (Serial & TCP/IP)

The data link layer uses a standard variable length frame format as defined in the IEC 870-5-1 Transmission Frame Formats document. The FT3 frame format class is suited for data transmission between stations that require medium information transfer rates and low residual error probability.

	Item	Remark
☐	Maximum Data Link Frame Size (octets): Transmitted: 292 Received:(must be 292)	fixed
☐	Data Link Layer Confirmation: configurable	if enabled link confirmation requested
☐	Maximum Data Link Retries: configurable	only valid if link layer confirmation enabled; valid range: 0 to 255
☐	Data Link Layer Timeout: configurable	only valid if link layer confirmation enabled; valid range: 0 to 10000 msecs

5.1.4.5 Transport Layer (Serial & TCP/IP)

Overview

The transport layer functions act as a pseudo-transport layer to the DNP data link layer. The pseudo transport layer function is specific for those messages that are larger than one Link Protocol Data Unit (LPDU) between primary and secondary stations. This pseudo-transport layer operates as the DNP data link user in a protocol stack consisting of only the DNP Data Link and DNP Application Layer.

Fragmentation of Transport Layer Frames

	Item	
☐	Fragmented transport frames in monitor direction	
n/i	Fragmented transport frames in control direction	not required
☐	Maximum transport fragment size: 250 octets	fixed



NOTE

The features and functions specified hereafter are valid for both protocol variants, serial and TCP/IP based communication

5.1.4.6 Application Layer

Fragmentation of Application Frames

	Procedure	Remark
☐	Fragmented application frames in monitor direction Maximum frame size: 2048 octets	Fragmented application frames will be supported in monitor direction.
☐	Fragmented application frames in control direction Maximum frame size: 2048 octets	

Application Layer Message Flow

	Item	Remark
☐	Application Layer Confirm configurable	if enabled application layer messages need confirmation (not recommended)

	Item	Remark
☐	Application Layer Retries configurable	only valid if application layer confirmation is enabled; valid range 0 to 255
☐	Application Layer Timeout configurable	only valid if application layer confirmation is enabled; valid range 1 to 10000 msec
☐	Unsolicited Response	allowed; not configurable

Application Layer Function Codes

	Code	Function	Remark
Transfer Function Codes			
☐	0	Confirm	
☐	1	Read	
☐	2	Write	
Control Function Codes			
☐	3	Select	
☐	4	Operate	
☐	5	Direct Operate	
☐	6	Direct Operate - No Acknowledgment	
Freeze Function Codes			
☐	7	Immediate Freeze	
☐	8	Immediate Freeze - No Acknowledgment	
Transfer Function Codes			
☐	9	Freeze and Clear	
☐	10	Freeze and Clear - No Acknowledgment	
n/i	11	Freeze and Time	
n/i	12	Freeze and Time - No Acknowledgment	
Application Function Codes			
☐	13	Cold Restart	
☐	14	Warm Restart	
☐	15	Initialize Data to Default	
n/i	16	Initialize Application	
n/i	17	Start Application	
n/i	18	Stop Application	
Configuration Function Codes			
n/i	19	Save Configuration	
☐	20	Enable Unsolicited Messages	
☐	21	Disable Unsolicited Messages	

	Code	Function	Remark
☐	22	Assign Class	
Time Synchronization Function Codes			
☐	23	Delay Measurement	
Reserve			
n/i	24 - 120	Reserved for future use	
n/i	121 - 128	Reserved for testing only	
Response Function Codes			
☐	0	Confirm	
☐	129	Response	
☐	130	Unsolicited Messages	

5.1.4.7 Sending Control Operations

The control code indicates the control function to perform. The applicability of this code depends on the type of hardware used in the end device.

	Item	Remark
n/i	Count > 1	On standard digital control command Count is always = 1
☐	Variable count	On analog setpoints that are converted to digital commands with variable count: Count is set according to the analog value. The analog value is converted through a characteristic curve into a suitable count value
☐	Command with configurable ON / OFF time	Standard digital control command
☐	Command with variable ON / OFF time	On analog setpoints converted to digital commands with variable ON/ OFF time. ON/OFF time is set according to the analog value. The analog value is converted through a characteristic curve into a suitable ON/OFF time value
☐	Pulse On - operation configurable	
☐	Pulse Off - operation configurable	
☐	Latch On - operation configurable	
☐	Latch Off - operation configurable	
n/i	Queue	
n/i	Clear Queue	

5.1.4.8 Data Objects and Functions

	Object (Group, Variation)	Request Function (decimal)	Qualifier (hex)	Response Function	Qualifier
☐	Single-Bit Binary Input (1, 0)	1, 22	00, 01, 06		

	Object (Group, Variation)	Request Function (decimal)	Qualifier (hex)	Response Function	Qualifier
☐	Single-Bit Binary Input (1, 1)	1	00, 01, 06	129, 130	00, 01
☐	Binary Input with Status (1, 2)	1	00, 01, 06	129, 130	00, 01
☐	Binary Input with Status (2, 0)	1	06, 07, 08		
☐	Binary Input Change Without Time (2, 1)	1	06, 07, 08	129, 130	17, 28
☐	Binary Input Change With Time (2, 2)	1	06, 07, 08	129, 130	17, 28
☐	Binary Input Change With Relative Time (2, 3)	1	06, 07, 08	129, 130	17, 28
☐	Binary Output (10, 0)	1	00, 01, 06		
n/i	Binary Output (10, 1)				
☐	Binary Output Status (10, 2)	1	00, 01, 06	129, 130	00, 01
n/i	Control Block (12, 0)				
☐	Control Relay Output Block (12, 1)	3, 4, 5, 6	17, 28	129	Echoes Request
n/i	Pattern Control Block (12, 2)				
n/i	Pattern Mask (12, 3)				
☐	Binary Counter (20, 0)	1, 7, 8, 9, 10, 22	00, 01, 06		
☐	32-Bit Binary Counter (20, 1)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Binary Counter (20, 2)	1	00, 01, 06	129, 130	00, 01
☐	32-Bit Delta Counter (20, 3)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Delta Counter (20, 4)	1	00, 01, 06	129, 130	00, 01
☐	32-Bit Binary Counter Without Flag (20, 5)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Binary Counter Without Flag (20, 6)	1	00, 01, 06	129, 130	00, 01
☐	32-Bit Delta Counter Without Flag (20, 7)	1	00, 01, 06	129, 130	00, 01

	Object (Group, Variation)	Request Function (decimal)	Qualifier (hex)	Response Function	Qualifier
☐	16-Bit Delta Counter Without Flag (20, 8)	1	00, 01, 06	129, 130	00, 01
☐	Frozen Counter (21, 0)	1, 22	00, 01, 06	129, 130	00, 01
☐	32-Bit Frozen Counter (21, 1)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Frozen Counter (21, 2)	1	00, 01, 06	129, 130	00, 01
☐	32-Bit Frozen Delta Counter (21, 3)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Frozen Delta Counter (21, 4)	1	00, 01, 06	129, 130	00, 01
☐	32-Bit Frozen Counter with Time of Freeze (21, 5)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Frozen Counter with Time of Freeze (21, 6)	1	00, 01, 06	129, 130	00, 01
☐	32-Bit Frozen Delta Counter with Time of Freeze (21, 7)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Frozen Delta Counter with Time of Freeze (21, 8)	1	00, 01, 06	129, 130	00, 01
☐	32-Bit Frozen Counter Without Flag (21, 9)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Frozen Counter Without Flag (21, 10)	1	00, 01, 06	129, 130	00, 01
☐	32-Bit Frozen Delta Counter Without Flag (21, 11)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Frozen Delta Counter Without Flag (21, 12)	1	00, 01, 06	129, 130	00, 01
☐	Counter Change Events (22, 0)	1	06, 07, 08		
☐	32-Bit Counter Change Event Without Time (22, 1)	1	06, 07, 08	129, 130	17, 28

	Object (Group, Variation)	Request Function (decimal)	Qualifier (hex)	Response Function	Qualifier
☐	16-Bit Counter Change Event Without Time (22, 2)	1	06, 07, 08	129, 130	17, 28
☐	32-Bit Delta Counter Change Event Without Time (22, 3)	1	06, 07, 08	129, 130	17, 28
☐	16-Bit Delta Counter Change Event Without Time (22, 4)	1	06, 07, 08	129, 130	17, 28
☐	32-Bit Counter Change Event With Time (22, 5)	1	06, 07, 08	129, 130	17, 28
☐	16-Bit Counter Change Event With Time (22, 6)	1	06, 07, 08	129, 130	17, 28
☐	32-Bit Delta Counter Change Event With Time (22, 7)	1	06, 07, 08	129, 130	17, 28
☐	16-Bit Delta Counter Change Event With Time (22, 8)	1	06, 07, 08	129, 130	17, 28
☐	Frozen Counter Events (23, 0)	1	06, 07, 08	129, 130	
☐	32-Bit Frozen Counter Event Without Time (23, 1)	1	06, 07, 08	129, 130	17, 28
☐	16-Bit Frozen Counter Event Without Time (23, 2)	1	06, 07, 08	129, 130	17, 28
☐	32-Bit Frozen Delta Counter Event Without Time (23, 3)	1	06, 07, 08	129, 130	17, 28
☐	16-Bit Frozen Delta Counter Event Without Time (23, 4)	1	06, 07, 08	129, 130	17, 28
☐	32-Bit Frozen Counter Event With Time (23, 5)	1	06, 07, 08	129, 130	17, 28
☐	16-Bit Frozen Counter Change With Time (23, 6)	1	06, 07, 08	129, 130	17, 28

	Object (Group, Variation)	Request Function (decimal)	Qualifier (hex)	Response Function	Qualifier
☐	32-Bit Frozen Delta Counter Event With Time (23, 7)	1	06, 07, 08	129, 130	17, 28
☐	16-Bit Frozen Delta Counter Event With Time (23, 8)	1	06, 07, 08	129, 130	17, 28
☐	Analog Input (30, 0)	1	00, 01, 06		
☐	32-Bit Analog Input (30, 1)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Analog Input (30, 2)	1	00, 01, 06	129, 130	00, 01
☐	32-Bit Analog Input Without Flag (30, 3)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Analog Input Without Flag (30, 4)	1	00, 01, 06	129, 130	00, 01
☐	Short Floating Analog Input (30, 5)	1	00, 01, 06	129, 130	00, 01
☐	Long Floating Analog Input (30, 6)1)	1	00, 01, 06	129, 130	00, 01
☐	Frozen Analog Input (31, 0)	1, 22	00, 01, 06		
☐	32-Bit Frozen Analog Input (31, 1)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Frozen Analog Input (31, 2)	1	00, 01, 06	129, 130	00, 01
☐	Analog Change Event (32, 0)	1	06, 07, 08		
☐	32-bit Analog Change Event Without Time (32, 1)	1	06, 07, 08	129, 130	17, 28
☐	16-bit Analog Change Event Without Time (32, 2)	1	06, 07, 08	129, 130	17, 28
☐	32-bit Analog Change Event With Time (32, 3)	1	06, 07, 08	129, 130	17, 28
☐	16-bit Analog Change Event With Time (32, 4)	1	06, 07, 08	129, 130	17, 28

	Object (Group, Variation)	Request Function (decimal)	Qualifier (hex)	Response Function	Qualifier
☐	Short Floating Analog Change Event (32, 5)1)	1	06, 07, 08	129, 130	17, 28
☐	Long Floating Analog Change Event (32, 6)	1	06, 07, 08	129, 130	17, 28
☐	Short Floating Analog Change Event with Time (32, 7)	1	06, 07, 08	129, 130	17, 28
☐	Long Floating Analog Change Event with Time(32, 8)	1	06, 07, 08	129, 130	17, 28
☐	Frozen Analog Event (33, 0)	1	06, 07, 08		
☐	32-Bit Frozen Analog Event w/o Time (33, 1)	1	06, 07, 08	129, 130	17, 28
☐	16-Bit Frozen Analog Event w/o Time(33, 2)	1	06, 07, 08	129, 130	17, 28
☐	32-Bit Frozen Analog Event with Time (33, 3)	1	06, 07, 08	129, 130	17, 28
☐	16-Bit Frozen Analog Event with Time (33, 4)	1	06, 07, 08	129, 130	17, 28
☐	Short Floating Frozen Analog Event (33, 5)	1	06, 07, 08	129, 130	17, 28
☐	Long Floating Frozen Analog Event (33, 6)1)	1	06, 07, 08	129, 130	17, 28
☐	Short Floating Frozen Analog Event with Time (33, 7)	1	06, 07, 08	129, 130	17, 28
☐	Short Floating Frozen Analog Event with Time (33, 7)	1	06, 07, 08	129, 130	17, 28
☐	Long Floating Frozen Analog Event with Time (33, 8)1)	1	06, 07, 08	129, 130	17, 28
n/i	16-Bit Analog Input Reporting Deadband Object (34, 1)				

	Object (Group, Variation)	Request Function (decimal)	Qualifier (hex)	Response Function	Qualifier
n/i	32-Bit Analog Input Reporting Deadband Object (34, 2)				
n/i	Floating Point Analog Input Reporting Deadband Object (34, 3)				
☐	Analog Output Status (40, 0)	1	00, 01, 06		
☐	32-Bit Analog Output Status (40, 1)	1	00, 01, 06	129, 130	00, 01
☐	16-Bit Analog Output Status (40, 2)	1	00, 01, 06	129, 130	00, 01
☐	Short Floating Point Analog Output Status (40, 3)	1	00, 01, 06	129, 130	00, 01
☐	Long Floating Point Analog Output Status(40, 4)1)	1	00, 01, 06	129, 130	00, 01
n/i	32-Bit Analog Output Block (41, 0)				
☐	32-Bit Analog Output Block (41, 1)	3, 4, 5, 6	17, 28	129	Echoes Response
☐	16-Bit Analog Output (41, 2)	3, 4, 5, 6	17, 28	129	Echoes Response
☐	Short Floating Point Analog Output Block (41, 3)	3, 4, 5, 6	17, 28	129	Echoes Response
☐	Long Floating Point Analog Output Block (41, 4)1)	3, 4, 5, 6	17, 28	129	Echoes Response
n/i	Time and Date (50, 0)				
☐	Time and Data (50, 1)	2 1	07 01	129	07
n/i	Time and Data with Interval (50, 2)				
n/i	Time and Date CTO (51, 0)				

	Object (Group, Variation)	Request Function (decimal)	Qualifier (hex)	Response Function	Qualifier
☐	Time and Data CTO (51, 1)			129, 130	07 quantity = 1
☐	Unsynchronized Time and Data CTO (51, 2)			129, 130	07 quantity = 1
n/i	Time Delay Coarse (52, 0)				
☐	Time Delay Coarse (52, 1)			129	07 quantity = 1
☐	Time Delay Fine (52, 2)			129	07 quantity = 1
☐	Class Zero Data (60, 1)	1	06		
☐	Class One Data (60, 2)	1 20, 21, 22	06, 07, 08 06		
☐	Class Two Data (60, 3)	1 20, 21, 22	06, 07, 08 06		
☐	Class Three Data (60, 4)	1 20, 21, 22	06, 07, 08 06		
n/i	File Identifier (70, 1)				
n/i	Internal Indications (80, 1)				
n/i	Storage Object (81, 1)				
n/i	Device Profile (82, 1)				
n/i	Private Registration Object (83, 1)				
n/i	Private Registration Object Descriptor (83, 2)				
n/i	Application Identifier (90, 1)				
n/i	Short Floating Point (100, 1)				
n/i	Long Floating Point (100, 2)				
n/i	Extended Floating Point (100, 3)				
n/i	Small Packed Binary-Coded Decimal (101, 1)				
n/i	Medium Packed Binary-Coded Decimal (101, 2)				
n/i	Large Packed Binary-Coded Decimal (101, 3)				

5.1.4.9 Special Communication Features

Communication Statistics

Communication Statistic Counters

Table 5-1 Summary of Communication Statistic Counters for both Link & Interface

Info ID	Label Statistics Counter	Description
901	Messages Sent Counter	All messages sent through CFE
902	Messages Received Counter	All good messages received by CFE
903	Messages Receive Error Counter	Summary of errors (CRC, length, pulse, etc.)
904	Message Timeout Counter	CFE did not receive an expected response on send
905	Scan Overrun Counter	Number of scans not sent because the previous scan was not finished
906	Scan Request Counter	Number of scan requests processed by protocol master
907	Command Request Counter	Number of command requests processed by protocol master

Legend:

Info ID Number used in IMM editor
 Label Statistics Name used in IMM editor or in trace output
 Counter

Communication Errors

Table 5-2 Summary of Communication Errors

Error No	Error Designation
-1	P-Stack: Service is unknown
-3	P-Stack: Request has timed out
-4	P-Stack: Link to Master failed
-6	P-Stack: Network buffer overflow
-7	P-Stack: Bad security bytes
-8	P-Stack: Network or application block not in sequence
-9	P-Stack: Requested service is in use
-10	P-Stack: Requested service is idle
-11	P-Stack: Requested service is not supported
-12	P-Stack: Object does not exist
-15	P-Stack: Variation on object is not supported
-19	P-Stack: Object can not be found
-20	P-Stack: Bad parameter passed to function
-21	P-Stack: Function is not known
-23	P-Stack: Invalid object
-28	P-Stack: Network I/O active
-35	P-Stack: RTU/device is running (during add/remove)
-36	P-Stack: Negative confirmation (NAK)
-37	P-Stack: Service implemented, but not functioning
-40	P-Stack: Request was rejected
-41	P-Stack: Not connected
-42	P-Stack: Request aborted

Error No	Error Designation
-43	P-Stack: Remote restart
-44	P-Stack: Data link
-45	P-Stack: Cannot broadcast
-46	P-Stack: No client
-47	P-Stack: Address error
-48	P-Stack: Direction is wrong
-49	P-Stack: No buffer
-50	P-Stack: Invalid point (index)
-51	P-Stack: No context for function
-52	P-Stack: Line is offline
-53	P-Stack: Device is offline
-55	P-Stack: Wrong request COUNT
-56	P-Stack: Wrong request ARGS
-57	P-Stack: Wrong request object
-58	P-Stack: Wrong request value qualifier
-61	P-Stack: Wrong request FUNCTION
-62	P-Stack: Wrong request STEP
-64	P-Stack: Invalid message type
-66	P-Stack: Invalid message
-69	P-Stack: Error in offline mode
-70	P-Stack: Error in listen mode
-73	P-Stack: Invalid indexing
-77	P-Stack: Local address error
-1001	PM: This function or object is not supported
-1002	PM: Any configuration error, which does not accept the item for the PM
-2001	I/O-driver: The hardware detected a framing error
-2002	I/O-driver: The hardware detected a break condition
-2003	I/O-driver: An error occurred during communications with the device
-2004	I/O-driver: The requested mode is not supported
-2005	I/O-driver: A character-buffer overrun has occurred, next character is lost
-2006	I/O-driver: A input buffer overflow has occurred
-2007	I/O-driver: The hardware detected a parity error
-2008	I/O-driver: The application tried to transmit a character, but the output buffer was full
-2009	I/O-driver: Transmission is waiting for the CTS (clear-to-send) signal to be sent
-2010	I/O-driver: Transmission is waiting for the DSR (data-set-ready) signal to be sent
-2011	I/O-driver: The end-of-file (EOF) character has been received
-2013	I/O driver: There is a character queued for transmission that has come to the communications device by a higher prior function. The communications device transmits such a character ahead of other characters in the devices output buffer
-2014	I/O driver: Transmission is waiting because the XOFF character was received
-2015	I/O driver: Transmission is waiting because the XOFF character was transmitted
-2020	PM: Communication to RTU not established
-2021	PM: Error during transmission
-2022	PM: Error during receiving
-2023	PM: Communication (Read, Write) was aborted; normal condition after block or switch-over
-2024	PM: The port did not react (to read or write)

Error No	Error Designation
-2025	PM: There was data expected but no data in buffer
-2026	PM: The port could not be initialized, opened or created
-2027	PM: An action at the port failed due to timeout
-3101	PM: The communication of the interface is stopped
-3301	PM: RTU/Device was not found in DB of PM
-3302	PM: The communication to RTU/Device is stopped, action performed requires a started device
-3303	PM: RTU/Device is offline
-3350	PM: The command from DM was not completely processed or send by PM
-3401	PM: Datapoint was not found in RTU/Device DB
-10001	PM: Transaction (request plus answer) could not be finished successfully
-10002	PM: No valid reuest/command (GetNextTransaction failed) in PM
-10101	PM: No valid object type (RTU, Datapoint) found.
-10102	PM: Invalid RTU or Datapoint (SW) Index
-10103	PM: Property not provided by PM

Legend:

P-Stack Protocol Stack (protocol-specific low-level software)
PM Protocol Module (protocol specific interface software)

Last Error

Info ID 900 is used to transport the last error occurred (Error No -1 through -nnnnn, 0= no error) as a counter value to the SCADA area for the following purposes:

- Displaying
- Logging
- Archiving

This numerically encoded information behaves like an event. It appears and does not disappear when the error condition is over.

Listening Mode

	Description	Remark
☐	Listening Mode	User parameters

Monitoring of Standby Channels / Standby Channel Information Buffering

Not available

5.1.5 Interoperability Check List for Protocol CDC Type II

5.1.5.1 Overview

CDC Type II is not a standard protocol in Spectrum Power 5. The implementation is not tested with Spectrum Power 5. No support is provided for this protocol with Spectrum Power 5.

This section summarizes the protocol CDCII specification and defines the set of messages, which are implemented in the respective Protocol Module.

The selected parameters are crossed in the white boxes.



NOTE

In addition, the full specification of a system may require individual selection of certain parameters for certain parts of the system, such as the individual selection of scaling factors for individually addressable measured values.

5.1.5.2 Network Configurations

	Configuration	Remark
☐	Point-to Point	redundant communication links (yes no)
n/i	Multiple Point-to Point	
☐	Multipoint-Partyline	redundant communication links (yes no)
☐	Multipoint-Star	
n/i	Data Concentrator	
☐	Multipoint-Ring	
☐	Dial in	
☐	Dial out	

5.1.5.3 Physical Layer

Transmission Speed (common for both directions)

	Speed	Remark
n/i	50 bit/s	Not Supported by ASE BCOM8 board
n/i	75 bit/s	Not Supported by ASE BCOM8 board
n/i	110 bit/s	Not Supported by ASE BCOM8 board
n/i	134 bit/s	Not Supported by ASE BCOM8 board
☐	150 bit/s	
☐	200 bit/s	
☐	300 bit/s	
☐	600 bit/s	
☐	1200 bit/s	
☐	1800 bit/s	
☐	2400 bit/s	
☐	4800 bit/s	
☐	9600 bit/s	
☐	19200 bit/s	

5.1.5.4 Link Layer

Message Format

	Procedure	Remark
■	Four byte header	Standard
■	Fourteen-byte data blocks	Standard
■	Start bit, stop bit, checksum per header and per data block	Standard

Link Transmission Procedure

	Description	Remark
■	Unbalanced, Master (half duplex)	
n/i	Unbalanced slave	

Frame Length

	Description	Remark
■	Maximum data length 255 words	

5.1.5.5 Application Layer

Process Information in Monitor Direction

	Description	Remark
☐	0: Scan 1 (2 bit CD, 1 bit CD, simple status, analogs)	
☐	1: Scan 2 (2 bit CD, 1 bit CD, simple status)	
☐	2: Scan 3 (all data within a range of sequence numbers)	
n/i	25: SOE event dump	Model 560/565 only
☐	26: SOE point status dump (range of PPU numbers)	Model 560/565 only
☐	33 / 47: Scan 3X table 1 to 15 (list of sequence numbers)	
☐	128: Repeat Scan1	
☐	129: Repeat Scan 2	
☐	130: Repeat Scan 3	
☐	161 / 175 Repeat scan 3X table 1 to 15	

System Information in Monitor Direction

	Description	Remark
n/i	28: SOE point enable/disable dump	Model 560/565 only

Parameters in Monitor Direction

	Description	Remark
--	-------------	--------

n/i	19: Memory read	
-----	-----------------	--

Process Information in Control Direction

	Description	Remark
☐	3: Direct setpoint for analog output	
n/i	3: Direct setpoint for digital generator control	DGC not in Power 5
☐	4: No operation	
☒	5: Control trip select	
☐	6: Control close select	
☐	8: Operate	
☐	9: Reset (cancel control)	
n/i	11: Direct trip	Direct commands not supported
n/i	12: Direct close	Direct commands not supported

System Information in Control Direction

	Description	Remark
☐	10: Accumulator freeze	
n/i	13: Reset pulse accumulator	
☐	27: SOE point enable/disable setup	Model 560/565 only
☐	29: SOE time synchronization	Model 560/565 only
☐	30: SOE calculated time correction	Model 560/565 only
☐	31: SOE forced time correction	Model 560/565 only

Parameters in Control Direction

	Description	Remark
☐	20: memory write	For 3X table download
☐	24: SOE size setup	Model 560/565 only

5.1.5.6 Basic Application Functions

Initialization

	Description	Remark
☐	Table 3X download	
☐	SOE size setup	Model 560/565 only
☐	Time synchronization	Model 560/565 only

Periodic Scans

	Description	Remark
☐	Type 1	
☐	Type 2	
☐	Type 3	
☐	Type 3X	
☐	SOE point status change	Model 560/565 only
☐	SOE event dump	Model 560/565 only

Controls

	Description	Remark
☐	SBO trip/close	
n/i	Direct trip/close	RTU option
☐	Direct setpoint for analog output	
n/i	Direct setpoint for digital generator control	DGC not in Power 5

Counters

	Description	Remark
☐	Read running and frozen accumulators	
☐	RTU Freeze	
☐	Line Freeze	
n/i	Reset accumulators	

5.1.5.7 Special Communication Features**Communication Statistics****Communication Statistic Counters**

Table 5-3 Summary of Communication Statistic Counters

Info ID	Label Statistics Counter	Description	Remark
901	Messages Sent Counter	All messages sent through CFE	Link & Interface
902	Messages Received Counter	All good messages received by CFE	Link & Interface
903	Messages Receive Error Counter	Summary of errors (CRC, length, pulse, etc.)	Interface only
904	Message Timeout Counter	CFE did not receive an expected response on send	Link & Interface
905	Scan Overrun Counter	Number of scans not sent because the previous scan was not finished	Link & Interface

Info ID	Label Statistics Counter	Description	Remark
906	Scan Request Counter	Number of scan requests processed by protocol master	Link & Interface
907	Command Request Counter	Number of command requests processed by protocol master	Link & Interface

Legend:

Info ID Number used in IMM editor
Label Statistics Counter Name used in IMM editor or in trace output

Communication Errors

Table 5-4 Summary of Communication Errors

Error No	Error Designation
001	Error Data Length
002	Failure Interface
003	Failure Device
004	Invalid Message
005	Error Msg Checksum
013	Error OutputBuf Overflow
016	Error Control
017	Error SW Logic

Last Error

Last Error (Info ID 900) is used to transport the last error occurred (Error No 001 through nnn, 0= no error) as a counter value to the SCADA area for the following purposes:

- Displaying
- Logging
- Archiving

This information has the behavior of an event (only appearing, no disappearing)

Listening Mode

Not available.

Monitoring of Standby Channels / Standby Channel Information Buffering

Not available.

5.1.6 Interoperability Check List for Protocol TG800

5.1.6.1 Overview

This section summarizes the protocol TG800 specification and defines the set of messages, which are implemented in the respective Protocol Module. The selected parameters should be crossed in the white boxes.



NOTE

In addition, the full specification of a system may require individual selection of certain parameters for certain parts of the system, such as the individual selection of scaling factors for individually addressable measured values.

5.1.6.2 Network Configurations

	Configuration	Remark
☐	Point-to Point	redundant communication links (yes no)
n/i	Multiple Point-to Point	
☐	Multipoint-Partyline	redundant communication links (yes no)
n/i	Multipoint-Star	
☐	Data Concentrator	
n/i	Multipoint-Ring	
n/i	Dial in	
n/i	Dial out	
n/i	Modem Bank	

5.1.6.3 Physical Layer

Transmission Speed (common for both directions)

	Speed	Remark
n/i	50 bit/s	
☐	75 bit/s	
☐	110 bit/s	
☐	134 bit/s	
☐	150 bit/s	
☐	200 bit/s	
☐	300 bit/s	
☐	600 bit/s	
☐	1200 bit/s	
☐	1800 bit/s	
☐	2400 bit/s	
☐	4800 bit/s	
☐	9600 bit/s	
☐	19200 bit/s	

5.1.6.4 Link Layer

Message Format

	Procedure	Remark
■	IEC standard TC57 for class format FT 1.2	
■	Long block (variable length) short block, single character	

	Procedure	Remark
☐	1 start bit, 8 data bits, 1 stop bit (Standard)	
n/i	1 start bit, 8 data bits, no parity, 1 stop bit (GSM Modem in RTU)	
☒	Security bit	

Link Transmission Procedure

	Description	Remark
☒	Unbalanced, Master (half duplex)	Both full- and half-duplex are supported.
n/i	Unbalanced slave	

Frame Length

	Description	Remark
☒	Maximum frame length = 237 octets	

Address field of the line

	Description	Remark
☒	One octet	
n/i	One octet + one octet address field extension (hierarchy)	
n/i	One octet + two octets address field extension (hierarchy)	

5.1.6.5 Application Layer

Process Information in Monitor Direction

	Description	Remark
☐	IAC ¹⁾ 1: Indications (without time tag)	
☐	IAC 2: Indications with time tag (SOE) (=ZFM)	
☐	IAC 3: Measurand (on event)	
n/i	IAC 4: Measurand with relative time from protection equipment	
n/i	IAC 6: Indication with relative time from protection equipment.	
n/i	IAC 8: SOE indication for logging TG709 SOE	
☐	IAC 10:Counter value called up and archived (single current)	
n/i	IAC 11: Counter value called up and archived (double current)	
n/i	IAC 12: Counter values after check request, before setting.	
☐	IAC 15:GA indications (block of indications)	

	Description	Remark
☐	IAC 16: Analogue GA measurands (blocks of measurands)	
☐	IAC 17: Digital GA measurands (blocks of measurands)	
n/i	IAC 18: Counter value after GA request	
n/i	IAC 166: TG800 Transfer of Disturbance Data According to VDEW/ZVEI IEC 103	

1) IAC = German: **Information Art Character** (Information type character)

System Information in Monitor Direction

	Description	Remark
n/i	IAC 15: Block with system indications	
n/i	IAC 25: Own station number	
n/i	IAC 35: Request for time from the CS	
☐	IAC 158: GA request to the CS: Indications	
☐	IAC 159: GA request to the CS: Measurands	
☐	IAC 168: Restart Subsystem	
☐	IAC 169: Subsystem Change Over	
n/i	IAC 170: TG 809 Fault Report Text Transfer to Station Module	

Parameters in Monitor Direction

	Description	Remark
n/i	IAC 175: Parameters	

Process Information in Control Direction

	Description	Remark
☐	IAC 1: Indications in command direction (without time tag)	
☐	IAC 2: indications in command direction (with time tag)	
☐	IAC 3: Measurands (on event) in command direction	
n/i	IAC 15: GA indications in command direction	
n/i	IAC 16: Analogue GA measurands in command direction	
n/i	IAC 17: Digital GA measurands in command direction	
☐	IAC 128: Pulse commands	
n/i	IAC 129: Permanent commands	
n/i	IAC 130: Stop permanent command	
☐	IAC 131: Set Point Value (VW ¹⁾)	
☐	IAC 133: PBN Set Point Value	

	Description	Remark
☐	IAC 134: PBN command	
n/i	IAC 166: TG800 Transfer of disturbance data According to VDEW/ ZVEI-IEC 103	

¹⁾ VW = German: **Vorgabewert**

System Information in Control Direction

	Description	Remark
☐	IAC 135: Synchronisation time	
n/i	IAC 140: Counter value freeze	
n/i	IAC 141: Counter value freeze, reset	
☐	IAC 142: Counter value freeze, transmit	
☐	IAC 143: Counter value freeze ,reset,transmit	
n/i	IAC 144: Counter value transmit	
n/i	IAC 145: GA request: Counter value	
n/i	IAC 146: Single CV Call up before CV setting	
n/i	IAC 148: Single Counter Value from the MS during CV setting	
☐	IAC 160: GA Request to Slaves: Indications.	
☐	IAC 161: GA Request to slaves: Measurands	
n/i	IAC 162: GA Request: System Indication Block	
n/i	IAC 163: System Indication Block Acknowledgement	
n/i	IAC 165: Command: send own station number	
n/i	IAC 168: Restart Subsystem	
n/i	IAC 169: Subsystem Change Over	

Parameters in Control Direction

	Description	Remark
n/i	IAC 175: Parameters	

5.1.6.6 Basic Application Functions

Station Initialization

	Description	Remark
n/i	Remote initialization	not supported

General Interrogation

	Description	Remark
☐	Indications	
☐	Measurands	

	Description	Remark
n/i	Counter values	

System Information in Monitor Direction

	Description	Remark
n/i	Detailed system indications	

System Information in Control Direction

	Description	Remark
☐	Clock Synchronisation	

Command Transmission

	Description	Remark
☐	Commands	
☐	Setpoints	
☐	PBN Commands	
☐	PBN Setpoints	

Transmission of Integrated Totals

	Description	Remark
n/i	Counter value freeze	
n/i	Counter value freeze, reset	
☐	Counter value freeze, transmit	
☐	Counter value freeze, reset, transmit	
n/i	Counter value transmit	
n/i	Counter value line freeze	
n/i	Counter value line freeze, reset	
☐	Counter value line freeze, reset, transmit	
n/i	Counter value line transmit	
n/i	GA Request: Counter value	
n/i	Counter value Setting	
n/i	Counter value call up groups	

Parameter Loading

Not available.

5.1.6.7 Special Communication Features**Communication Statistics**

Not available.

Listening Mode

Available.

Monitoring of Standby Channels / Standby Channel Information Buffering

Not available.

5.1.7 Interoperability Check List for Protocol TG8979

5.1.7.1 Overview

TG8979 is not a standard protocol in Spectrum Power 5. The implementation is not tested with Spectrum Power 5. No support is provided for this protocol with Spectrum Power 5.

This section summarizes the protocol TG8979 specification and defines the set of messages, which are implemented in the respective Protocol Module. The selected parameters should be crossed in the white boxes.

5.1.7.2 Network Configurations

	Configuration	Remark
☐	Point-to Point	
n/i	Multiple Point-to Point	
☐	Multipoint-Partyline	
n/i	Multipoint-Star	
n/i	Data Concentrator	
n/i	Multipoint-Ring	
n/i	Dial in	
n/i	Dial out	
n/i	Modem Bank	

5.1.7.3 Physical Layer

Transmission Speed (common for both directions)

	Speed	Remark
n/i	50 bit/s	
☐	75 bit/s	
☐	110 bit/s	
☐	134 bit/s	
☐	150 bit/s	
☐	200 bit/s	
☐	300 bit/s	
☐	600 bit/s	
☐	900 bit/s	
☐	1800 bit/s	
☐	2400 bit/s	
☐	4800 bit/s	

	Speed	Remark
☐	9600 bit/s	
☐	19200 bit/s	

5.1.7.4 Link Layer

Message Format

	Procedure	Remark
☒	Asynchronous, 1 start bit, 8data bits, no parity, 1 stop bit	
n/i	Synchronous, 8 data bits	
☒	16-bit CRC	
☒	First character is 0xff	
☒	Single data block in control direction	
☒	Multiple data blocks in monitor direction	
n/i	Macro command in control direction	
☒	Short response in monitor direction	

Link Transmission Procedure

	Description	Remark
☒	Unbalanced, Master	Both full and half duplex are supported.
n/i	Unbalanced slave	

Frame Length

	Description	Remark
☒	Maximum message length 140 bytes without leading 0xff and CRC	

5.1.7.5 Application Layer

Process Information in Monitor Direction

	Description	Remark
☐	0: Analog change report	
☐	1: Analog force report	
☐	2: Analog group change report	
☐	3: Analog group force report	
☐	5: ADC reference force report	
☐	6: Indication change report	
☐	7: Indication force report	

	Description	Remark
☐	8: SOE change report	
☐	9: SOE force report	
n/i	11: Digital input force report	
n/i	12: Accumulator change report	
☐	13: Accumulator force report	
☐	14: SOE log change report	

System Information in Monitor Direction

	Description	Remark
n/i	47: Table read	
n/i	51: Spontaneous report sequence number	
☐	63: Exception report	

Parameters in Monitor Direction

	Description	Remark
☒	31: RTU configuration	Data not validated
n/i	39: Firmware configuration	

Process Information in Control Direction

	Description	Remark
☐	20: Analog output	May be packed
☐	21: SBO select	
☐	22: SBO operate	
n/i	23: Digital output	
☐	25: Pulse output	May be packed
n/i	26: Pulse train output	May be packed
n/i	28: SBO immediate execute	
n/i	36: Accumulator preset	

System Information in Control Direction

	Description	Remark
☐	24: Accumulator freeze	Line or RTU
n/i	30: Restart RTU	
☐	32: Time synchronization	RTU, not line
☐	33: Time bias	
☐	37: Continuation request	
n/i	38: Repeat last message	

	Description	Remark
n/i	48: Table write	Not standard

Parameters in Control Direction

	Description	Remark
☐	34: Analog deadbands	Hardcoded values
☐	35: Analog group define	
n/i	50: Spontaneous report interval	

5.1.7.6 Basic Application Functions

Station Initialization

	Description	Remark
☒	Configuration check	Data not validated
☐	Analog deadband download	Hardcoded values
☐	Analog group download	
☐	Time synchronisation and bias measurement	

Data Scans

	Description	Remark
☐	Analog change	
☐	Analog force	
☐	Analog group change	
☐	Analog group force	
☐	Status change	
☐	Status force	
☐	SOE change	
☐	SOE force	
☐	SOE log	
n/i	Macro functions	
n/i	Spontaneous reporting	

Controls

	Description	Remark
☐		
n/i		
n/i		

	Description	Remark
☐		May be packed
n/i		May be packed
☐		May be packed

Counters

	Description	Remark
☐	Read accumulators	
☐	RTU Freeze	
☐	Line Freeze	
☐	Preset accumulators	

5.1.7.7 Special Communication Features

Communication Statistics

Communication Statistic Counters

Table 5-5 Summary of Communication Statistic Counters

Info ID	Label Statistics Counter	Description	Remark
901	Messages Sent Counter	All messages sent through CFE	Link & Interface
902	Messages Received Counter	All good messages received by CFE	Link & Interface
903	Messages Receive Error Counter	Summary of errors (CRC, length, pulse, etc.)	Interface only
904	Message Timeout Counter	CFE did not receive an expected response on send	Link & Interface
905	Scan Overrun Counter	Number of scans not sent because the previous scan was not finished	Link & Interface
906	Scan Request Counter	Number of scan requests processed by protocol master	Link & Interface
907	Command Request Counter	Number of command requests processed by protocol master	Link & Interface

Legend:

Info ID Number used in IMM editor
Label Statistics Name used in IMM editor or in trace output
Counter

Communication Errors

Table 5-6 Summary of Communication Errors

Error No	Error Designation
001	Error Data Length
002	Failure Interface
003	Failure Device
004	Invalid Message

Error No	Error Designation
005	Error Msg Checksum
013	Error OutputBuf Overflow
016	Error Control
017	Error SW Logic

Last Error

Info ID 900 is used to transport the last error occurred (Error No 001 through nnn, 0 = no error) as a counter value to the SCADA area for the following purposes:

- Displaying
- Logging
- Archiving

This numerically encoded information behaves like an event. It appears and does not disappear when the error condition is over.

Listening Mode

	Description	Remark
☐	Listening Mode	User parameters

Monitoring of Standby Channels / Standby Channel Information Buffering

Not available.

5.1.8 Interoperability Check List for Protocol TG065 /TD065**5.1.8.1 Overview**

TG065/TD065 is not a standard protocol in Spectrum Power 5. The implementation is not tested with Spectrum Power 5. No support is provided for this protocol with Spectrum Power 5.

5.1.8.2 Network Configurations

	Configuration	Remark
☐	Point-to Point	redundant communication links (yes no)
n/i	Multiple Point-to Point	
☐	Multipoint-Partyline	redundant communication links (yes no)
n/i	Multipoint-Star	
n/i	Data Concentrator	
n/i	Multipoint-Ring	
n/i	Dial in	
n/i	Dial out	

5.1.8.3 Physical Layer**Transmission Speed (common for both directions) Link Layer**

	Speed	Remark
☐	50 bit/s	
☐	75 bit/s	

	Speed	Remark
☐	110 bit/s	
n/i	134 bit/s	
☐	150 bit/s	
☐	200 bit/s	Standard Baud rate.
☐	300 bit/s	
☐	600 bit/s	Standard Baud rate.

Message Format

	Procedure	Remark
☒	1 start bit, 8 data bits, odd parity, 1 stop bit (Standard)	
☒	Security bit	

Link Transmission Procedure

	Description	Remark
☒	Unbalanced, Master (half duplex)	
n/i	Unbalanced, Slave	

Frame Length

	Description	Remark
☒	Maximum frame length = 32 octets	

Address Field of the Link

	Description	Remark
☒	One octet (US# 1 -61)	
☒	One octet (US# 0) use for BAA ¹⁾	

¹⁾ BAA = (German) **Befehl an alle** (Broadcast command)

5.1.8.4 Application Layer

Process Information in Monitor Direction

	Description	Remark
☐	IAC 0: Measurand 9-Bits (single or group transmission)	
☐	IAC 1: Indications (without time tag, single or group transmission)	
☐	IAC 2: TG065 Counter value called up 2-Byte)	
☐	IAC 2: TG065 Counter value called up 3-Byte)	Does not exist for TD065.

	Description	Remark
☐	IAC 3: Measurand 12-Bits (single or group transmission)	Does not exist for TD065.
n/i	IAC 1: Telemetred positions (transmitted as ind. and then transformed to mea)	
n/i	IAC 2: FAB ¹⁾ Alarmbytes	
n/i	IAC 2: FAB Counter value called up 4-Decade (BCD)	
n/i	IAC 2: FAB Counter value called up 6-Decade (BCD)	
n/i	IAC 2: FAB Sonderinformation (Special Information)	

¹⁾ FAB = Programmable telecounting device of Landis & Gyr

System Information in Monitor Direction

Not relevant.

Parameters in Monitor Direction

Not relevant.

Process Information in Control Direction

	Description	Remark
☐	IAC0: Set point value (VW ¹⁾)	for TD065 IAC1: (-255...+255)
n/i	IAC2: Counter values	
☐	IAC3: Pulse commands	
n/i	IAC3: Raise/lower commands	

¹⁾ VW = German: **Vorgabewert**

System Information in Control Direction

	Description	Remark
☐	IA 1 TG065: stop continuous command	
☐	IA 2 TG065: freeze and transmit all Counter values	
☐	IA 3 TG065: General interrogation Indications	
☐	IA 4 TG065: Transmit all Counter values	
☐	IA 5 TG065: freeze and reset and transmit all Counter values	
n/i	IA 6 TG065: Stop measurand transmission	
n/i	IA 7 TG065: Start measurand transmission	
n/i	IA 1 FAB: FAB answer only with short acknowledge	
n/i	IA 2 FAB: FAB answers normally	
n/i	IA 3 FAB: React: retransmit whole RUPU ¹⁾ again	

	Description	Remark
n/i	IA 4 FAB: freeze and transmit all Counter values	
n/i	IA 5 FAB: freeze and reset and transmit all Counter values	
n/i	IA 6 FAB: Array element Setpoint Value telegram	
n/i	IA 7 FAB: Time telegram	
n/i	IA 8 FAB: Time Synchronisation telegram	
n/i	IA 9-15 FAB: free definable	

¹⁾ RUPU = German **Rundpuffer** (Circular buffer)

Parameters in Control Direction

Not relevant.

5.1.8.5 Basic Application Functions

Station Initialization

Not relevant.

General Interrogation

	Description	Remark
☐	Indications	

System Information in Monitor Direction

	Description	Remark
☐	Byte number 0 is reserved for the system indication byte	
	Bit 0: US new initialized	
	Bit 1: Error in command output	
	Bit 2: Error in measurement input HW	
	Bit 3: Error in counter value input HW	
	Bit 4: Error in setpoint output HW	
	Bit 5: General interrogation running	

System Information in Control Direction

	Description	Remark
n/i	FAB: Clock Synchronization	

Command Transmission

	Description	Remark
☐	Commands	
☐	Setpoints	
n/i	Commands to all RTUs	

Transmission of Integrated Totals

	Description	Remark
☐	Counter value freeze, transmit	
☐	Counter value freeze, reset, transmit	
☐	Counter value transmit	
☐	Counter value line freeze, transmit (BAA)	
n/i	Counter value line freeze, reset, transmit	
n/i	Counter value line transmit	
n/i	Counter value Setting	

5.1.8.6 Special Communication Features**Communication Statistics**

Not available.

Listening Mode

Not available.

Monitoring of Standby Channels / Standby Channel Information Buffering

Not available.

5.1.9 Interoperability Check List for Protocol Sinaut 8FW**5.1.9.1 Overview**

Sinaut 8FW is not a standard protocol in Spectrum Power 5. The implementation is not tested with Spectrum Power 5. No support is provided for this protocol with Spectrum Power 5.

This section summarizes the protocol Sinaut 8FW specification and defines the set of messages, which are implemented in the respective Protocol Module.

The selected subset should be crossed in the white boxes.

**NOTE**

In addition, the full specification of a system may require individual selection of certain parameters for certain parts of the system, such as the individual selection of scaling factors for individually addressable measured values.

5.1.9.2 Network Configurations

	Configuration	Remark
☐	Point-to Point	redundant communication links (yes no)
n/i	Multiple Point-to Point	
☐	Multipoint-Partyline	redundant communication links (yes no)
n/i	Multipoint-Star	
n/i	Data Concentrator	
n/i	Multipoint-Ring	
n/i	Dial in	

	Configuration	Remark
n/i	Dial out	

5.1.9.3 Physical Transmission Parameter

Com-Port Pin Assignment

Pin assignment V.24/ V.28 (RS232)	Signal levels	Com-Port Pin no.	Applied
TxD	V.24/V.28	2	yes
DTR	V.24/V.28	20	no
RTS	V.24/V.28	4	yes
GND		1 / 7	yes
RxD	V.24/V.28	3	yes
DSR	V.24/V.28	6	no
CTS	V.24/V.28	5	yes
DCD	V.24/V.28	8	no
RC	V.24/V.28	17	no

Modulation Types

	Type	Remark
☐	DPDM	needs ASE BCOM8-56k board
n/i	PCM (UART)	not supported

Transmission Rates for DPDM Modulation

	Speed	Remark
☐	50 bit/s	
☐	75 bit/s	
n/i	100 bit/s	
☐	150 bit/s	
☐	200 bit/s	
☐	300 bit/s	
☐	600 bit/s	
n/i	1200 bit/s	

Transmission Parameter for DPDM Modulation

	Type	Range of values (default value)
	Long / short pulse ratio (Receiver)	2.0 - 5.0 (2.2)
	Long / short pulse ratio (Transmitter)	2.0 - 5.0 (2.2)
	Max Bit Distortion	0 . . .50 (40)
	Hamming Distance (Receiver)	d = { 2, 4, 6 } (4)
	Hamming Distance (Transmitter)	d = { 2, 4 ,6 } (4)

5.1.9.4 Link Parameters

Transmission Mode / Parameter

	Procedure	Data Range (default)
	balanced mode (spontaneous)	
☐	test cycle time	0 - 10000 ms (0)
	unbalanced mode (polling)	
☐	number of interrogation lists	0 . . . 4 (0)

Station Address

	Description	Data Range / Remark
☐	Station address 0, 7, 8 (bit)	0 : (no station address, valid only for balanced mode) 7 bit-address: 1 .. 127, (obligatory for party line) 8 bit-address: 1 .. 255, (valid only for balanced mode)

MSC Processing

	Description	Data Range (Default)
☐	MSC active	0 = no 1= yes (1)
☐	MSC starting command / overflow acknowledgement command repeat after n messages	n= 1 .. 5 (3)
☐	MSC Maximum Gap (n)	n= 1 .. 25 (2)
☐	GI messages increments MSC	0 = no, 1= yes (1)
☐	MSC storage acknowledgement after n messages	n= 1 .. 30 (10)
☐	MSC number of repetition of a repeat command	n= 1 .. 9 (3)
☐	MSC timeout for repetition of a repeat command	n= 1000 .. 60000 ms (2000)

5.1.9.5 Application Data Messages

Data Messages in Monitoring Direction

	Message Reference	Telegram Address	Remark
	MD01	4x bit pattern 8bits	
n/i	MD02	1x bit pattern 32bits	
n/i	MD03	8x indications (1byte)	single / double point
☐	MD04	32x indications (4byte)	single / double point
☐	MD05	64x indications (8byte)	single / double point
☐	MD06a	8x indications (1byte) w. time (10ms)	single / double point

	Message Reference	Telegram Address	Remark
☐	MD06b	8x indications (1byte) w. time (1ms)	single / double point
☐	MD07	1x measured value 9bit	
☐	MD08	4x measured value 9bit	
☐	MD09	8x measured value 9bit	
☐	MD10	2x measured value 12bit	
☐	MD11	4x measured value 12bit	
☐	MD12	32x fleeting (4byte)	
☐	MD13	8x fleeting (1byte) w. time	
☐	MD14a	4x transformer tap BDC coded	
☐	MD14b	4x transformer tap binary coded	
☐	MD15	1x metered value binary coded	
☐	MD16	1x metered value dual coded	
☐	MD17	4x measured value 12 bit w. time (sec)	

Data Messages in Control Direction

	Message Reference	Description	Remark
☐	CD01	8x bit control output (1byte)	single / double command
☐	CD02	1x analog setpoint 9bits	
n/i	CD03	4x analog setpoint 9bits	
n/i	CD04	8x analog setpoint 9bits	
n/i	CD05	1x digital setpoint 8bits	
n/i	CD06	1x digital setpoint 16bits	
n/i	CD07	1x digital setpoint 32bits	

Command Qualifier 0-15

	Description	Remark
☐	Short duration pulse	circuit breaker
☐	Long duration pulse	isolator
☐	Persistent output	

Housekeeping Messages in Monitoring Direction

	Message Reference	Sinaut 8FW-Record-No: Description	Remark
☐	MH01	512: Test record	
n/i	MH02	515: Single error record Substation	visible only within trace

	Message Reference	Sinaut 8FW-Record-No: Description	Remark
☐	MH03	515: Single error record Master Station	only FWA internal
☐	MH04	767: Acknowledgement check back before operate	
n/i	MH05	768: Acknowledgement threshold value operation	
n/i	MH06	769: Acknowledgement smoothing factor operation	
☐	MH07	770: Acknowledgement synchronization	accepted but not evaluated
n/i	MH08	770: Acknowledgement setting minutes	
n/i	MH09	770: Acknowledgement setting date	
n/i	MH10	772: Acknowledgement to start-up interrogation	
n/i	MH11	773: Negative acknowledgement : remote parameter setting	
n/i	MH12	780: Memory contents transfer : 4 bytes or 6 bytes	
☐	MH13	781 : Error bit pattern transfer (ZFBIT)	
☐	MH14	782: Cause of last STOP in the substation	
☐	MH15	1012: Error messages from faulty substations for polling operation with multi-polling-levels	
☐	MH16	1016: System error Bit Map "ZBBIT"	
☐	MH17	1020: System error Bit Map "ZBBIT" FWA 1024	
n/i	MH18	1022: Negative acknowledgement "single or continuous interrogation"	
n/i	MH19	1023: Acknowledgement "end of continuous interrogation"	
☐	MH20	Reply to poll message	own message format (depending on the station)

Housekeeping Messages in Control Direction

	Message Reference	Record-No: Description	Remark
☐	CH01	512: Test record	
☐	CH02	513: MSC acknowledgement (repeat request)	
☐	CH03	514: Startup acknowledgement	
☐	CH04	519: Reset command	
n/i	CH05	520: Single interrogation	
☐	CH06	521: Group interrogation	
n/i	CH07	522: Continuous interrogation (count)	
n/i	CH08	523: Continuous interrogation (on)	

	Message Reference	Record-No: Description	Remark
n/i	CH09	524: Continuous interrogation (off)	
n/i	CH10	525: Timed lists ON	
n/i	CH11	526: Timed lists OFF	
n/i	CH12	530: Receiver enable/disable	only FWA internal
n/i	CH13	531: Station enable/disable	only FWA internal
n/i	CH14	768: Threshold for measured values	
n/i	CH15	769: Smoothing factor for measured values	
☐	CH16	770: Synchronization of precision time (Stell1)	
n/i	CH17	770: Set minutes (Stell2)	
n/i	CH18	770: Set date (Stell3)	
n/i	CH19	771: Measured value selection	
n/i	CH20	772: Startup interrogation	
n/i	CH21	773: Remote parameter setting	
☐	CH22	779: Interposing test	
n/i	CH23	780: Memory interrogation	
☐	CH24	781: Error bit pattern interrogation	
n/i	CH25-31	790 - 796: Threshold operation FW512	
☐	CH32	Polling (special record)	

5.1.9.6 Basic Application Functions

Scan Control

	Description	Data Range (Default)
☐	Quick Check Cycle	0 - 65535 ms (2000)
☐	Quick Check TimeOut, Quick Scan TimeOut	0 - 65535 ms (2000)

Command Functions

	Description	Remark
☐	direct operation	
☐	check back before operate	
☐	clock sync	interval: 0 = no sync, 1 min, 10 min

5.1.9.7 Special Communication Features

Communication Statistic Counters

Table 5-7 Summary of Communication Statistic Counters

Info ID	Label Statistics/Error	Interface	Link	Description
901	Messages Sent Counter	Y	Y	All messages sent through CFE
902	Messages Received Counter	Y	Y	All good messages received by CFE

Info ID	Label Statistics/Error	Interface	Link	Description
903	Messages Receive Error Counter	Y	N	Summary of errors (CRC, length, pulse, etc.)
904	Message Timeout Counter	Y	Y	CFE did not receive an expected response on send

Legend:

Info ID	Number used in IMM editor
Label Statistics Counter	Name used in IMM editor or in trace output
Interface	Y(es): evaluated for the interface, N(o): not available for the interface
Link	Y(es): evaluated for every link/interface, N(o): not available per link

Listening Mode

	Description	Remark
☐	Listening Mode	User parameters

Monitoring of Standby Channels / Standby Channel Information Buffering

	Description	Remark
☒	Monitoring	
☐	Buffering	Buffer configurations with user parameters

5.1.9.8 Data Acquisition

Information object address structure

Subsystem	Message Number	Position Number
0 - 7	0 - 1023	0 - 7
8 bit starting at bit 24	16 bit starting at bit 8	8 bit starting at bit 0

Position numbers dependent on the type of information (and for some exceptions on the message number)

Info-Type	Position number	Message number	Message number offset	Comment
M_BO	0	~	1 Byte	MD 1
M_BO	0	0 - 1008 *1	4 Byte	MD 2
M_BO	0	515		MH 3
M_BO	0	1012	4 Byte	MH 15
M_SP	0 - 7	0 - 511	1 Byte	MD 3/4/5/12
M_SP w. time	0 - 7	0 - 511	1 Byte	MD 6/13
M_SP	0 - 7	1012 - 1023	1 Byte	MD 15/16/17
M_SP	0 - 31		781	MH 13
M_DP	0,2,4,6	0 - 511	1 Byte	MD 3/4/5
M_DP w. time	0,2,4,6	0 - 511	1 Byte	MD 6
M_ME	0,2,4,6	0 - 1010 *1	2 Byte	MD 7/8/9/10/11
M_ST (BCD)	0	~	1 Byte	MD 14
M_ST (BIN)	0	~	1 Byte	MD 14 (dual/binary)
M_ITR (BCD)	0	0 - 1008 *1	4 Byte	MD 15
M_ITR (BIN)	0	0 - 1008 *1	4 Byte	MD 16 (dual/binary)
C_SC	0	~	1 Byte	CD 1

Info-Type	Position number	Message number	Message number offset	Comment
C_DC	0,2,4,6	~	1 Byte	CD 1
C_SE	0	~	1 Byte	CD 2/3/4
C_BO_8	0	~	1 Byte	CD5 1byte
C_BO_16	0	0 - 1010 *1	2 Byte	CD6 2byte
C_BO_32	0	0 - 1010 *1	2 Byte	CD7 4byte

*1 only even message numbers

M_ monitor: BO bit string, SP single point, DP double point, ME measurement, ST step position, IT count

C_ commands: SC single command, DC double command, SE setpoint analog, BO setpoint digital

5.1.10 Interoperability Check List for Protocol IEC60870-5-103

5.1.10.1 Overview

IEC60870-5-103 is not a standard protocol in Spectrum Power 5. The implementation is not tested with Spectrum Power 5. No support is provided for this protocol with Spectrum Power 5.

The companion standard defined presents sets of parameters and alternatives from which subsets have to be selected to implement particular telecontrol systems. Certain parameter values, such as the number of octets in the COMMON ADDRESS of ASDUs represent mutually exclusive alternatives. This means that only one value of the defined parameters is admitted per system. Other parameters, such as the listed set of different process information in command and in monitor direction allow the specification of the complete set or subsets, as appropriate for given applications. This clause summarizes the parameters of the previous clauses to facilitate a suitable selection for a specific application. If a system is composed of equipment originating from different manufacturers it is necessary that all partners agree on the selected parameters.

The selected parameters should be crossed in the white boxes.



NOTE

In addition, the full specification of a system may require individual selection of certain parameters for certain parts of the system, such as the individual selection of scaling factors for individually addressable measured values.

5.1.10.2 Network Configuration

	Configuration	Remark
☐	Point-to Point	redundant communication links (yes no)
n/i	Multiple Point-to Point	redundant communication links (yes no)
☐	Multipoint-Partyline	redundant communication links (yes no)
n/i	Multipoint-Star	
n/i	Data Concentrator	
n/i	Multipoint-Ring	
n/i	Dial in /Dial out	
n/i	Modem Bank	

5.1.10.3 Physical Layer

Electrical Interface

	Type	Remark
☐	EIA RS232	no modem control signals, no pre- / post delay

	Type	Remark
☐	EIA RS485	realized through external hardware

Optical Interface (by External Hardware)

	Type	Remark
☐	Glass fibre	
☐	Plastic fibre	
☐	BFOC/2,5 type connector	
☐	F-SMA type connector	

Transmission Speed

	Speed	Remark
☐	1200 bit/s	
☐	2400 bit/s	
☐	4800 bit/s	
☐	9600 bit/s	
☐	19200 bit/s	
☐	38400 bit/s	

5.1.10.4 Link Layer**Link Transmission Procedure**

	Procedure	Remark
n/i	Balanced Transmission	
☒	Unbalanced Transmission	

Address Field of the Link

	Description	Remark
n/i	not present	only for balanced transmission the link address may be omitted
☒	1 Octet	always the same as Common Address of ASDU
n/i	2 Octets	

5.1.10.5 Application Layer**Transmission Mode for Application Data**

Mode 1 (least significant octet first), as defined in clause 4.10 of IEC 870-5-4, is used exclusively in this standard.

Common Address of ASDU

	Description	Remark
■	1 Octet	
n/i	2 Octets	

Function Type & Information Number

	Description	Remark
■	2 Octets	1st octet: Function type 2nd octet: Information Number

Cause of Transmission

	Description	Remark
■	1 Octet	CoT, P/N, T Telegrams with Test bit T not evaluated!
n/i	2 Octets	2nd octet is foreseen as Originator Address (not supported)

Selection of Standard ASDUs

Process Information in Monitor Direction

Compatible Range:

	Type Ident	Description	Remark
☐	<1>	Time-tagged message	
☐	<2>	Time-tagged message with relative time	relative time not evaluated
☐	<3>	Measurands I	
☐	<4>	Time-tagged measurands with relative time	relative time not evaluated
n/i	<5>	Identification message	
☐	<6>	Time synchronization	
☐	<8>	Termination of General Interrogation	
☐	<9>	Measurands II	
n/i	<10>	Generic data	
n/i	<11>	Generic identification	
n/i	<23>	List of recorded disturbances	
n/i	<26>	Ready for transmission of disturbance data	
n/i	<27>	Ready for transmission of a channel	
n/i	<28>	Ready for transmission of tags	
n/i	<29>	Transmission of tags	
n/i	<30>	Transmission of disturbance data	
n/i	<31>	End of transmission	

Private Range

	Type Ident	Description	Remark
☐	<205>	Time-tagged metering value	

Specification of the Private Type Time-Tagged Metering Value

Metering values (For example, kWh) are not defined in the IEC103 standard. There are no compatible data units available, which are suitable for the transmission of metered values. Some SIPROTEC 4 relays offer the possibility to transmit metered values via the IEC interface. For this reason, private data unit 205 is defined for the transmission of metered values.

The relay manuals give detailed information about the amount of information available with the function types and information numbers. One metering value per data unit is transmitted.

Structure of the Time-Tagged Metering ASDU Type <205>

Description	Value
Type ID	205
Var. Structure Qualifier	0x81
Cause of Transmission	1 or 7
Common Address of ASDU	1...255
Function Type	
Information Number	1...255
Information Object Type <205>, see below	
Time Tag CP32Time2a	

Information Object Type <205>

Byte1	Metered Value							
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰
Byte2	Metered Value							
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁰⁹	2 ⁰⁸
Byte3	Metered Value							
	2 ²³	2 ²²	2 ²¹	2 ²⁰	2 ¹⁹	2 ¹⁸	2 ¹⁷	2 ¹⁶
Byte3	Ext Err	Int Err	Stor Bit	Metered Value				
				Sign	2 ²⁷	2 ²⁶	2 ²⁵	2 ²⁴
bit	7	6	5	4	3	2	1	0

Legend:

Ext Err External Error: not used (always 0)
 Int Err Internal Error: Set after start-up or overflow, otherwise 0
 Stor Bit Storage bit: Changed after every counter freeze, starting at 0

Process information in Control Direction

Compatible Range:

	Type Ident	Description	Remark
☐	<6>	Time synchronization	
☐	<7>	Initiation of General Interrogation	
n/i	<10>	Generic data	
☐	<20>	General command	
n/i	<21>	Generic command	
	<24>	Order for disturbance data transmission	
	<25>	Acknowledgement for disturbance data transmission	

Private Range:

	Type Ident	Description	Remark
☐	<255>	Metered value freeze	

Specification of the Private Type Metered Value Freeze

Data unit 205 (metering value) is frozen and sent spontaneously but some equipments require a freeze command on request. After freezing, the values are transmitted automatically. The master system sends the following freeze ASDU:

Structure of the Metering Freeze Command, ASDU Type <255>

Description	Value
Type ID	255
Var. Structure Qualifier	0x00
Cause of Transmission	0
Common Address of ASDU	1...255
Function Type	
Information Number	1...255
Information Object irrelevant	

Cause of Transmission

	CoT	Cause	
☐	<1>	Spontaneous	
☐	<2>	Cyclic	
☐	<3>	Reset Frame Count Bit (FCB)	
☐	<4>	Reset communication unit (CU)	
☐	<5>	Start / restart	
☐	<6>	Power on	
☐	<7>	Test mode	
☐	<8>	Time synchronization	
☐	<9>	General Interrogation	
☐	<10>	Termination of General Interrogation	
☐	<11>	Local operation	
☐	<12>	Remote operation	
☐	<20>	Positive acknowledgement of command	
☐	<21>	Negative acknowledgement of command	
n/i	<31>	Transmission of disturbance data	
n/i	<40>	Positive acknowledgement of generic write command	
n/i	<41>	Negative acknowledgement of generic write command	
n/i	<42>	Valid data response to generic read command	

	CoT	Cause	
n/i	<43>	invalid data response to generic read command	
n/i	<44>	Generic write command	

**NOTE**

Causes 1 through 12 are accepted but not evaluated / indicated by Power 5

Quality Descriptors Evaluated by Spectrum Power 5

	QoI	Cause	Remark
☐	<OV>	overflow	Measured values
☐	<er>	invalid	Measured values

5.1.10.6 Basic Application Functions

	Description	Remark
☐	Test mode	
☐	Blocking of monitoring direction	
n/i	Disturbance data	
n/i	Generic serviced	
☐	Private data*	

**NOTE**

*) Private data as specified in Selection of Standard ASDUs is supported

5.1.10.7 Special Communication Features**Communication Statistics**

Not available.

Listening Mode

Not available.

Monitoring of Standby Channels / Standby Channel Information Buffering

Not available.

5.1.11 Interoperability Check List for Protocol MODBUS**5.1.11.1 Overview**

The MODBUS protocol standard presents sets of parameters and alternatives from which subsets have to be selected to implement particular telecontrol systems. Certain parameter values, represent mutually exclusive alternatives. This means that only one value of the defined parameters is admitted per system. Other parameters, such as the listed set of different process information in command and in monitor direction allow the specification of the complete set or subsets, as appropriate for given applications. If a system is composed of

equipment originating from different manufacturers it is necessary that all partners agree on the selected parameters.

The selected parameters should be crossed in the white boxes.



NOTE

In addition, the full specification of a system may require individual selection of certain parameters for certain parts of the system, such as the individual selection of scaling factors for individually addressable measured values.

5.1.11.2 Network Configurations

	Configuration	Remark
☐	Point-to Point	redundant communication links (yes no)
☐	Multipoint-Partyline	redundant communication links (yes no)
☐	TCP/IP	redundant communication links (yes no)

5.1.11.3 Physical Layer

Transmission Speed (common for both directions)

Unbalanced interchange

Circuit V.24/ V.28 Standard

	Speed	Remark
n/i	100 bit/s	
n/i	100 bit/s	
☐	300 bit/s	
☐	600 bit/s	
☐	1200 bit/s	
☐	2400 bit/s	
☐	4800 bit/s	
☐	9600 bit/s	
☐	19200 bit/s	
☐	38400 bit/s	
n/i	56000 bit/s	
n/i	64000 bit/s	

5.1.11.4 Transmission Modes

ASCII Mode

	Procedure	Remark
■	Asynchronous Serial Character Mode, one hexadecimal character contained in each ASCII character of the message. 1 start bit, 7 data bits, least significant bit sent first. 1 bit for even/odd parity; no bit for no parity 1 stop bit if parity is used; 2 bits for no parity	
■	Unbalanced, Master-Slave communication	
■	Begin: One character (8 bits) messages always begin with a colon (:) character (ASCII 3A hex)	
■	Device Address: 2 characters	Range 1 ... 247
■	Point Address: 4 characters	Range 0 ... 65535
■	Function: 2 characters	
■	Data: n characters. Max possible 500 characters - based on 125 registers each 4 characters (32 bits)	
■	Longitudinal Redundancy Check (LRC) - 2 characters (16 bits)	
■	End: 2 characters (16 bits) messages always end with a carriage return - line feed (CRLF) pair (ASCII 0D0A hex)	
■	Maximum message length: The maximum amount of data that each controller can request or send in a master query, or return in a slave response is based on the Controller Model and function. In general the maximum length of the entire message must not exceed 256 bytes	

RTU Mode

	Procedure	Remark
■	Asynchronous Serial Binary Mode, two hexadecimal characters contained in each 8- bit field of the message. 1 start bit, 8 data bits, least significant bit sent first 1 bit for even/odd parity; no bit for no parity 1 stop bit if parity is used; 2 bits for no parity	
■	Unbalanced, Master-Slave communication	
■	Start: T1 - T2 - T3 - T4 Messages start with a silent interval of at least 3.5 character times	
■	Device Address: 8 bits	
■	Point Address: 16 bits	
■	Function: 8 bits	

	Procedure	Remark
■	Data: n x 8 bits Max possible 250 bytes - based on 125 registers each 2 bytes (16 bits)	
■	Cyclical Redundancy Check (CRC) - 16 bits	
	End: T1 - T2 - T3 - T4 following the last transmitted character, a similar interval of at least 3.5 character times marks the end of the message	
■	Maximum Message length: The maximum amount of data that each controller can request or send in a master query, or return in a slave response is based on the Controller Model and function. In general the maximum length of the entire message must not exceed 256 bytes	

TCP/IP Mode

	Procedure	Remark
■	MODBUS Client-Server communication over TCP/IP Registered port 502	
■	MODBUS Application Protocol Header (MBAP) - 7 bytes: Transaction Identifier - 2 bytes Protocol Identifier - 2 bytes (MODBUS identified by 0) Length - 2 bytes Unit Identifier (address) - 1 byte	
■	Device Address: 8 bits	
■	Data: n x 8 bits Max. possible 250 bytes based on 125 registers each 2 bytes (16 bits)	
■	Maximum message length: 249 bytes (includes MBAP, max. data)	

5.1.11.5 Application Layer

Process Information in Monitor Direction

	Description	Remark
☐	01. Read Coil Status	Single and double information
☐	02. Read Input Status	Single and double information
☐	03. Read Holding Registers	Signed 16 bit, 32bit and short float values
☐	04. Read Input Registers	Signed 16 bit, 32bit and short float values
n/i	20. Read General Reference	
n/i	23. Read/Write 4X Registers	

Process Information in Control Direction

	Description	Remark
☐	05. Force Single Coil	Single command
☐	06. Preset Single Register	Setpoint 16 bit signed

	Description	Remark
☒	15. Force Multiple Coils	Automatically used in case of double commands and two addresses are needed
☒	16. Preset Multiple Registers	Automatically used if the setpoint value is a 32 bit signed or short float value and therefore needs two addresses
n/i	21. Write General Reference	
n/i	22. Mask Write 4X Registers	

5.1.11.6 Basic Application Functions

Data Scans

	Description	Remark
☐	01 Read Coil Status	
☐	02 Read Input Status	
☐	03 Read Holding Registers	
☐	04 Read Input Registers	

Controls

	Description	Remark
☒	Direct Commands	For additional information, refer to Interoperability Check List for Protocol MODBUS

5.1.11.7 Special Communication Features

Communication Statistics

Not available.

Listening Mode

Not available.

Monitoring of Standby Channels / Standby Channel Information Buffering

Not available.

5.1.12 Interoperability Check List for Protocol SNMP

5.1.12.1 Overview

This section summarizes the SNMP protocol specification and defines the functionality, which is implemented in the respective Protocol Module.

5.1.12.2 Network Configuration

	Configuration	Remark
☐	Point-to-Point	redundant communication links (yes no)
n/i	Multiple Point-to Point	

	Configuration	Remark
☐	Multipoint Partyline	redundant communication links (yes no)
n/i	Multipoint-Star	

5.1.12.3 Message Types

	PDU Type	Name	Remark
☐	0	GetRequest	
n/i	1	GetNextRequest	
☐	2	GetResponse	
n/i	3	SetRequest	
n/i	4	Trap for SNMP v1	
n/i	5	GetBulkRequest	
n/i	6	InformRequest	
n/i	7	Trap for SNMP v2	Trap message is not evaluated but can be used to trigger a General Interrogation
n/i	8	Report	

5.1.12.4 SNMP Version

	Version	Name	Description
☐	0	SNMP v1	SNMP Version 1
☐	2	SNMP v2	SNMP Version 2
n/i	3	SNMP v3	SNMP Version 3

5.1.13 PICS and PIXIT for Spectrum Power 5 IEC61850 Client

5.1.13.1 Protocol Implementation Conformance Statement (PICS)

Introduction

A Protocol Implementation Conformance Statement (PICS) provides details of the protocol services supported by the SICAM PAS protocol implementation.

This PICS refers to the template in IEC 61850-7-2 and 61850-8-1.



NOTE

The conformance statements PICS and PIXIT for the Power 5 implementation of the IEC 61850 Client is based on the compliance statement of SICAM PAS. The format and the structure of this is taken over from SICAM PAS unchanged, the contents is partly adapted for Power 5 purpose only. The Server functionality is NOT provided.

ACSI Conformance Statement

General

This chapter refers to the ACSI Conformance Statement template of IEC61850, Part 7-2, Annex A.1.

ACSI Basic Conformance Statement

The ACSI models conformance statement is defined in the following table.

Table 5-8 Basic Conformance Statement

		Client/ Subscriber		Server/Publisher		Value/ Comments
Client-Server Roles		m/o/-	Supp	m/o/-	Supp	
B11	Server side (of TWO-PARTY-APPLICATION-ASSOCIATION)	--	--	C1	N	
B12	Client side of (TWO-PARTY-APPLICATION-ASSOCIATION)	c1	Y	--	--	
SCSMs supported						
B21	SCSM: IEC 61850-8-1 used		Y		N	
B22	SCSM: IEC 61850-9-1 used		N		N	
B23	SCSM: IEC 61850-9-2 used		N		N	
B24	SCSM: other		N		N	
Generic substation event model (GSE)						
B31	Publisher side	--	--	O	N	
B32	Subscriber side	O	N	--	--	
Transmission of sampled value model (SVC)						
B41	Publisher side	--	--	O	N	
B42	Subscriber side	O	N	--	--	
O - Optional M - Mandatory Y - Yes (supported) N - No (not supported)						

ACSI Models Conformance Statement

The ACSI models conformance statement is defined in the following table.

Table 5-9 ACSI Models Conformance Statement

		Client/Subscriber		Server/Publisher		Value/ Comments
Only Client side supported (B12)		m/o/-	Supp	m/o/-	Supp	
M1	Logical device	c2	Y			
M2	Logical node	c3	Y			
M3	Data	c4	Y			
M4	Data set	c5	Y			
M5	Substitution	O	N			
M6	Setting group control	O	Y			
	Reporting					
M7	Buffered report control	O	Y			
M7-1	sequence-number	O	Y			
M7-2	report-time-stamp	O	O			
M7-3	reason-for-inclusion	O	R			
M7-4	data-set-name	O	O			

		Client/Subscriber		Server/Publisher		Value/ Comments
Only Client side supported (B12)		m/o/-	Supp	m/o/-	Supp	
M7-5	data-reference	O	O			
M7-6	buffer-overflow	O	O			
M7-7	entryID	O	R			
M7-8	BufTim	O	O			
M7-9	IntgPd	O	O			
M7-10	GI	O	O			
M8	Unbuffered report control	O	Y			
M8-1	sequence-number	O	Y			
M8-2	report-time-stamp	O	O			
M8-3	reason-for-inclusion	O	R			
M8-4	data-set-name	O	O			
M8-5	data-reference	O	O			
M8-6	buffer-overflow	O	O			
M8-7	IntgPd	O	O			
M8-8	GI	O	Y			
	Logging					
	Logging	O	N			
M9	Log control	O	N			
M9-1	IntgPd	O	N			
M10	Log	O	N			
	Control					
M11	Control	M	Y			
	GSE					
M12	GOOSE	O	N			
M13	GSSE	O	N			
	SVC					
M14	Multicast SVC	O	N			
M15	Unicast SVC	O	N			
	Time					
M16	Time	O	Y			PAS is NTP Client and NTP server
	Filetransfer					
M17	Filetransfer	O	Y			
c2 – shall be ‘M’ if support for LOGICAL-NODE model has been declared c3 – shall be ‘M’ if support for DATA model has been declared c4 – shall be ‘M’ if support for DATA-SET, Substitution, Report, Log Control, or Time model has been declared c5 – shall be ‘M’ if support for Report, GSE, or SMV models has been declared O – Optional M – Mandatory Y – Yes (supported) N – No (not supported) R – Required						

ACSI Service Conformance Statement

Table 5-10 ACSI Service Conformance Statement

		AA:	Client/Subscriber		Server/Publisher		Value/ Comments
Client role		TP/MC	m/o/-	Supp	m/o/-	Supp	
S1	ServerDirectory	TP		Y	M	-	
Application association (clause 7)			m/o/-	Supp	m/o/-	Supp	
S2	Associate		M	Y	M	-	
S3	Abort		M	Y	M	-	
S4	Release		M	Y	M	-	
Logical Device (clause 8)			m/o/-	Supp	m/o/-	Supp	
S5	LogicalDeviceDirectory	TP	M	Y	M	-	
Logical Node (clause 9)			m/o/-	Supp	m/o/-	Supp	
S6	LogicalNodeDirectory	TP	M	Y	M	-	
S7	GetAllDataValues	TP	O	N	M	-	
Data (clause 10)			m/o/-	Supp	m/o/-	Supp	
S8	GetDataValues	TP	M	Y	M	-	
S9	SetDataValues	TP	O	Y	O	-	
S10	GetDataDirectory	TP	O	Y	M	-	
S11	GetDataDefinition	TP	O	Y	M	-	
Data set (clause 11)			m/o/-	Supp	m/o/-	Supp	
S12	GetDataSetValues	TP	O	N	M	-	
S13	SetDataSetValues	TP	O	N	O	-	
S14	CreateDataSet	TP	O	Y	O	-	
S15	DeleteDataSet	TP	O	Y	O	-	
S16	GetDataSetDirectory	TP	O	Y	O	-	
Substitution (clause 12)			m/o/-	Supp	m/o/-	Supp	
S17	SetDataValues	TP	O	N	O	-	
SG Control (clause 13)			m/o/-	Supp	m/o/-	Supp	
S18	SelectActiveSG	TP	O	Y	O	-	
S19	SelectEditSG	TP	O	N	O	-	
S20	SetSGValues	TP	O	N	O	-	
S21	ConfirmEditSGValues	TP	O	N	O	-	
S22	GetSGValues	TP	O	N	O	-	
S23	GetSGCBValues	TPO	O	N	O	-	
Reporting (clause 14)			m/o/-	Supp	m/o/-	Supp	
Buffered report control block (BRCB)		TP	c6	Y	c6	-	
S24	Report	TP	c6	Y	c6	-	
S24-1	data-change (dchg)	TP	c6	Y	c6	-	
S24-2	qchg-change (qchg)	TP	c6	Y	c6	-	
S24-3	data-update (dupd)	TP	c6	Y	c6	-	
S25	GetBRCBValues	TP	c6	Y	c6	-	

		AA:	Client/Subscriber		Server/Publisher		Value/ Comments
Client role		TP/MC	m/o/-	Supp	m/o/-	Supp	
S26	SetBRCBValues	TP	c6	Y	c6	-	
Unbuffered report control block (URCB)							
S27	Report	TP	c6	Y	c6	-	
S27-1	data-change (dchg)	TP	c6	Y	c6	-	
S27-2	qchg-change (qchg)	TP	c6	Y	c6	-	
S27-3	data-update (dupd)	TP	c6	Y	c6	-	
S28	GetBRCBValues	TP	c6	Y	c6	-	
S29	SetBRCBValues	TP	c6	Y	c6	-	
c6 – shall declare support for at least one (BRCB or URCB)							
Logging (clause 14)			m/o/-	Supp	m/o/-	Supp	
Log control block							
S30	GetLCBValues	TP	M	N	M	-	
S31	SetLCBValues	TP	O	N	M	-	
Log						-	
S32	QueryLogByTime	TP	c7	N	M	-	
S33	QueryLogAfter	TP	c7	N	M	-	
S34	GetLogStatusValues	TP	M	N	M	-	
c7 – shall declare support for at least one (QueryLogByTime or QueryLogByEntry)							
Generic substation event model (GSE) (clause 14.3.5.3.4)			m/o/-	Supp	m/o/-	Supp	
GOOSE-CONTROL-BLOCK							
S35	SendGOOSEMessage	MC	c8	N	c8	-	
S36	GetGoReference	TP	O	N	c9	-	
S37	GetGOOSEElement-Number	TP	O	N	O	-	
S38	GetGoCBValues	TP	O	N	O	-	
S39	SetGoCBValues	TP	O	N	O	-	
GSSE-CONTROL-BLOCK							
S40	SendGSSEMessage	MC	c8	N	c8	-	
S41	GetGsReference	TP	O	N	c9	-	
S42	GetGSSEElement-Number	TP	O	N	c9	-	
S43	GetGsCBValues	TP	O	N	O	-	
S44	SetGsCBValues	TP	O	N	O	-	
c8 – shall declare support for at least one (SendGOOSEMessage or SendGSSEMessage)							
c9 – shall declare support if TP association is available							
Transmission of sampled value model (SVC) (clause 16)			m/o/-	Supp	m/o/-	Supp	
Multicast SVC							

		AA:	Client/Subscriber		Server/Publisher		Value/ Comments
Client role		TP/MC	m/o/-	Supp	m/o/-	Supp	
S45	SendMSVMessage	MC	c10	N	c10	-	
S46	GetMSVCBValues	TP	O	N	N	-	
S47	SetMSVCBValues	TP	O	N	N	-	
Unicast SVC							
S48	SendUSVMessage	TP	c10	N	c10	N	
S49	GetUSVCBValues	TP	O	N	O	N	
S50	SetUSVCBValues	TP	O	N	O	N	
c10 – shall declare support for at least one (SendMSVMessage or SendUSVMessage)							
Control (clause 17.5.1)			m/o/-	Supp	m/o/-	Supp	
S51	Select	TP	M	Y	O	-	
S52	SelectWithValue	TP	M	Y	O	-	
S53	Cancel	TP	O	Y	O	-	
S54	Operate	TP	M	Y	M	-	
S55	Command-Termination	TP	M	Y	O	-	
S56	TimeActivated-Operate	TP	O	Y	O	-	
File transfer (clause 20)			m/o/-	Supp	m/o/-	Supp	
S57	GetFile	TP	O	Y	M	-	
S58	SetFile	TP	O	N	O	-	
S59	DeleteFile	TP	O	N	O	-	
S60	GetFileAttribute-Values	TP	O	Y	M	-	
Time (5.5)			m/o/-	Supp	m/o/-	Supp	
T1	Time resolution of internal clock			0,1us 2 ⁻²¹			nearest negative power of 2 insec- onds
T2	Time accuracy of internal clock			0,1ms 2 ⁻¹⁰			nearest negative power of 2 insec- onds
				Y			T0 (10ms)
				Y			T1 (1ms)
				Y			T2 (100µ s)
				N			T3 (25µs)
				N			T4 (4µs)
				N			T5 (1µs)
T3	Supported time stamp resolution			from 2 ⁻⁰ to 2 ⁻³¹			Nearest value of 2 ⁻⁰ in seconds according to 5.5.3.7.3.3
O – Optional M – Mandatory Y – Yes (supported) N – No (not supported)							

MMS Conformance Statement

General

This chapter refers to the MMS Conformance Statement template of IEC61850.

Profile Conformance

For the following Clause, the following definitions apply:

- m: mandatory support. The item shall be implemented.
- c: conditional support. The item shall be implemented if the stated condition exists.
- o: optional support. The implementation may decide to implement the item.
- x: excluded. The implementation shall not implement this item.
- i: out-of-scope. The implementation of the item is not within the scope of this standard.
- F/S: Functional Standard. Should be applied.
- Base: Shall be applied in any application claiming conformance to this standard.

A-Profile Support

Table 5-11 Pics for A-Profile Support

A-Profile		Client		Server		Note
		F/S		F/S		
A1	Client/Server	c1	Y	c1	N	
A2	GOOSE/GSE Management	c2	N	c2	N	
A3	GSSE	c3	N	c3	N	
A4	Time-Sync	c4	Y	c4	N	
c1 - Shall be 'm' if support for any service specified in Table 2 are declared within the ACSI basic conformance statement. c2 - Shall be 'm' if support for any service specified in Table 6 are declared within the ACSI basic conformance statement. c3 - Shall be 'm' if support for any service specified in Table 9 are declared within the ACSI basic conformance statement. c4 - Support for at least one other A-Profile shall be declared (example, in A1-A3) in order to claim conformance to IEC 61850-8-1.						

T-Profile Support

Table 5-12 Pics for T-Profile Support

T-Profile		Client		Server		Note
		F/S		F/S		
T1	TCP/IP	c1	Y	c1	N	
T2	OSI	c2	N	c2	N	
T3	GOOSE/GSE	c3	N	c3	N	
T4	GSSE	c4	N	c4	N	
T5	Time-Sync	o	Y	o	Y	
c1 - Shall be 'm' if support for A1 is declared. Otherwise, shall be 'i'. c2 - Shall be 'o' if support for A1 is declared. Otherwise, shall be 'i'. c3 - Shall be 'm' if support for A2 is declared. Otherwise, shall be 'i'. c4 - Shall be 'm' if support for A3 is declared. Otherwise, shall be 'i'.						

MMS Conformance

MMS conformance is in accordance with ISO/ISP 14226-2.

Character Sets

- MMS identifier
The MMS Identifier is constrained to BasicIdentifier.
- MMSString
The MMSString is constrained to use the ISO646String character set.
- ObjectName
The MMS ObjectName is constrained to use BasicIdentifier.

MMS Service Conformance**Initiate Conformance**

The following tables specify the conformance of the initiate service.

Table 5-13 MMS InitiateRequest General Parameters

InitiateRequest	Client-CR				Server-CR		
	Base	F/S	Value/ Range	Realized	Base	F/S	Value/ Range
InitiateRequest				Y			
localDetailCalling	m	m		Y			
proposedMaxServOut- standing Calling	m	m	1 or greater	Y			
proposedMaxServOus- tanding Called	m	m	1 or greater	Y			
initRequestDetail	m	m		Y			
InitiateRequestDetail	m	m		Y			
proposedVersionNumber	m	m	1 or higher	Y			
proposedParameterCBB	m	m		Y			
servicesSupportedCalling	m	m		Y			
additionalSupportedCalling	c1	x		Y			
additionalCbbSupported- Calling	c1	x		Y			
privilegeClassIdentity- Calling	c1	x		Y			
1 - Conditional upon Parameter CBB CSPI.							

Table 5-14 MMS InitiateResponse General Parameters

InitiateResponse	Client-CR				Server-CR		
	Base	F/S	Value/ Range	Realized	Base	F/S	Value/ Range
InitiateResponse				Y			
localDetailCalled	m	m		Y			
negotiatedMaxServOut- standing Calling	m	m	1 or greater	Y			
negotiatedMaxServOus- tanding Called	m	m	1 or greater	Y			
initResponseDetail	m	m		Y			
InitiateResponseDetail	m	m		Y			

negotiatedVersionNumber	m	m	Shall be 2.1	Y			
negotiatedParameterCBB	m	m		Y			
servicesSupportedCalled	m	m		Y			
additionalSupportedCalled	c1	x		N			
additionalCbbSupported-Called	c1	x		N			
privilegeClassIdentity-Called	c1	x		N			
c1 - Conditional upon Parameter CBB CSPI.							

Services Supported

The following table defines the service support requirement, and restrictions, for this standard.

Table 5-15 MMS Service Supported Conformance Table

MMS service supported CBB	Client-CR				Server-CR		
	Base	F/S	Request or Recieve/ Realized		Base	F/S	Value/ Range
status	o	o		N			
getNameList	o	o	Req	Y			
identity	o	o		N			
rename	o	o		N			
read	o	o	Req	Y			
write	o	o	Req	Y			
getVariableAccessAttributes	o	o	Req	Y			
defineNamedVariable	o	o		N			
defineScatteredAccess	o	i		N			
getScatteredAccessAttributes	o	i		N			
deleteVariableAccess	o	o		N			
defineNamedVariableList	o	o	Req	Y			
getNamedVariableListAttributes	o	o	Req	Y			
deleteNamedVariableList	o	o	Req	Y			
defineNamedType	o	i		N			
getNamedTypeAttributes	o	i		N			
deleteNamedType	o	i		N			
input	o	i		N			
output	o	i		N			
takeControl	o	i		N			
relinquishControl	o	i		N			
defineSemaphore	o	i		N			
deleteSemaphore	o	i		N			
reportPoolSemaphoreStatus	o	i		N			
reportSemaphoreStatus	o	i		N			
initiateDownloadSequence	o	i		N			
downloadSegment	o	i		N			

terminateDownloadSequence	o	i		N			
initiateUploadSequence	o	i		N			
uploadSegment	o	i		N			
terminateUploadSequence	o	i		N			
requestDomainDownload	o	i		N			
requestDomainUpload	o	i		N			
loadDomainContent	o	i		N			
storeDomainContent	o	i		N			
deleteDomain	o	i		N			
getDomainAttributes	o	o		N			
createProgramInvocation	o	i		N			
deleteProgramInvocation	o	i		N			
start	o	i		N			
stop	o	i		N			
resume	o	i		N			
reset	o	i		N			
kill	o	i		N			
getProgramInvocationAttributes	o	i		N			
obtainFile	o	c9		N			
defineEventCondition	o	i		N			
deleteEventCondition	o	i		N			
getEventConditionAttributes	o	i		N			
reportEventCondition-Status	o	i		N			
alterEventConditionMonitoring	o	i		N			
triggerEvent	o	i		N			
defineEventAction	o	i		N			
deleteEventAction	o	i		N			
alterEventEnrollment	o	i		N			
reportEventEnrollment-Status	o	i		N			
getEventEnrollmentAttributes	o	i		N			
acknowledgeEventNotification	o	i		N			
getAlarmSummary	o	i		N			
getAlarmEnrollmentSummary	o	i		N			
readJournal	o	c13		N			
writeJournal	o	i		N			
initializeJournal	o	i		N			
reportJournalStatus	o	i		N			
createJournal	o	i		N			
deleteJournal	o	i		N			
fileOpen	o	c8	REQ	Y			

fileRead	o	c8	REQ	Y			
fileClose	o	c8	REQ	Y			
fileRename	o	i		N			
fileDelete	o	c9		N			
fileDirectory	o	c11	REQ	Y			
unsolicitedStatus	o	i		N			
informationReport	o	c7	RCV	Y			
eventNotification	o	i		N			
attachToEventCondition	o	i		N			
attachToSemaphore	o	i		N			
conclude	m	m	REQ	Y			
cancel	o	o	REQ	Y			
getDataExchangeAttributes	o	c10		N			
exchangeData	o	c10		N			
defineAccessControlList	o	c10		N			
getAccessControlListAttributes	o	c10		N			
reportAccessControlledObjects	o	c10		N			
deleteAccessControlList	o	c10		N			
alterAccessControlList	o	c10		N			
reconfigureProgramInvocation	o	c10		N			

c1 - Shall be 'm' if logical device or logical node model support is declared in ACSI basic conformance statement.
c2 - Shall be 'm' if logical node model support is declared in ACSI basic conformance statement or if support for the MMS write service is declared.
c3 - Shall be 'm' if ACSI support for SetDataValues service is declared or implied.
c4 - Shall be 'm' if logical node model support is declared in ACSI basic conformance statement.
c5 - Shall be 'm' if data set support is declared in the ACSI basic conformance statement.
c6 - Shall be 'm' if support for defineNamedVariableList is declared.
c7 - Shall be 'm' if support for ACSI Report or ACSI command termination is declared.
c8 - Shall be 'm' if support for ACSI GetFile is declared.
c9 - Shall be 'm' if support for ACSI SetFile is declared.
c10 - Shall not be present since MMS minor version is declared to be 1.
c11 - Shall be 'm' if support for ACSI GetFileAttributeValues is declared.
c12 - Shall be 'm' if support for the ACSI log model is declared.
c13 - Shall be 'm' if support for the ACSI QueryLogByTime or QueryLogAfter is declared.
c14 - Shall be 'm' if support for the ACSI logical device model is declared.

Table 5-16 MMS Parameter Conformance Building Block (CBB)

MMS Parameter CBB	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
STR1	o	o		Y			
STR2	o	o		Y			
NEST	1	1	5	Y			

VNAM	o	o		Y			
VADR	o	o		Y			
VALT	o	o		Y			
Bit 5	x	x		--			
TPY	o			Y			
VLIS	o	c1o		Y			
Bit 8	x	x		--			
Bit 9	x	x		--			
CEI	o	i		N			
ACO	o	c4		--			
SEM	o	c4		--			
CSR	o	c4		--			
CSNC	o	c4		--			
CSPLC	o	c4		--			
CSPI	o	c4		--			
c1 - Shall be 'm' if ACSI logical node model support declared c2 - Shall be five(5) or greater if ACSI logical node model support is declared c3 - Shall be 'm' if ACSI data set, reporting, GOOSE or logging model support is declared c4 - Shall not be present. Receiving implementations shall assume not supported							

GetNameList Conformance

The following table defines the conformance of the GetNameList service.

Table 5-17 GetNameList Conformance Statement

GetNameList	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
ObjectClass	m	m		Y			
ObjectScope	m	m		Y			
DomainName	o	o		Y			
ContinueAfter	o	m		Y			
Response+							
List Of Identifier	m	m		Y			
MoreFollows	m	m		Y			
Response-							
Error Type	m	m		Y			

**NOTE**

Object class 'vmd' (formerly VMDSpecific in MMS V1.0) shall not appear. If a request contains this Object-Class, an MMS Reject shall be issued.

Variable Access Conformance**AlternateAccessSelection**

Table 5-18 AlternateAccessSelection Conformance Statement

AlternateAccessSelection	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range

accessSelection	o	o		N			
component	o	o		N			
index	o	i		N			
indexRange	o	i		N			
allElements	o	i		N			
alternateAccess	o	o		N			
selectAccess	o	o		N			
component	o	o		N			
index	o	i		N			
indexRange	o	i		N			
allElements	o	i		N			

VariableAccessSpecification

The following table defines the conformance of the variable access.

Table 5-19 VariableAccessSpecification Conformance Statement

VariableAccessSpecifica- tion	Client-CR				Server-CR		
	Base	F/S	Value/ Range	Realized	Base	F/S	Value/ Range
listOfVariable	o	o		Y			
variableSpecification	o	o		Y			
alternateAccess	o	o		N			
variableListName	o	o		Y			

VariableSpecification

The following table defines the conformance of the variable specification.

Table 5-20 VariableSpecification Conformance Statement

VariableSpecification	Client-CR				Server-CR		
	Base	F/S	Value/ Range	Realized	Base	F/S	Value/ Range
name	o	o		Y			
address	o	o		Y			
variableDescription	o	o		Y			
scatteredAccessDescrip- tion	o	x		N			
invalidated	o	x		N			

Read

The following table defines the conformance of the read service.

Table 5-21 Read Conformance Statement

Read	Client-CR				Server-CR		
	Base	F/S	Value/ Range	Realized	Base	F/S	Value/ Range
Request							
specificationWithResult	o	o		N			
variableAccessSpecifica- tion	m	m		Y			
Response							

variableAccessSpecification	o	o		N			
listOfAccessResult	m	m		Y			

Write

The following table defines the conformance of the write service.

Table 5-22 Write Conformance Statement

Write	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
variableAccessSpecification	m	m		Y			
listOfData	m	m		Y			
Response							
failure	m	m		Y			
success	m	m		Y			

InformationReport

The following table defines the conformance of the InformationReport service.

Table 5-23 InformationReport Conformance Statement

InformationReport	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
variableAccessSpecification	m	m		Y			
listOfAccessResult	m	m		Y			

GetVariableAccessAttributes

The following table defines the conformance of the GetVariableAccessAttributes service.

Table 5-24 GetVariableAccessAttributes Conformance Statement

GetVariableAccessAttributes	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
name	o	o		Y			
address	o	o		N			
Response							
mmsDeletable	m	m		Y			
address	o	x		N			
typeSpecification	m	m		Y			

DefinedNameVariableList

The following table defines the conformance of the DefinedNameVariableList service.

Table 5-25 DefinedNameVariableList Conformance Statement

DefinedNameVariableList	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
variableListName	m	m		Y			
listOfVariable	m	m		Y			
variableSpecification	m	m		Y			
alternateAccess	o	i		N			
Response	m	m		Y			

GetNamedVariableListAttributes

The following table defines the conformance of the GetNamedVariableListAttributes service.

Table 5-26 GetNamedVariableListAttributes Conformance Statement

GetNamedVariableListAttributes	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
ObjectName	m	m		Y			
Response							
mmsDeletable	m	m		Y			
listOfvariable	m	m		Y			
variableSpecification	m	m		Y			
alternateAccess	o	m		Y			

DeleteNamedVariableList

The following table defines the conformance of the DeleteNamedVariableList service.

Table 5-27 DefinedNameVariableList Conformance Statement

DefinedNameVariableList	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
Scope	m	m		Y			
listOFVariableListName	m	m		Y			
domainName	o	m		Y			
Response				m			
numberMatched	m	m		Y			
numberDeleted	m	m		Y			
DeleteNamedVariableList-Error	m	m		Y			

Journal Management Services

The Power 5 61850 Client is compatible with journal management services but does not use them.

File Management Services

FileDirectory

The following table defines the conformance of the FileDirectory service.

Table 5-28 FileDirectory Conformance Statement

FileDirectory	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
fileSpecification	o	o		Y			
continueAfter	o	o		Y			
Response+							
listOfDirectoryEntry	m	m		Y			
MoreFollows	m	m		Y			

FileOpen

The following table defines the conformance of the FileOpen service.

Table 5-29 FileOpen Conformance Statement

FileOpen	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
filename	m	m		Y			
initialPosition	o	o		N			
Response+							
frsmID	m	m		Y			
fileAttributes	m	m		Y			

FileRead

The following table defines the conformance of the FileRead service.

Table 5-30 FileRead Conformance Statement

FileRead	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
frsmID	m	m		Y			
Response+							
fileData	m	m		Y			
moreFollows	m	m		Y			

FileClose

The following table defines the conformance of the FileClose service.

Table 5-31 FileClose Conformance Statement

FileClose	Client-CR				Server-CR		
	Base	F/S	Value/Range	Realized	Base	F/S	Value/Range
Request							
frsmID	m	m		Y			
Response+							

Goose Services

GOOSE Conformance Statement

The Power 5 61850 Client is compatible to devices with GOOSE services but does not publish or subscribe to GOOSE messages.

GSSE Conformance Statement

The Power 5 61850 Client is compatible to devices with GSSE services but does not publish or subscribe to GSSE messages.

5.1.13.2 Protocol Implementation Extra Information for Testing (PIXIT)

Introduction

This document specifies the protocol implementation extra information for testing (PIXIT) of the IEC 61850 interface in the client system: **SICAM PAS** with version **6.0**, further referred to as **client**.

Together with the PICS and MICS, the PIXIT forms the basis for a conformance test according to IEC 61850-10.

The following chapters specify the PIXIT for each applicable ACSI service model as structured in IEC 61850-10 and the **Conformance Test Procedures for Client System with IEC 61850-8-1 interface**.

5.1.13.3 Model Specific PIXIT

PIXIT for Configuration

Table 5-32 PIXIT for Configuration

Description	Value / Clarification
Describe how the client handles nameplate configuration revision mismatches	The client will display the configuration mismatch and will stop the connection.
Describe how the client handles report control block configuration revision mismatches	As the RCB configuration revision may not be maintained correctly in some server implementations, the client instead checks all relevant RCB values and the dataset elements at startup to assure compatible configurations.

PIXIT for Association Model

Table 5-33 PIXIT for Association Model

Description	Value / Clarification
Maximum number of servers a Power 5 61850 Client can connect to depending on the performance of the used server and other running applications on the server	150-200 connected IEC61850 servers per CFE server pair; observe the restriction of a maximum of 90 interface processes per CFE server. More are possible if the CFE server performance allows it.
Lost connection detection time range (default range of TCP_KEEPAIVE is 1 – 20 seconds)	20 seconds
Lost (abort) connection retry time	30 seconds
Is authentication supported?	N
What is the maximum and minimum MMS PDU size?	Max MMS PDU size: 30000 Min MMS PDU size: depends on server
What is the typical startup time after a power supply interrupt?	Windows Boot without Login + (Number of Devices X 1 SECOND)

If the server does not behave as expected, the client will abort the connection. Typical issues are:

- Configuration revisions in Nameplate read from SCD file and from Server do not match.
- IED or LD names read from SCD file and from Server do not match.

- RCB cannot be initialized (RCB is already reserved or write failures).
- GI is enabled but is not received.
- No response for an outstanding request for more than 20 secs.

PIXIT for Server Model

Table 5-34 PIXIT for Server Model

Description	Value / Clarification
Maximum object identification length	32/32 64/64 will be supported with PAS V6.0 SP3
Does client support autodescription?	Yes, details see below.
What analogue value (MX) quality bits are used in the client?	Y / Good, Y / Invalid, Y / Reserved, Y / Questionable Y / Overflow Y / OutofRange Y / BadReference Y / Oscillatory Y / Failure Y / OldData Y / Inconsistent Y / Inaccurate Y / Process Y / Substituted N / Test Y / OperatorBlocked
Which status value (ST) quality bits are used in the client?	Y / Good, Y / Invalid, Y / Reserved, Y / Questionable Y / Overflow Y / OutofRange Y / BadReference Y / Oscillatory Y / Failure Y / OldData Y / Inconsistent Y / Inaccurate Y / Process Y / Substituted N / Test Y / OperatorBlocked
Describe how to view/display quality values	The quality values are converted into Power 5 quality format as needed. Use an RT Explorer to display the quality information.

Description	Value / Clarification
Describe how to force a SetDataValues request	The client allows writing all values with a control model like CSWI and LEDRes. Additionally the active setting group can be written. Use an RT Explorer to force the write request.
Describe how to force a GetAllDataValues request	The data values are received using information reports. A command GI Command to force an additional general interrogation is available.
Describe how the client behaves in case of: - GetDataDefinition response- - GetLogicalDeviceDirectory response- - GetAllDataValues response- - GetDataValues response- - SetDataValues response-	The client will stop communication with IED. Some SetDataValues requests are allowed to fail like negative operate responses or writing an outdated entryID to a buffered report control block.

The client supports autodescription of the server to verify the content of the server ICD / SCD file against the IED. The following autodescriptions are supported.

- GetServerDirectory
- GetLogicalDeviceDirectory
- GetLogicalNodeDirectory
- GetDataDirectory
- GetDataDefinition

The view of the IED will not be updated. The model of the IED will not be updated.
In case of mismatching datamodels the client reacts in the following way.

Table 5-35 Reactions on Autodescription Mismatch

Mismatch	Reaction
Logical Device not found	Stop communication with IED
Additional Logical Device	The client continues normal operation
Logical Node not found	Missing DOs are shown as 'Not valid'
Additional Logical Node	The client continues normal operation
Data Object not found	Missing DO is shown as 'Not valid'
Additional Data Object	The client continues normal operation

PIXIT for Data Set Model

Table 5-36 PIXIT for Data Set Model

Description	Value / Clarification
Describe how to force a GetDataSetValues request	The client does not read the values using GetDataSetValues, information report is used instead.
Describe how to force a SetDataSetValues request	Describe how to force a SetDataSetValues
Describe how to force a DeleteDataSet request	If the server offers the DynDataSet service: create a list in IMM and set a valid MMS-Address. A corresponding DataSet will be created by the client on connect and will be deleted on disconnect

Description	Value / Clarification
Describe how the client handles following dataset mismatches between the SCL and the data sets exposed using MMS: - New dataset element - Missing dataset element - Reordered dataset elements in a dataset of a different data type - Reordered dataset elements in a dataset of the same data type	- Normal operation - Restart the connection if a Used value is missing - Normal operation - Normal operation
Describe how the client behaves in case of: - GetLogicalNodeDirectory(DATA-SET) response- - GetDataSetDirectory response-	- Restart if a DataSet containing an Used value is missing - Restart if a DataSet containing an Used value is missing
Does the client create: - Persistent datasets - Non-persistent datasets	- Y - Y
Describe how the client behaves in case of: - CreateDataSetDirectory response- - DeleteDataSet response-	- Restart the connection - DeleteDataSet is done on disconnecting, a negative response has no effect in this case

PIXIT for Substitution Model

This service is not supported as defined in PICS.

PIXIT for Setting Group Control Model

Table 5-37 PIXIT for Setting Group Control Model

Description	Value / Clarification
Describe how to change the active setting group	The server shall have an SG control block like <Ldname>\$LLN0\$SGCB With RT Explorer a controllable value with name c_<Ldname>\$LLN0\$SP\$SGCB will set the active setting group.
Describe how to get the actual setting group values	A value <Ldname>\$LLN0\$SP\$SGCB which indicates the current setting group is available in RT Explorer
Describe how to edit setting group values	Not supported
Describe how the client behaves in case of: - GetSGCBValues response - The configured SG is different then the actual setting group	- If the read request fails the client will restart the connection - The client displays the actual setting group

PIXIT for Reporting Model

Table 5-38 PIXIT for Reporting Model

Description	Value/ Clarification
Does the client search for RCB in all logical nodes? When not specify the logical nodes?	All logical nodes

Which dynamic RCB attributes are or can be configured by the client?	RptID / Y DataSet / Y Optional fields / Y Trigger conditions / Y Buffer time / Y Integrity period / Y
Does the client supports IED's with indexed and non-indexed report control blocks (RCB)?	Buffered RCB indexed / Y Buffered RCB not indexed / Y Unbuffered RCB indexed / Y Unbuffered RCB not indexed / Y
The supported trigger conditions are	integrity / Y data change / Y quality change / Y data update / Y general interrogation
The minimum required optional fields are	sequence-number / Y report-time-stamp / Y (only for BRCB) reason-for-inclusion / Y data-set-name / N data-reference / N buffer-overflow / Y (only for BRCB) entryID / Y (only for BRCB) conf-rev / N
Does the client support segmented reports?	Y
Does the client support pre-assigned RCB?	Y
Does the client support reported data set containing structured data objects or data attributes?	reporting of data object / Y reporting of data attributes / Y
Describe how the client does respond when an URCB is already reserved	A PAS list object represents an IEC61850 DataSet. If a list has a Report control block assigned, the client stops the connection if the RCB is already in use. If the list has no RCB assigned, then the client will search another unused RCB.
Describe how the client does respond when a BRCB is already reserved	A PAS list object represents an IEC61850 DataSet. If a list has a report control block assigned, the client stops the connection if the RCB is already in use. If the list has no RCB assigned, then the client will search another unused RCB.
Describe how the client does respond on a SetBRCBValues(EntryID) respond-	The client will purge the buffer and send a Sys_BufferLost value to HMI, TC, and Value-Viewer.

Describe how the client does respond when a report has an unknown: dataset, RptId, unexpected number of dataset entries, and/or unexpected data type format entries	ReportSettings are Dyn : Client will write correct RCB attribute and start normal ReportSettings are Fix or Conf : Dataset: restart connection RptId: restart connection Data type: restart connection Service DynDataSet= TRUE : Num dataset entries: Client will correct the DataSet and start normal Service DynDataSet= FALSE : Num dataset entries: Client will start normal in case of additional entries and it will restart in case of missing entries
Describe how the client detects reporting configuration changes (mismatches). Does it check the configuration revision attributes and/or does it check the dataset elements?	As the RCB configuration revision may not be maintained correctly in some server implementations, the client instead checks all relevant RCB values and the dataset elements at startup to assure compatible configurations.
Describe how to force the client to change the RCB buffer-time	In IMM, each list offers a Buffer Time attribute. A specific firmware version (or later) is required for data buffering in the device. The firmware version is read from the device during operation and compared with the version indicated in the parameter. If the firmware version of the device is very old and does not support data buffering, the buffering feature is deactivated.
additional items:	
Integrity period	Configurable >=1 second;
Dynamic URCB reservation after an abort of the client/ server association	Dynamic URCB reservation after an abort of the client/ server association
Configured RCB reservation after an abort of the client/ server association	Reservation of the RCB is done on the configured RCB.

PIXIT for Logging Model

This service is not supported as defined in PICS.

PIXIT for Generic Substation Events Model

This service will not be supported as defined in PICS.

PIXIT for Transmission of Sample Values Model

This service will not be supported as defined in PICS.

PIXIT for Control Model

Table 5-39 PIXIT for Control Model

Description	Value / Clarification
-------------	-----------------------

What control models are supported?	Y / status-only Y / direct-with-normal-security N* / sbo-with-normal-security Y / direct-with-enhanced-security Y / sbo-with-enhanced-security * sbo-with-normal-security will be supported with PAS V6.0 SP3
Is Time activated operate (operTm) supported?	N
Is operator-many supported?	N
Can the client set the test flag?	N
What are the conditions for the time (T) attribute in the SelectWithValue and/or Operate request?	Time attribute is not relevant.
Is operator-many supported?	N
Is pulse configuration supported?	N
What check conditions can be set?	Y / synchrocheck Y / interlock-check By default interlock-check is enabled for every command. Interlock checking can be disabled by a separate internal marker command. By default synchrocheck is enabled for all commands to circuit breaker position. Synchro checking can be disabled by a separate internal marker command. For SIPROTEC relays synchro-checking is only enabled, when activated in DIGSI. For additional information, refer to Control Requirements.
Which originator categories are supported and what is the originator identification?	Y / remote Y / bay Y / station The identification string is: "remote via ControlClient"
Describe if and how the client sets/increments the ctlNum	Describe if and how the client sets/increments the ctlNum
What does the client when it receives a LastApplicationError and describe how to view the additional cause?	The client waits for the server response and converts the additional cause to a corresponding AddCause in HMI, TC, and ValueViewer.
What does the client when it receives a Select, SelectWithValue or Operate respond negative?	The client waits for the server response and converts the additional cause to a corresponding AddCause in HMI, TC, and ValueViewer. The cause OfTransmission will be SelectConfNeg or CommandConfNeg.
Can the client change the control model using online services?	N
What does the client when the ctlModel is not initialized in the SCL?	For security reasons the client always reads the actual control model from the server, the configured ctlModel is informative.

5.1.13.4 Additional Information

Mapping of IEC61850 Reason/TriggerCondition to Power 5

Table 5-40 Mapping of IEC61850 Reason/TriggerCondition

IEC 61850 Reason	PAS CauseOfTransmission	Comment
< Buffered Report >	Buffered	For Buffered Reports
Reserved	NotDefinedCoT	TriggerCondition
Datachange	Spontaneous	TriggerCondition
Qualitychange	Spontaneous	TriggerCondition
Dataupdate	Spontaneous	TriggerCondition
Integrity	Periodic	TriggerCondition
General Interrogation	GeneralInterrogation	TriggerCondition
Reserved bit 1	NotDefinedCoT	TriggerCondition
Reserved bit 2	NotDefinedCoT	TriggerCondition

For example, a buffered report is an event during lost association which is reported when the association is restored.

If more than one Reason is present:Buffered overrides Spontaneous overrides Periodic overrides GI.

Behavior if IED is not Consistent with ICD File

Table 5-41 Behavior if IED is not Consistent with ICD File

Difference	Result	Comment
ConfigRev in nameplate does not match ICD or SCD file	Reconnect every 60 seconds until the ConfigRev matches	
IED name or LogicalDevice name do not match	Reconnect every 60 seconds until the names match	
Preconfigured DataSet which is used in CFE is missing	Reconnect every 60 seconds until the DataSet is available	
Logical Node or DataObject is missing in IED (DataSet is preconfigured in ICD or SCD)	Reconnect every 60 seconds until the DO is available	
Logical Node or DataObject is missing in IED (DataSet is NOT preconfigured in ICD or SCD)	One reconnect, the validity of missing DOs will remain not valid	
Common Data Class CDC is different but DO name is same	Additional attributes are tolerated, missing attributes like dirGeneral remain not valid in CFE	
OptFlds	Reconnect every 60 seconds if minimum OptFlds or OptFids from ICD cannot be read or written	

Requirements

Reporting Requirements

Table 5-42 Optional Fields Needed by Power 5 61850 Client

Use Case	Optional Fields (BRCB.OptFlds)	Comment
Unbuffered Reporting	reason-for-inclusion Sequence-number	Client needs these bits and will disconnect if value is wrong and write fails
Buffered Reporting	reason-for-inclusion sequence-number entryId buffer-overflow report-time-stamp	Client needs these bits and will disconnect if value is wrong and write fails

Table 5-43 RCB Attributes Written by Power 5 61850 Client

Use Case	BRCB attribute	Comment
Unbuffered Reporting (pre-configured)		
Unbuffered Reporting	RptId	If RptIDis empty or not unique, a unique string is written
	Resv	Always written
	RptEna	Always written
	Datset	Written only - If value is different - And ReportSettings are Dyn
	OptFlds	Written only - If value is different - And ReportSettings are Dyn
	TrgOps	Written only - If value is different - And ReportSettings are Dyn
	IntgPd	Written only - If value is different - And ReportSettings are Dyn
	GI	Written only if GI is enabled in TrgOps
	BufTm	Written only if SIEMENS IED, BufTm can be enabled for other vendors by extending Manufacturer and SW Revision in PAS UI-Config and importing it into IMM
Buffered Reporting (Preconfigured)		
	RptId	If RptIDis empty or not unique, a unique string is written
	ResvTms	Only written if present in IEC61850 Edition2 IEDs
	RptEna	Always written

Use Case	BRCB attribute	Comment
	Datset	Written only - If value is different - And ReportSettings are Dyn
	OptFlds	Written only - If value is different - And ReportSettings are Dyn
	TrgOps	Written only - If value is different - And ReportSettings are Dyn
	IntgPd	Written only - If value is different - And ReportSettings are Dyn
	GI	Written only if GI is enabled in TrgOps
	BufTm	Written only if SIEMENS IED, BufTm can be enabled for other vendors by extending Manufacturer and SW Revision in PAS UI-Config and importing it into IMM
	EntryID	Always written
	PurgeBuf	Written only if buffered reporting cannot enter SYNC state.

Control Requirements

Table 5-44 Control Requirements

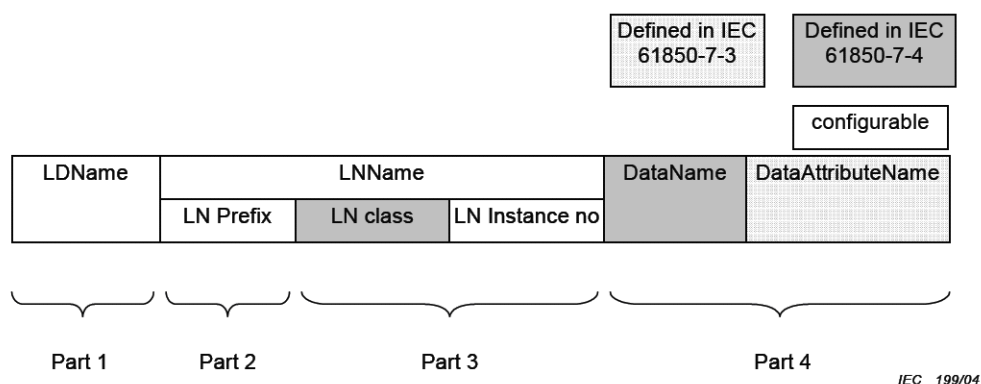
Issue	Comment
Commands with enhanced security	- If the command value is equal to the monitoring value, the client rejects the command with the additional cause "StateAlreadyReached" - If the command value is not inside the range of allowed values, the client rejects the command with the additional cause "PlausibilityError" - If the value was successfully written, the client waits for a corresponding feedback message. - If no feedback is detected during the configured monitoring time, the client will return to the idle state.
Tap change commands with enhanced security	Same conditions as for normal commands apply. Unlike the case above the client waits for an increment or decrement of the controlled tap position. If no feedback is detected during the configured monitoring time, the client will return to the idle state.

Issue	Comment
NoLock and NoSync qualifier	As the NoLock and NoSync command qualifiers are not common in other protocols, these bits are implemented as separate marker commands. The NoLock marker command is created for each control object. The NoSync marker command is only created for circuit breakers.

5.1.13.5 61850 Addressing Scheme

The IEC 61850 standard defines at least 2 addressing schemes for sending and retrieving of information between partners, servers, and clients on the network.

The addressing scheme derived from the SCD reflects the hierarchical structure of the information.



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An example would be: G7SA6314PROT\$PDIS1\$ST\$StrCA

This address is defined from naming attributes of the hierarchical structure.

IED::name = "G7SA6314"	
LDevice::inst = "PROT"	Logical Device - Protection
LN::prefix = ""	Empty
LN::lnClass = "PDIS"	Logical node class
LN::inst = "1"	Instance number
DO::name = "StrCA"	Data Object
DA::fc = "ST"	Data attribute - functional constraint

The address is the equivalent to a telegram address for conventional RTU information. It will be stored with CFE data point.

5.2 Interoperability Lists for Slave Protocols

5.2.1 Interoperability Check List for Protocol IEC-60870-5-101/Slave

5.2.1.1 Overview

The companion standard presents sets of parameters and alternatives from which subsets have to be selected to implement particular telecontrol systems. Certain parameter values, such as the number of octets in the COMMON ADDRESS of ASDUs represent mutually exclusive alternatives. This means that only one value of the defined parameters is admitted per system. Other parameters, such as the listed set of different process information in command and in monitor direction allow the specification of the complete set or subsets, as appropriate for given applications. If a system is composed of equipment originating from different manufacturers it is necessary that all partners agree on the selected parameters.

The selected parameters should be crossed in the white boxes.



NOTE

In addition, the full specification of a system may require individual selection of certain parameters for certain parts of the system, such as the individual selection of scaling factors for individually addressable measured values.

5.2.1.2 Network Configuration

	Configuration	Remark
☒	Point-to Point	
☐	Multiple Point-to Point	
☐	Multipoint-Partyline	
☐	Multipoint-Star	
☐	Data Concentrator	
☐	Multipoint-Ring	
☐	Dial in / Dial out	
☐	Modem Bank	

5.2.1.3 Physical Layer

Transmission Speed (common for both directions)

Unbalanced Interchange

Circuit V.24 / V.28 Standard

	Speed	Remark
☐	110 bit/s	
☐	200 bit/s	
☐	300 bit/s	
☐	600 bit/s	
☐	1200 bit/s	
☐	2400 bit/s	
☐	4800 bit/s	
☐	9600 bit/s	

	Speed	Remark
☐	19200 bit/s	
☐	38400 bit/s	
n/i	56800 bit/s	
n/i	64000 bit/s	

5.2.1.4 Link Layer

Link Transmission Procedure

Frame format FT 1.2, single character¹ (=E5) and the fixed time out interval are used exclusively in this companion standard.

	Procedure	Remark
☐	Balanced Transmission	
n/i	Send / No reply	
☐	Unbalanced Transmission	
n/i	max. length of the message (number of octets)	Monitoring direction: 255 Command direction: irrelevant (no blocking)
☐	Repetition of messages	Default: 1 Done by the primary station (unbalanced mode: master)
☐	ACD Bit Handling	
n/i	DFC Bit Handling in monitor direction	

Address Field of the Link

	Description	Remark
☐	Not present	Only for balanced transmission the link address may be omitted
☐	1 Octet	Default
☐	2 Octets	
n/i	Structured	
☒	Unstructured	

5.2.1.5 Application Layer

Transmission Mode for Application Data

Mode 1 (least significant octet first), as defined in clause 4.10 of IEC 60870-5-4, is used exclusively in this standard.

Common Address of ASDU

	Description	Remark
☐	1 Octet	
☐	2 Octets	
n/i	Structured	

	Description	Remark
☒	Unstructured	

Information Object Address

	Description	Remark
☐	1 Octet	
☐	2 Octets	
☐	3 Octets	Default
☐	Structured	
☐	Unstructured	Default

Cause of Transmission

	Description	Remark
☐	1 Octet	T, P/N, CoT, Telegrams with Test bit T not evaluated!
☐	2 Octets	2nd octet is foreseen as Originator Address (not evaluated for incoming and zero for outgoing messages)

Selection of Standard ASDUs

Process Information in Monitor Direction

	Type Ident	ASDU	Description	Remark
☐	<1>	M_SP_NA_1	Single-point information	
☐	<2>	M_SP_TA_1	Single-point information with time tag	
☐	<3>	M_DP_NA_1	Double-point information	
☐	<4>	M_DP_TA_1	Double-point information with time tag	
☐	<5>	M_ST_NA_1	Step position information	Positive values only!
☐	<6>	M_ST_TA_1	Step position information with time tag	ditto
n/i	<7>	M_BO_NA_1	Bitstring of 32 bit	
n/i	<8>	M_BO_TA_1	Bitstring of 32 bit with time tag	
☐	<9>	M_ME_NA_1	Measured value, normalized value	recommended type
☐	<10>	M_ME_TA_1	Measured value, normalized value with time tag	
☐	<11>	M_ME_NB_1	Measured value, scaled value	
☐	<12>	M_ME_TB_1	Measured value, scaled value with time tag	
☐	<13>	M_ME_NC_1	Measured value, short floating point value	

	Type Ident	ASDU	Description	Remark
☐	<14>	M_ME_TC_1	Measured value, short floating point value with time tag	
☐	<15>	M_IT_NA_1	Integrated totals	
☐	<16>	M_IT_TA_1	Integrated totals with time tag	
n/i	<17>	M_EP_TA_1	Event of protection equipment with time tag	
n/i	<18>	M_EP_TB_1	Packed start events of protection equipment with time tag	
n/i	<19>	M_EP_TC_1	Packed output circuit information of protection equipment with time tag	
n/i	<20>	M_PS_NA_1	Packed single-point information with status change detection	
n/i	<21>	M_ME_ND_1	Measured value, normalized value without quality descriptor	
☐	<30>	M_SP_TB_1	Single-point information with time tag CP56Time2a	
☐	<31>	M_DP_TB_1	Double-point information with time tag CP56Time2a	
☐	<32>	M_ST_TB_1	Step position information with time tag CP56Time2a	
n/i	<33>	M_BO_TB_1	Bitstring of 32 bit with time tag CP56Time2a	
☐	<34>	M_ME_TD_1	Measured value, normalized value with time tag CP56Time2a	
☐	<35>	M_ME_TE_1	Measured value, scaled value with time tag CP56Time2a	
☐	<36>	M_ME_TF_1	Measured value, short floating point value with time tag	
☐	<37>	M_IT_TA_1	Integrated totals with time tag CP56Time2a	
n/i	<38>	M_EP_TD_1	Event of protection equipment with time tag CP56Time2a	
n/i	<39>	M_EP_TE_1	Packed start events of protection equipment with time tag CP56Time2a	

	Type Ident	ASDU	Description	Remark
n/i	<40>	M_EP_TF_1	Packed output circuit information of protection equipment with time tag CP56Time2a	

Process information in Control Direction

	Type Ident	ASDU	Description	Remark
☐	<45>	C_SC_NA_1		
☐	<46>	C_DC_NA_1		
☐	<47>	C_RC_NA_1		
☐	<48>	C_RC_NA_1		
☐	<49>	C_SE_NB_1		
☐	<50>	C_SE_NC_1		
n/i	<51>	C_BO_NA_1		

Cause of Transmission

	CoT	Cause	Remark
☐	<1>	periodic, cyclic	
☐	<2>	background scan	
☐	<3>	spontaneous	
n/i	<5>	requested	
☐	<6>	activation	
☐	<7>	activation confirmation	
n/i	<8>	deactivation	
n/i	<9>	deactivation confirmation	
☐	<10>	activation termination	
n/i	<11>	return information caused by a remote command	
n/i	<12>	return information caused by a local command	
☐	<20>	interrogated by general interrogation	

Quality Descriptors Sent by Spectrum Power 5

	QoI	Cause	Remark
☐	<NT>	not topical	
☐	<IV>	invalid	for measured values, integrated totals, digitals
☐	<OV>	overflow	

	Qol	Cause	Remark
☐	<SB>	substituted	
n/i	<BL>	blocked	

System Information

System Information in Monitoring Direction

	Type Ident	ASDU	Description	Remark
☐	<70>	M_EI_NA_1	End of initialization	After sending TK70 to RCC a GI request is expected

System Information in Control Direction

	Type Ident	ASDU	Description	Remark
☒	<100>	C_IC_NA_1	General interrogation command	
☐	<101>	C_CI_NA_1	Counter interrogation command	
☐	<102>	C_RD_NA_1	Read command	
☐	<103>	C_CS_NA_1	Clock synchronization command	Content of clock synchronization message is not evaluated. Configurable: positive or negative acknowledgement when received
☐	<104>	C_TS_NA_1	Test command	
n/i	<105>	C_RP_NA_1	Reset process command	
n/i	<106>	C_CD_NA_1	Delay acquisition command	

Parameter Transfer in Control Direction

	Type Ident	ASDU	Description	Remark
n/i	<110>	P_ME_NA_P q KPA=1 KPA=2 KPA=3/4	Parameter of measured value, normalized value - threshold - smoothing factor - low / high limit	
n/i	<111>	P_ME_NB_1 KPA=1 KPA=2 KPA=3/4	Parameter of measured value, scaled value - threshold - smoothing factor - low / high limit	

	Type Ident	ASDU	Description	Remark
n/i	<112>	P_ME_NC_1 KPA=1 KPA=2 KPA=3/4	Parameter of measured value, short floating point value - threshold - smoothing factor - low / high limit	
n/i	<113>	P_AC_NA_1 QPA=3	Parameter activation - activation of cyclic transmission - deactivation of cyclic transmission	

File Transfer

n/i File Transfer in Monitoring Direction

n/i File Transfer in Control Direction

	Type Ident	ASDU	Description	Remark
n/i	<120>	F_FR_NA_1	File ready	
n/i	<121>	F_SR_AN_1	Section ready	
n/i	<122>	F_SC_NA_1	Call directory, select file, call file, call selection	
n/i	<123>	F_LS_NA_1	Last section	
n/i	<124>	F_AF_NA_1	Ack file, ack section	
n/i	<125>	F_SG_NA_1	Segment	
n/i	<126>	F_DR_TA_1	Directory	

Type Identifier and Cause of Transmission Assignments

The following chart is an extract from IEC60870-5 Companion Standard 101. It shows for each message Type ID the supported Cause of Transmission assignments (white boxes). Shaded boxes show combinations that are not supported.

IEC101SL		Cause of Transmission (CoT)																			
Type Identification		1	2	3	4	5	6	7	8	9	10	11	12	13	20	37	44				
1	M_SP_NA_1														36	41	47				
2	M_SP_TA_1																				
3	M_DP_NA_1																				
4	M_DP_TA_1																				
5	M_ST_NA_1																				
6	M_ST_TA_1																				
9	M_ME_NA_1																				
10	M_ME_TA_1																				
11	M_ME_NB_1																				
12	M_ME_TB_1																				
13	M_ME_NC_1																				
14	M_ME_TC_1																				
15	M_IT_NA_1																				
16	M_IT_TA_1																				
30	M_SP_TB_1																				
31	M_DP_TB_1																				
32	M_ST_TB_1																				
34	M_ME_TD_1																				
35	M_ME_TE_1																				
36	M_ME_TF_1																				
37	M_IT_TB_1																				
45	C_SC_NA_1																				
46	C_DC_NA_1																				
47	C_RC_NA_1																				
48	C_SE_NA_1																				
49	C_SE_NB_1																				
50	C_SE_NC_1																				
70	M_EI_NA_1																				
100	C_IC_NA_1																				
101	C_CI_NA_1																				
102	C_RD_NA_1																				
103	C_CS_NA_1																				
104	C_TS_NA_1																				
105	C_RP_NA_1																				

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Figure 5-6 CoT for every Message Type

5.2.1.6 Basic Application Functions

General Interrogation

	Type Ident	ASDU	Description	Remark
☐	<100>	C_IC_NA_1 (QOI=20)	global G I	the following COT parameters are applied: in control direction: <6> activation (C_IC ACT) in monitoring direction: <7> activation confirmation (C_IC ACTCON) <10> activation termination (C_IC ACTTERM)
☐	<100>	C_IC_NA_1 (QOI=21)	group 1	
☐	<100>	C_IC_NA_1 (QOI=2n)	group 2 ... 16	

Command Transmission

	Description	Remark
☐	Direct command transmission	
☒	Direct set-point command transmission	
☒	Command termination	
☐	Select and execute command	
☒	Select and execute set-point command	
☐	C_SE ACTTERM used	for data points known to the system

Qualifier for Command Transmission

	QOC	Description	Remark
☒	<0>	no additional definition	Preferably, duration of individual command output pulses are to be parameterized in the RTU
n/i	<1>	Short pulse duration (duration determined by a system parameter in the outstation)	
n/i	<2>	Long pulse duration (duration determined by a system parameter in the outstation)	
n/i	<3>	Persistent output	
n/i		(others to be defined)	

**NOTE**

Qualifier of Command (QOC) unequal zero can be configured on the RTU side of CFE

Counter Interrogation Commands

	Type Ident	ASDU	Description	Remark
n/i	<101>	C_CI_NA_1 ...(FRZ=0)	no freeze or reset	
n/i		...(FRZ=1)	counter freeze without reset	
n/i		...(FRZ=2)	counter freeze with reset	
n/i		...(FRZ=3)	counter reset	
n/i		...(RQT=0)	no counter request	
☐		...(RQT=1-4)	request counter group 1...4	
☐		...(RQT=5)	General request counter	

**NOTE**

Counter acquisition commands (freeze, reset) are preferably to be carried out by the RTU. The transmission of the counter values should be initialized by the RTU spontaneously after each freezing. Counter values that CFE receives from RTU are immediately forwarded to the RCC as far as they are configured for spontaneous transmission.

5.2.1.7 Special Communication Features

Communication Statistics

Not available.

Listening Mode

Not available.

Monitoring of Standby Channels / Standby Channel Information Buffering

Not available.

5.2.2 Interoperability Check List for Protocol IEC-60870-5-104/Slave

5.2.2.1 Overview

This companion standard presents sets of parameters and alternatives from which, subsets must be selected to implement particular telecontrol systems. Certain parameter values, such as the choice of structured or unstructured fields of the INFORMATION OBJECT ADDRESS of ASDUs represent mutually exclusive alternatives. Thus, only one value of the defined parameters is admitted per system. Other parameters, such as the listed set of different process information in command and in monitor direction allow the specification of the complete set or subsets, as appropriate for given applications. If a system is composed of equipment originating from different manufacturers it is necessary that all partners agree on the selected parameters.

The interoperability list is defined as in IEC 60870-5-104 and extended with parameters used in this standard. Parameters which are not valid for this companion standard are marked with n/i.



NOTE

In addition, the full specification of a system may require individual selection of certain parameters for certain parts of the system, such as the individual selection of scaling factors for individually addressable measured values.

5.2.2.2 Network Configurations

	Configuration	Remark
■	Point-to Point	Redundant communication links (yes no)

5.2.2.3 Device Function

Indicates the stations function

	Description	Remark
	Controlling station (Master)	Master function is standard, see respective Interoperability List for IEC104
■	Controlled station (Slave)	Specified by this document

5.2.2.4 Protocol Structure

The following table shows the structure of the end system

Selection of Application Functions of IEC 60870-5-5 according to IEC 60870-5-101	Initialization	User Process
Selection of Application Service Data Units of IEC 60870-5-101 and IEC 60870-5-104	Application (layer 7)	
APCI (Application Protocol Control Information) Transport Interface (User to TCP Interface)		

Selection of TCP/IP Protocol Suite (RFC 2200)	Transport (layer 4)
	Network (layer 3)
	Link (layer 2)
	Physical (layer 1)

5.2.2.5 Application Layer

Transmission Mode for Application Data

Mode 1 (least significant octet first), as defined in clause 4.10 of IEC 870-5-4, is used exclusively in this standard.

Common Address of ASDU

	Description	Remark
☒	2 Octets	
☐	structured	
☐	unstructured	

Information Object Address

	Description	Remark
☒	3 Octets	
☐	structured	
☐	unstructured	

Cause of Transmission

	Description	Remark
☒	2 Octets	1st octet: T, P/N, CoT Telegrams with Test bit T not evaluated 2nd octet: originator address (always zero to Master, not evaluated from Master)

Length of APDU

	Maximum length of APDU
--	------------------------

The maximum length of the APDU is 253 (default). The maximum length may be reduced per system.

Selection of Standard ASDUs

Process Information in Monitor Direction

	Type Ident	ASDU	Description	Remark
☐	<1>	M_SP_NA_1	Single-point information	
☐	<3>	M_DP_NA_1	Double-point information	
☐	<5>	M_ST_NA_1	Step position information	
n/i	<7>	M_BO_NA_1	Bitstring of 32 bit	

	Type Ident	ASDU	Description	Remark
☐	<9>	M_ME_NA_1	Measured value, normalized value	
☐	<11>	M_ME_NB_1	Measured value, scaled value	
☐	<13>	M_ME_NC_1	Measured value, short floating point value	
☐	<15>	M_IT_NA_1	Integrated totals	
n/i	<20>	M_PS_NA_1	Packed single-point information with status change detection	
n/i	<21>	M_ME_ND_1	Measured value, normalized value without quality descriptor	
☐	<30>	M_SP_TB_1	Single-point information with time tag CP56Time2a	
☐	<31>	M_DP_TB_1	Double-point information with time tag CP56Time2a	
☐	<32>	M_ST_TB_1	Step position information with time tag CP56Time2a	
n/i	<33>	M_BO_TB_1	Bitstring of 32 bit with time tag CP56Time2a	
☐	<34>	M_ME_TD_1	Measured value, normalized value with time tag CP56Time2a	
☐	<35>	M_ME_TE_1	Measured value, scaled value with time tag CP56Time2a	
☐	<36>	M_ME_TF_1	Measured value, short floating point value with time tag	
☐	<37>	M_IT_TA_1	Integrated totals with time tag CP56Time2a	
n/i	<38>	M_EP_TD_1	Event of protection equipment with time tag CP56Time2a	
n/i	<39>	M_EP_TE_1	Packed start events of protection equipment with time tag CP56Time2a	
n/i	<40>	M_EP_TF_1	Packed output circuit information of protection equipment with time tag CP56Time2a	

Process Information in Control Direction

	Type Ident	ASDU	Description	Remark
☐	<45>	C_SC_NA_1	Single command	
☐	<46>	C_DC_NA_1	Double command	

	Type Ident	ASDU	Description	Remark
☐	<47>	C_RC_NA_1	Regulating step command	
☐	<48>	C_SE_NA_1	Set point command, normalized value	
☐	<49>	C_SE_NB_1	Set point command, scaled value	
☐	<50>	C_SE_NC_1	Set point command, short floating point value	
n/i	<51>	C_BO_NA_1	Bitstring of 32 bits	
☐	<58>	C_SC_TA_1	Single command with time tag CP56Time2a	
☐	<59>	C_DC_TA_1	Double command with time tag CP56Time2a	
☐	<60>	C_RC_TA_1	Regulating step command with time tag CP56Time2a	
☐	<61>	C_SE_TA_1	Set point command, normalized value with time tag CP56Time2a	
☐	<62>	C_SE_TB_1	Set point command, scaled value with time tag CP56Time2a	
☐	<63>	C_SE_TC_1	Set point command, short floating point value with time tag CP56Time2a	
n/i	<64>	C_BO_TC_1	Bitstring of 32 bit with time tag CP56Time2a	

System Information

System Information in Monitoring Direction

	Type Ident	ASDU	Description	Remark
☐	<70>	M_EI_NA_1	End of initialization (COI)	After sending TK70 to RCC a GI request is expected

System Information in Control Direction

	Type Ident	ASDU	Description	Remark
☐	<100>	C_IC_NA_1	General Interrogation command	
☐	<101>	C_CI_NA_1	Counter interrogation command	
☐	<102>	C_RD_NA_1	Read command	
☐	<103>	C_CS_NA_1	Clock synchronization command	Positive or negative confirmation configurable. Time info ignored
☐	<105>	C_RP_NA_1	Reset process command	Command confirmed but not executed
☐	<107>	C_TC_TA_1	Test command with time tag CP56time2a	

Parameter Transmission in Control Direction

	Type Ident	ASDU	Description	Remark
n/i	<110>	P_ME_NA_1	Parameter of measured value, normalized value	
n/i	<112>	P_ME_NB_1	Parameter of measured value, scaled value	
n/i	<113>	P_ME_NC_1	Parameter of measured value, short floating point value	
n/i	<114>	P_AC_NA_1	Parameter activation	

File Transfer

	Type Ident	ASDU	Description	Remark
n/i	<120>	F_FR_NA_1	File ready	
n/i	<121>	F_SR_AN_1	Section ready	
n/i	<122>	F_SC_NA_1	Call directory, select file, call file, call selection	
n/i	<123>	F_LS_NA_1	Last section	
n/i	<124>	F_AF_NA_1	Ack file, ack section	
n/i	<125>	F_SG_NA_1	Segment	
n/i	<126>	F_DR_TA_1	Directory	

Type Identifier and Cause of Transmission Assignments

The following chart is an extract from IEC60870-5 Companion Standard 104. It shows for each message Type ID the supported Cause of Transmission assignments (white boxes). Shaded boxes show combinations that are not supported.

IEC104SL		Cause of Transmission (CoT)																			
Type Identification		1	2	3	4	5	6	7	8	9	10	11	12	13	20	37	44				
		36	41	47																	
1	M_SP_NA_1	*	*																		
3	M_DP_NA_1	*	*																		
5	M_TP_NA_1	*	*																		
9	M_ME_NA_1		*																		
11	M_ME_NB_1		*																		
13	M_ME_NC_1		*																		
15	M_IT_NA_1	*	*																		
30	M_SP_TB_1																				
31	M_DP_TB_1																				
32	M_ST_TB_1																				
34	M_ME_TD_1																				
35	M_ME_TE_1																				
36	M_ME_TF_1																				
37	M_IT_TB_1																				
45	C_SC_NA_1																				
46	C_DC_NA_1																				
47	C_RC_NA_1																				
48	C_SE_NA_1																				
49	C_SE_NB_1																				
50	C_SE_NC_1																				
58	C_SC_TA_1																				
59	C_DC_TA_1																				
60	C_RC_TA_1																				
61	C_SE_TA_1																				
62	C_SE_TB_1																				
63	C_SE_TC_1																				
70	M_EI_NA_1																				
100	C_IC_NA_1																				
101	C_CI_NA_1																				
102	C_RD_NA_1																				
103	C_CS_NA_1																				
105	C_RP_NA_1																				
107	C_TS_TA_1																				

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Figure 5-7 CoT for each Message Type (* Note: Standard does not mandate this - only for Legacy Control Center support)

5.2.2.6 Basic Application Functions

General

	Description	Remark
☐	Remote station initialization	
☐	Cyclic data transmission	
☐	Read procedure	
☐	Spontaneous transmission	

Double Transmission of Information Objects with Cause of Transmission Spontaneous

The following type identifications may be transmitted in succession caused by a single status change of an information object. The particular information object addresses for which double transmission is enabled are defined in a project-specific list.

	Description	Remark
☐	Single-point information	M_SP_NA_1 and M_SP_TB_1
☐	Double-point information	M_DP_NA_1 and M_DP_TB_1
☐	Step position information	M_ST_NA_1 and M_ST_TB_1
n/i	Bitstring of 32 bit3	M_BO_NA_1 and M_BO_TB_1 (if defined for a specific project)
n/i	Measured value, normalized value	M_ME_NA_1 and M_ME_TD_1
n/i	Measured value, scaled value	M_ME_NB_1 and M_ME_TE_1
n/i	Measured value, short floating point number	M_ME_NC_1 and M_ME_TF_1

Station Interrogation

	Description	Remark
☒	global GI	The following COT parameters are applied: in control direction: <6> activation (C_IC ACT) in monitoring direction: <7> activation confirmation (C_IC ACTCON) <10> activation termination (C_IC ACTTERM)
☐	group 1 ... 16	

Clock Synchronization

☐	Clock Synchronization commands from RCC (accepted but not evaluated)
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Command Transmission

	Description	Remark
☐	Direct command transmission	
☐	Direct set-point command transmission	
☐	Select and execute command	
☐	Select and execute set point command	
☐	C_SE ACTTERM used	
n/i	Supervision of maximum delay in command direction	

Qualifier for Command Transmission

	QOC	Description	Remark
☐	<0>	no additional definition	Preferably, duration of individual command output pulses are to be parameterized in the RTU
n/i	<1>	Short pulse duration (duration determined by a system parameter in the outstation)	

	QOC	Description	Remark
n/i	<2>	Long pulse duration (duration determined by a system parameter in the outstation)	
n/i	<3>	Persistent output	

**NOTE**

Qualifier of Command (QOC) unequal zero can be configured on the RTU side of CFE

Transmission of Integrated Totals

	Description	Remark
☐	Mode A	Local freeze with spontaneous transmission
☐	Mode B	Local freeze with counter interrogation
n/i	Mode C	Freeze and transmit by counter interrogation commands
n/i	Mode D	Freeze by counter interrogation command, frozen values reported

Parameter Loading

	Description	Remark
n/i	Threshold value	
n/i	Smoothing factor	
n/i	Low limit for transmission of measured values	
n/i	High limit for transmission of measured values	

Parameter Activation

	Description	Remark
n/i	Parameter activation	Act/deact of persistent cyclic or periodic transmission of the addressed object

Test Procedure

	Description	Remark
n/i	Test procedure	Act/deact of persistent cyclic or periodic transmission of the addressed object

File Transfer**File Transfer in Monitor Direction**

	Description	Remark
n/i	Transparent file	
n/i	Transmission of disturbance data of protection	
n/i	Transmission of sequences of events	
n/i	Transmission of sequences of recorded analogue values	

File Transfer in Control Direction

	Description	Remark
n/i	Transparent file	

Background Scan

	Description	Remark
☐	Background scan	

Definitions of Time-Outs

Parameter	Default Value	Meaning	Selected Value
t_0	30s	Time out of connection establishment	30s
t_1	15s	Time out of send or test ADPUs	15s
t_2	10s	Time out for acknowledges in case of no data messages $t_2 < t_1$	10s
t_3	20s	Time out for sending test frames in case of a long idle state	20s

Maximum range of values for all timeouts: 1 to 255 s, accuracy 1 s

Maximum number of Outstanding I Format ADPUs k and Latest Acknowledge

Parameter	Default Value	Meaning	Selected Value
k	12 ADPUs	Maximum difference receive sequence number to send state variable	<configurable>
w	8 ADPUs	Latest acknowledge after receiving w I-format ADPUs	<configurable>

Maximum range of values k: 1 to 32767 ADPUs, accuracy 1 APDU

Maximum range of values w: 1 to 32767 ADPUs, accuracy 1 APDU (Recommendation: w should not exceed 2/3 of k)

Port Number

Parameter	Value	Remarks
Portnumber	2404	Any port number can be configured. 2404 is default value and specified in IEC

RFC 2200 Suite

RFC 2200 is an official Internet Standard. It describes the state of standardization of protocols used in the internet as determined by the Internet Architecture Board (IAB). It offers a broad spectrum of actual standards used in the internet. The suitable selection of documents from RFC 2200 defined in this standard for given projects is chosen by the user of this standard.

☐	Ethernet
☐	Serial X.21
☐	Other selection

5.2.2.7 Special Communication Features

Communication Statistics

Not available.

Listening Mode

Not available.

Monitoring of Standby Channels / Standby Channel Information Buffering

Not available.